

Final logistics

A2 grades: could be released soon

Today: last lecture, not the last class

Wed: last programming labs

Next Mon: milestone day (similar to FP discussion)

Next Wed: last class, extended office hours

Uncertainty & Probability

CMSC471: Introduction to Information Visualization

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April 26, 2026

University of Maryland, College Park

<https://fumeng-yang.github.io/CMSC471>

What we talk about when we
talk about “uncertainty”...

"Uncertainty" can mean ...

Doubt

Risk

Variability

Error

Lack of Knowledge

Hedging

...

in domains like



Healthcare

Public policy (e.g., COVID)

Politics

New outlets

Weather / natural disaster

AI/ML/DL/LLM

....

U.S. pre-election **win** probabilities

```
graph TD; A[U.S. pre-election win probabilities] --> B[Based on polls, demographics, etc.]; C[NOT those you see on election night] --> D[Based on counted votes, demographics, etc.]
```

Based on **polls**,
demographics, etc.

NOT those you see on election night

Based on **counted votes**,
demographics, etc.

U.S. electoral system

Every **four** years for President and VP.

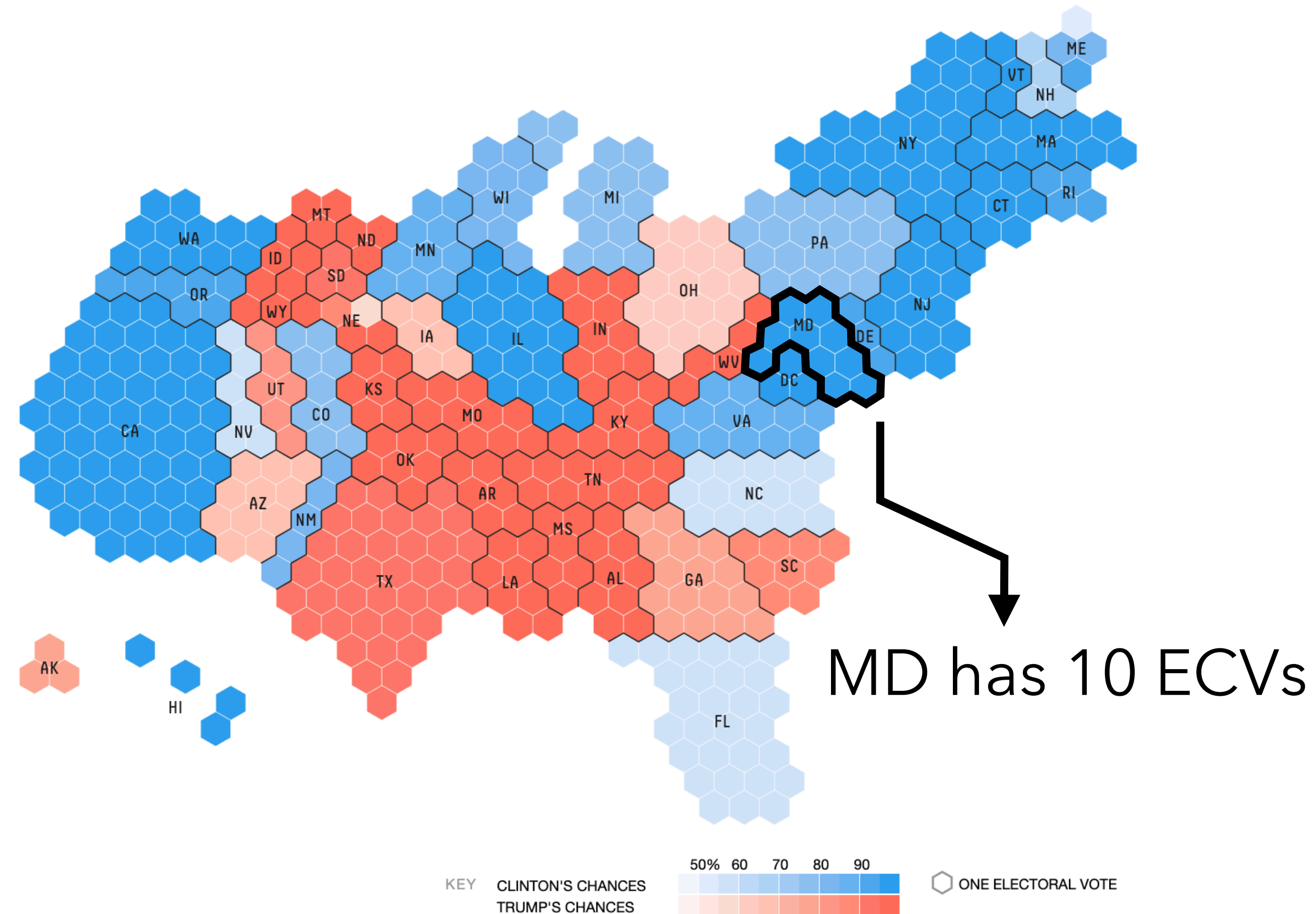
Citizens cast their votes for a slate of electors in **their state** pledged to their preferred candidate.

These electors then cast votes in the **electoral college**, which officially determines the winner.

A candidate must secure **a majority of** the 538 Electoral College votes—at least 270—to become president.

It's all about the 538 Electoral College votes

Here's a map of the country, with each state sized by its number of electoral votes and shaded by the leading candidate's chance of winning it.

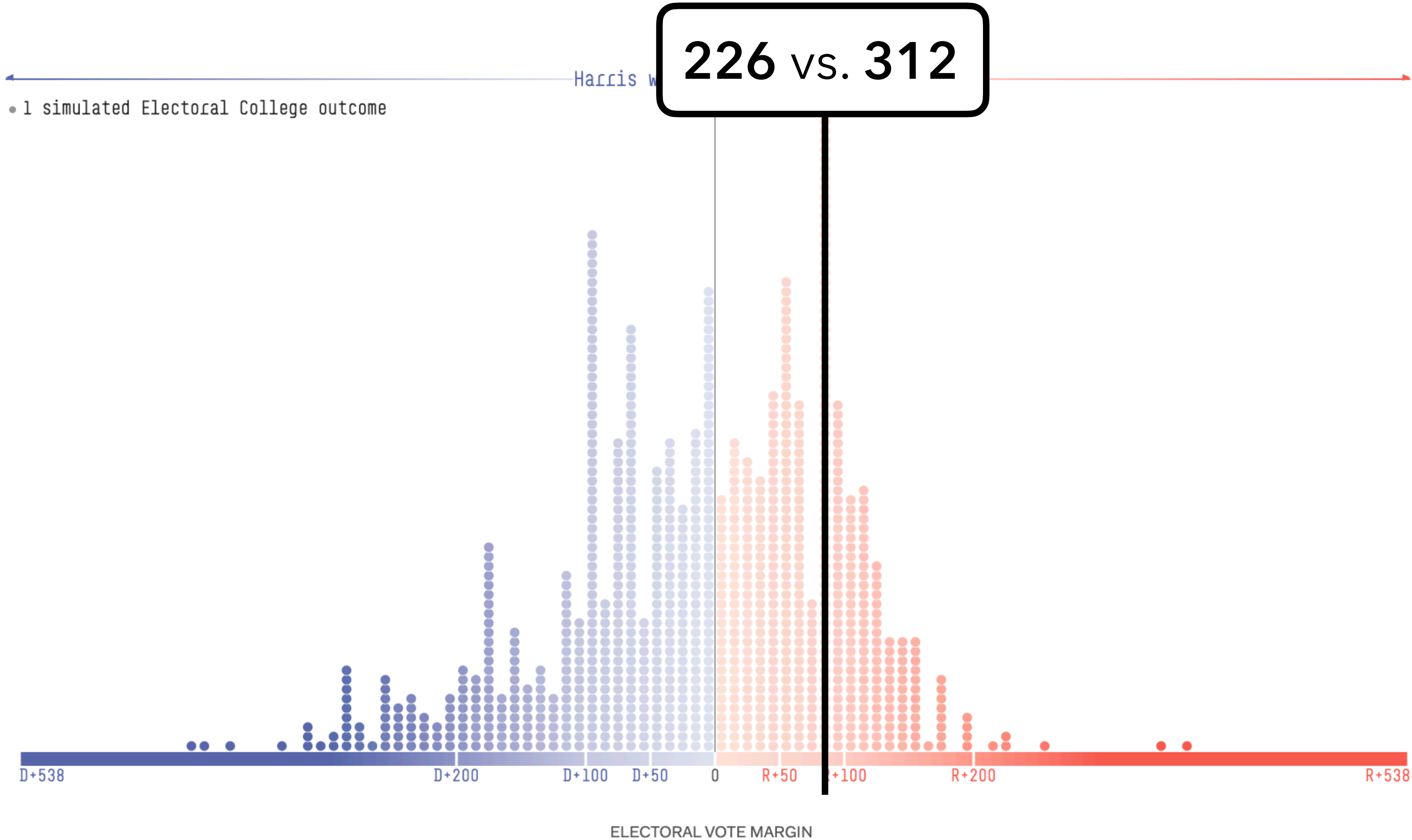


2024

Harris: **226** vs. Trump: **312**

Harris wins 50 times out of 100
in our simulations of the 2024 presidential election.
Trump wins 49 times out of 100.

There is a less than 1-in-100 chance of no Electoral College winner.



On the final day, Kamala Harris moves into a narrow lead

Last updated on November 5th 2024

226 vs. 312

National forecast

How this works

Key states

Michigan

Wisconsin

Pennsylvania

Nevada

Georgia

North Carolina

Simulated elections*



Kamala Harris wins
56 in 100 times

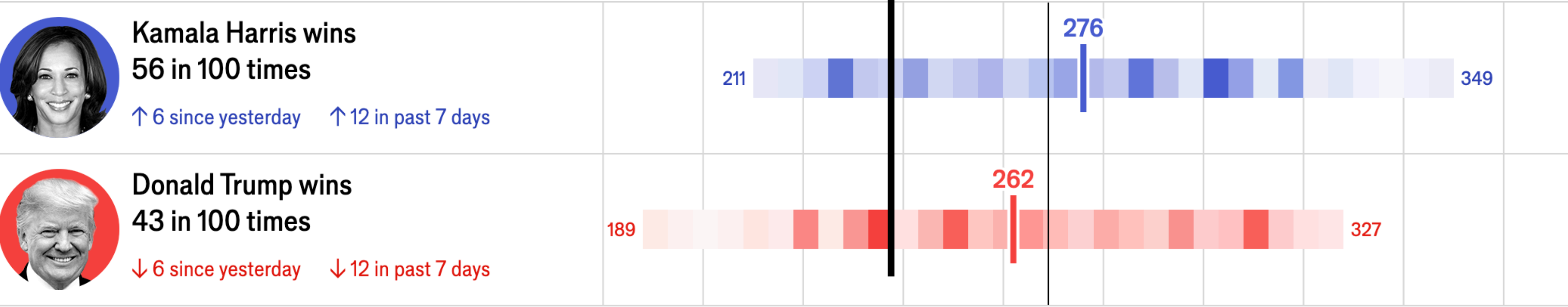
↑ 6 since yesterday ↑ 12 in past 7 days



Donald Trump wins
43 in 100 times

↓ 6 since yesterday ↓ 12 in past 7 days

Likely range Median



270 electoral votes to win

*The number of simulations each candidate wins approximates their chance of winning the election

<https://www.economist.com/interactive/us-2024-election/prediction-model/president>

2016

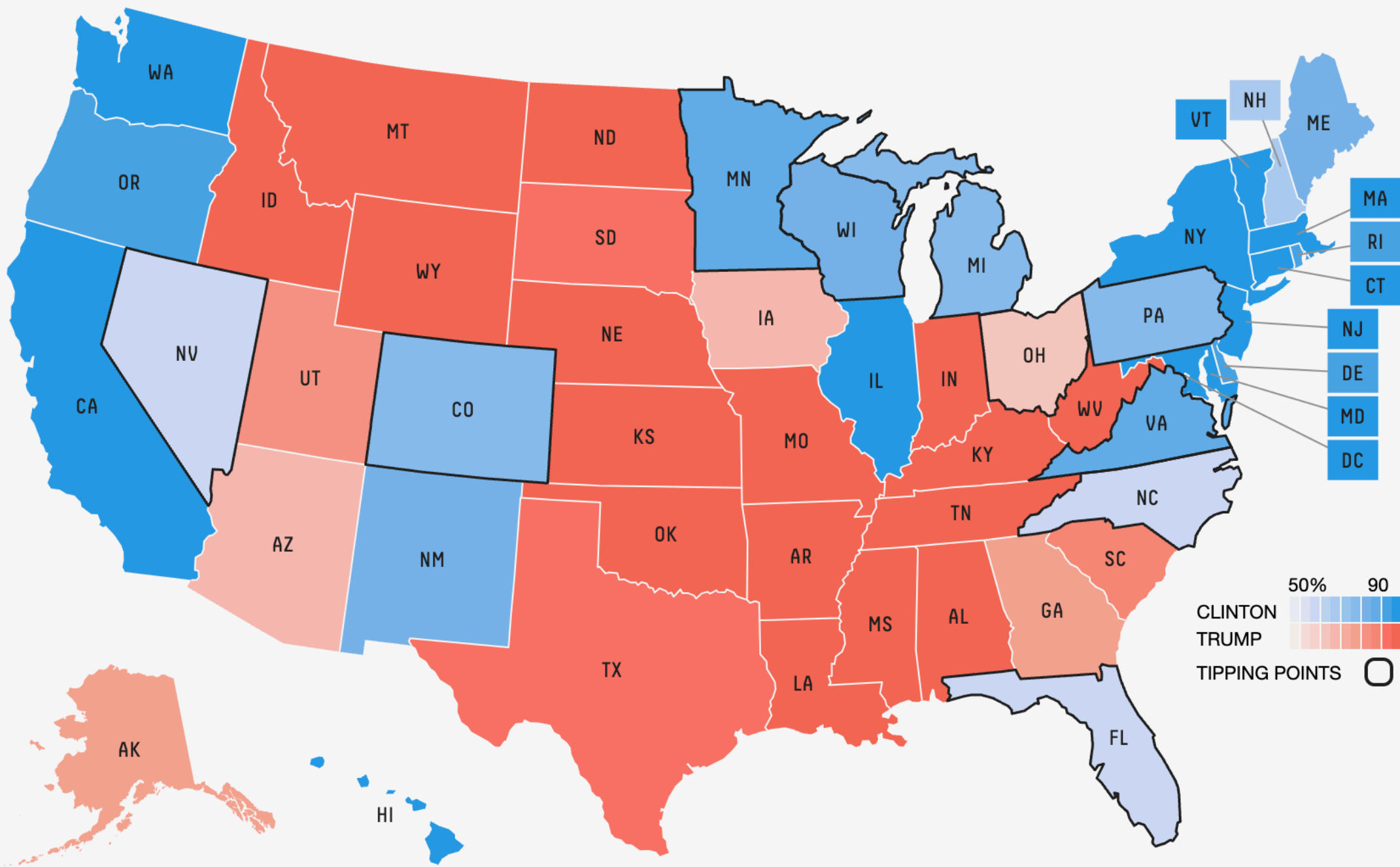
Clinton: **232** vs. Trump: **306**

2020

Biden: **306** vs. Trump: **232**

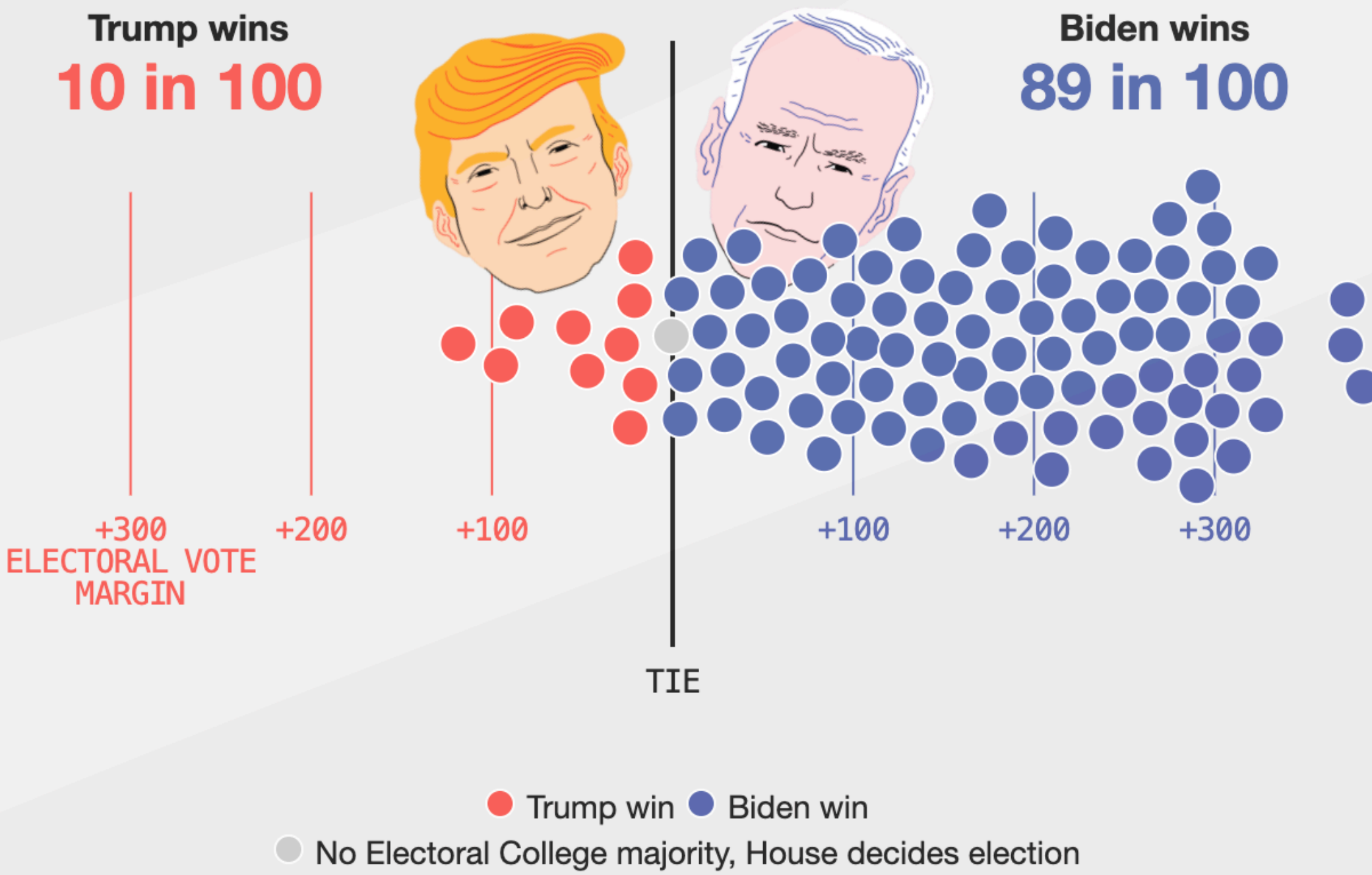
Who will win the presidency?

Chance of winning



Biden is *favored* to win the election

We simulate the election 40,000 times to see who wins most often. The sample of 100 outcomes below gives you a good idea of the range of scenarios our model thinks is possible.

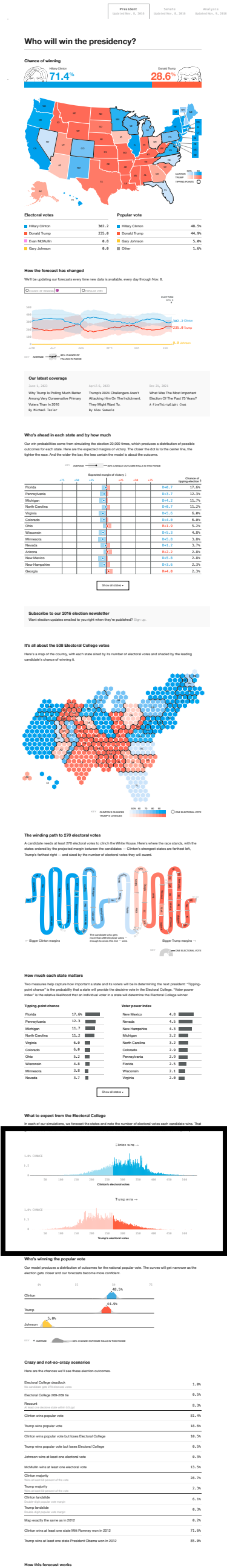
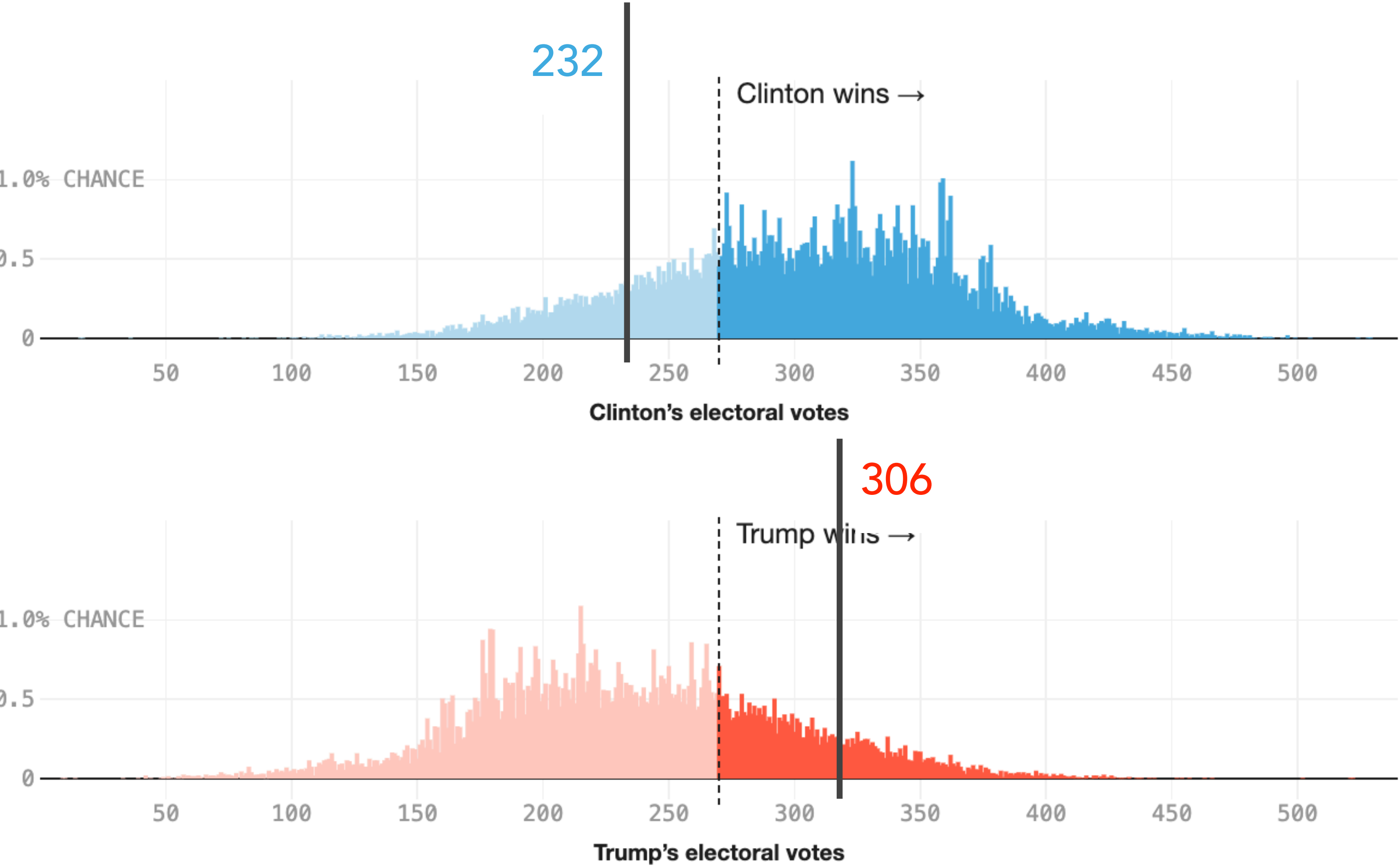


Is the 2016 forecast a bad one?
What about the 2020 forecast? 2024?

We may feel 70 vs. 30 is a huge difference
(70 is twice as much as 30?)

The real difference

In each of our simulations, we forecast the states and note the number of electoral votes each candidate wins. That gives us a distribution for each candidate, where the tallest bar is the outcome that occurred most frequently.



Does **305** vs. **235** feel like a huge difference?

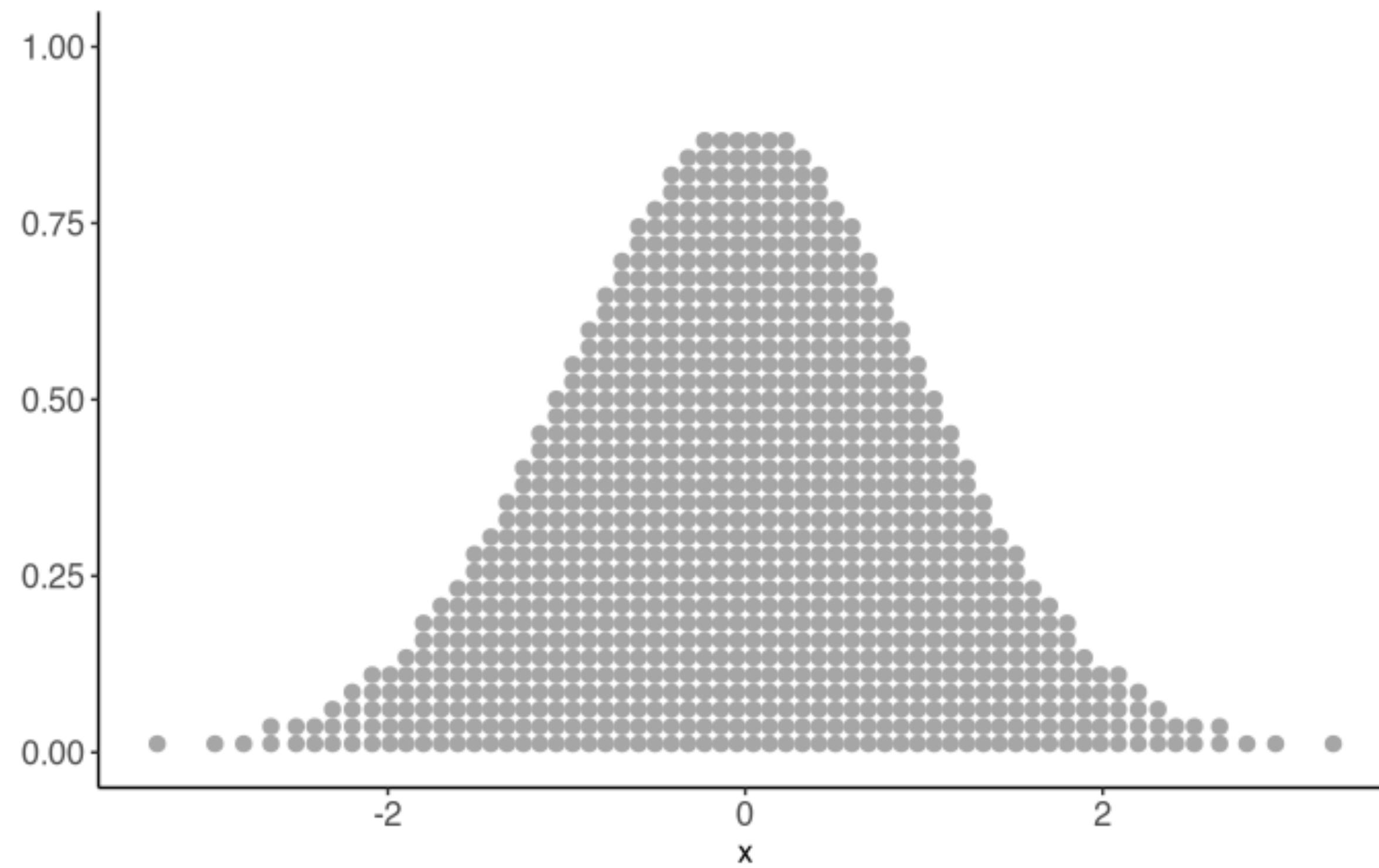
(Sidebar — what is probability?

A large sample of a measure

```
[1] -0.430268965 -1.319506017 -1.722383890 1.182518574 1.621082251 0.338481986 -0.142101289 0.134509120 -1.576111974
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[226] 0.333177899 1.443072674 1.845258117 1.550589875 0.327883169 0.921096591 1.060317897 -0.154773352 1.042888363
[235] -0.617355905 -0.956142986 0.283230695 1.792830694 0.312053322 0.905878812 -1.534120544 0.430268965 -1.064725869
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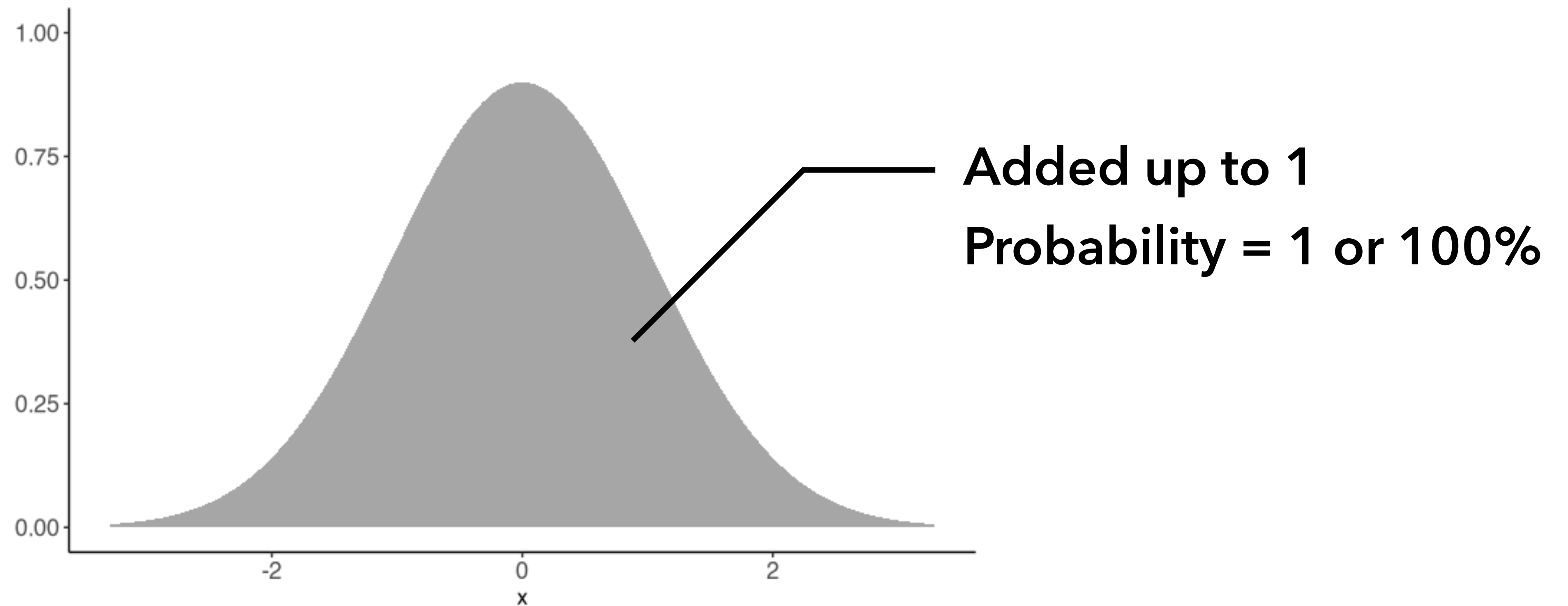
...

If we stack them up



If we have infinite samples

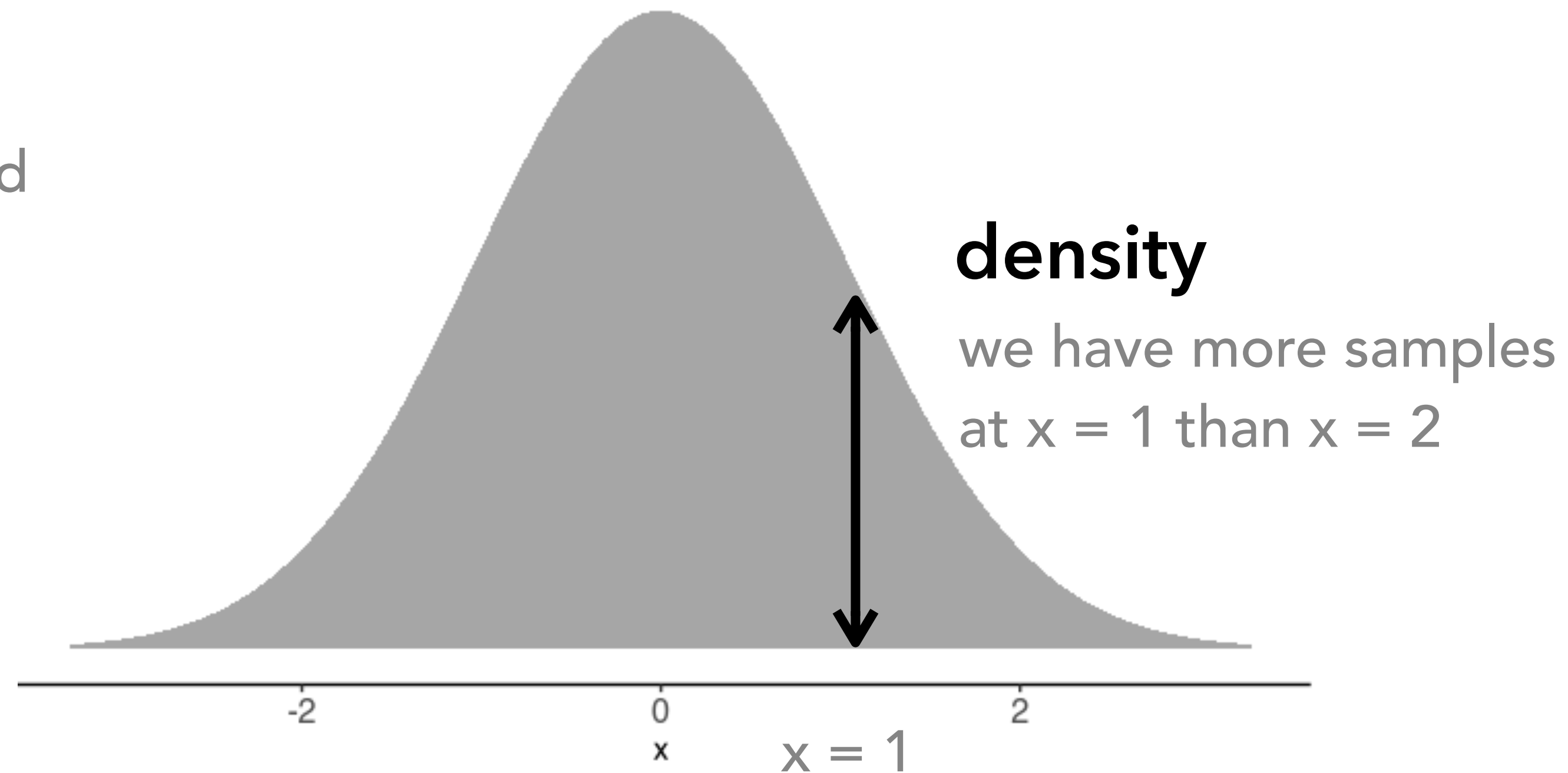
A probabilistic distribution



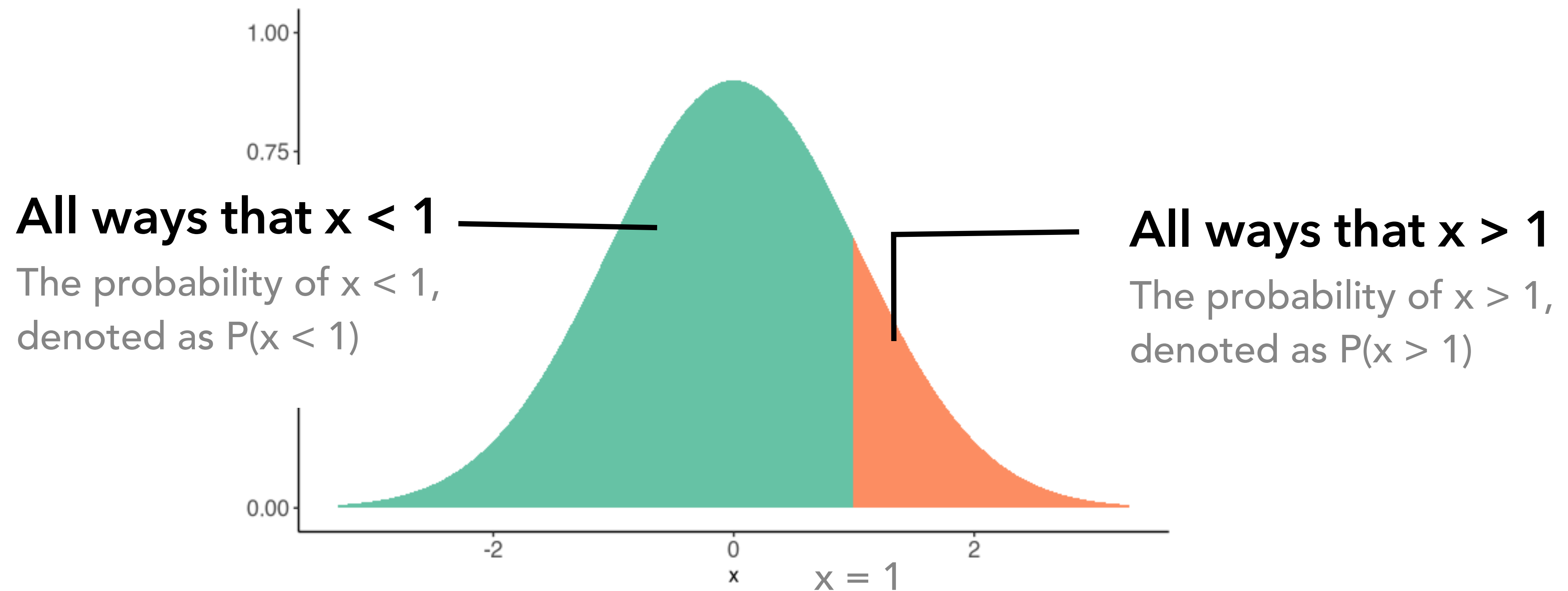
If we have infinite samples

A probabilistic distribution

So the y-axis is not a
meaningful quantity, and
sometimes not drawn



A probabilistic distribution

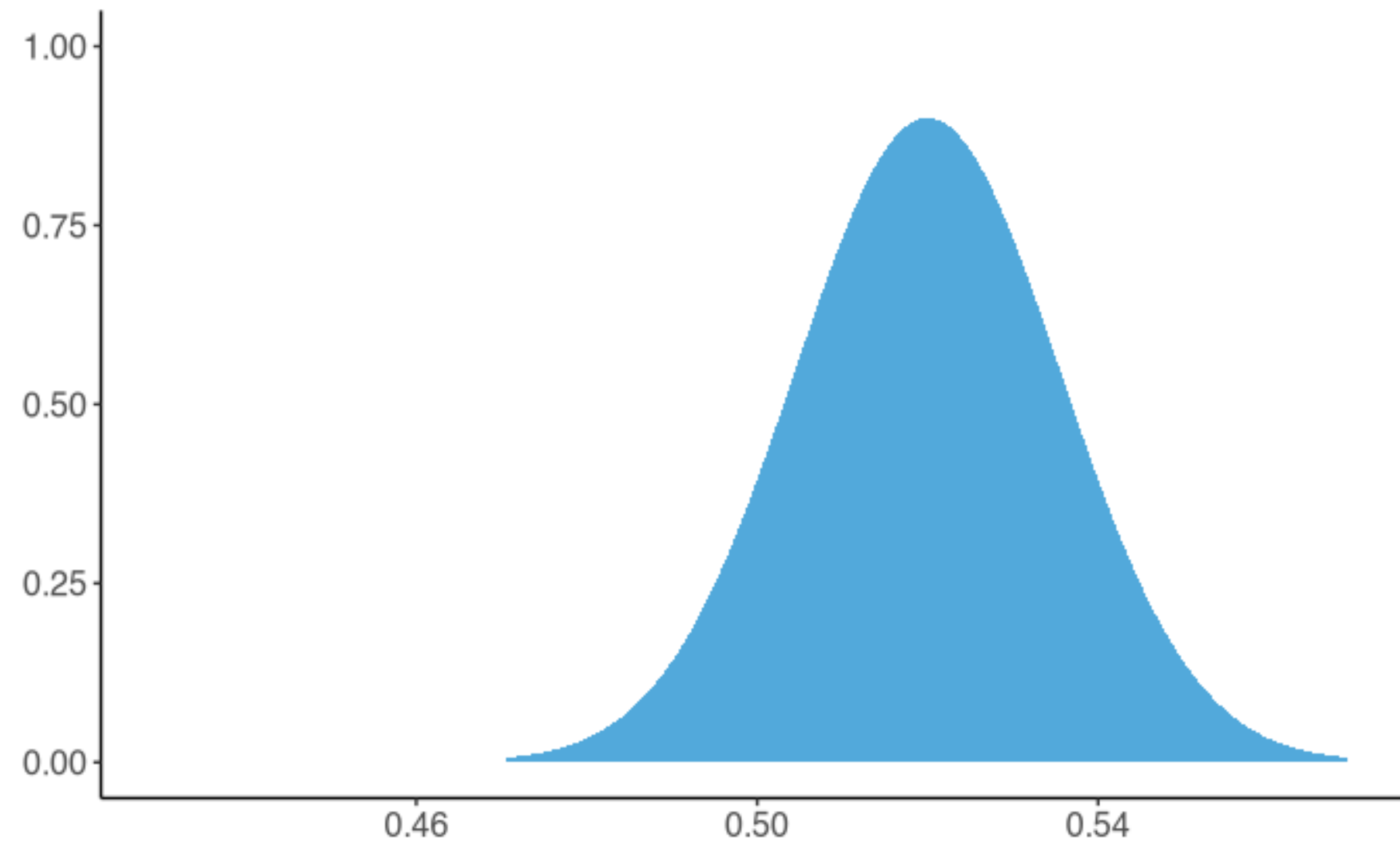


— back to win probability)

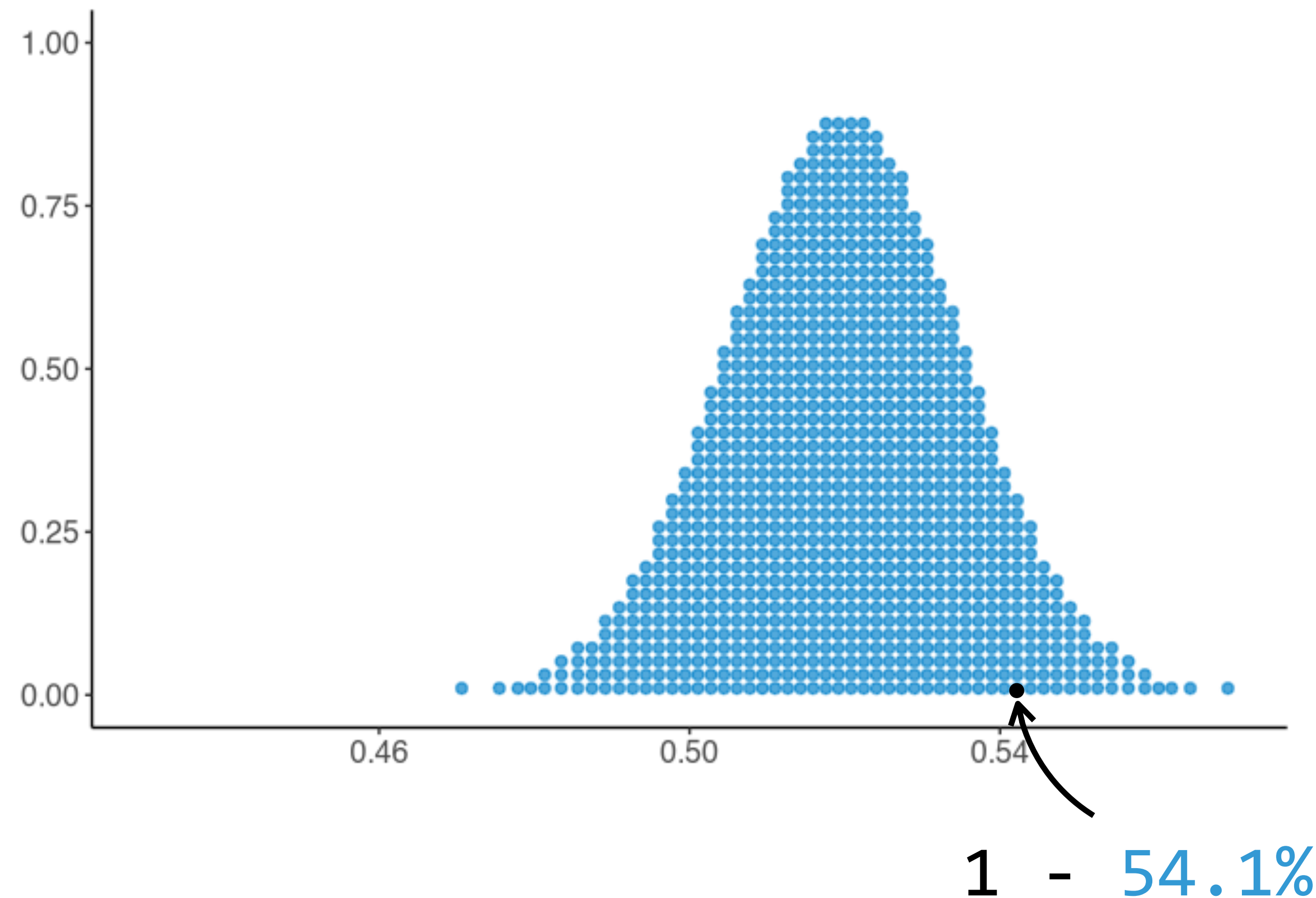
Let's simplify election college into vote share

consider a state election: ignore winner-take-all, the third party, etc.

Hypothetical Democratic vote share in Maryland



Hypothetical Democratic vote share in Maryland

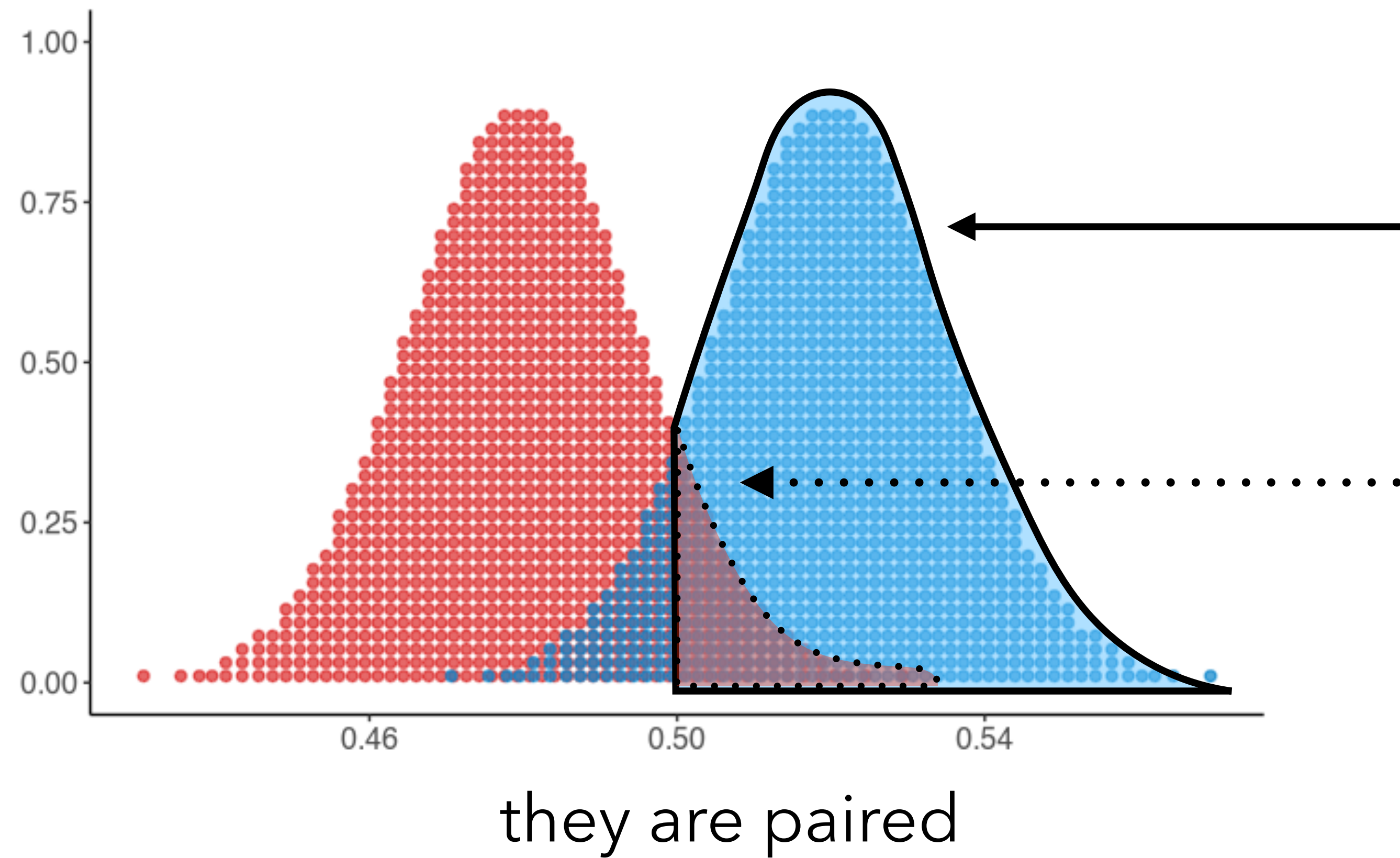


We draw a large sample (or a large-enough representative sample) from the distribution. Then, $1 - x$ to get Republican's vote share

This is a simplified version, we can use a joint distribution if we want the third-party votes

Hypothetical Democratic vote share in Maryland

Among all these pairs, we count ... and divided by the total number



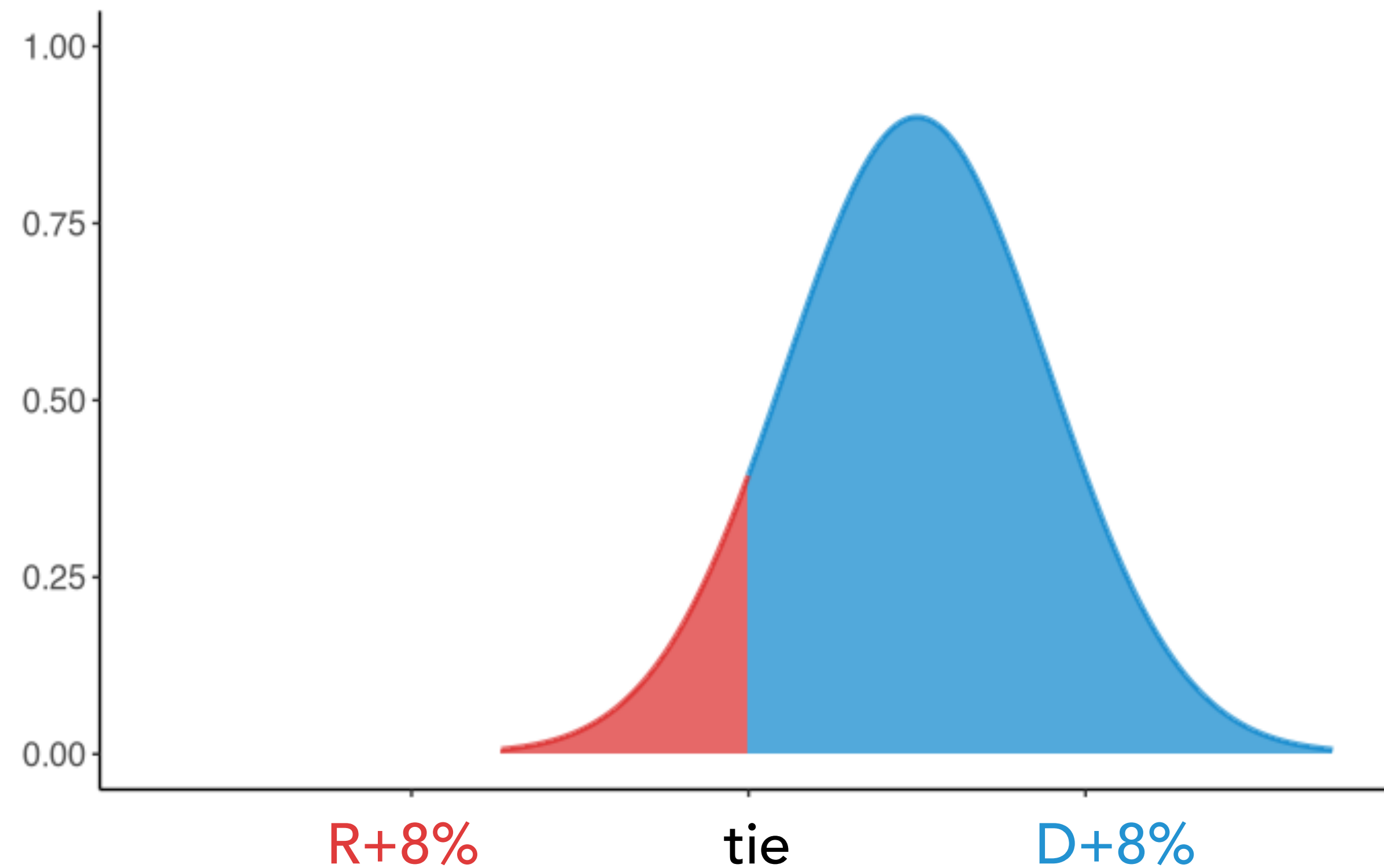
the number of D win
 $P(x_d > 50\%) = 90.9\%$

the number of R win
 $P(x_r > 50\%) = 9.1\%$

the number of tie
 $P(x_r = x_d) = 0$

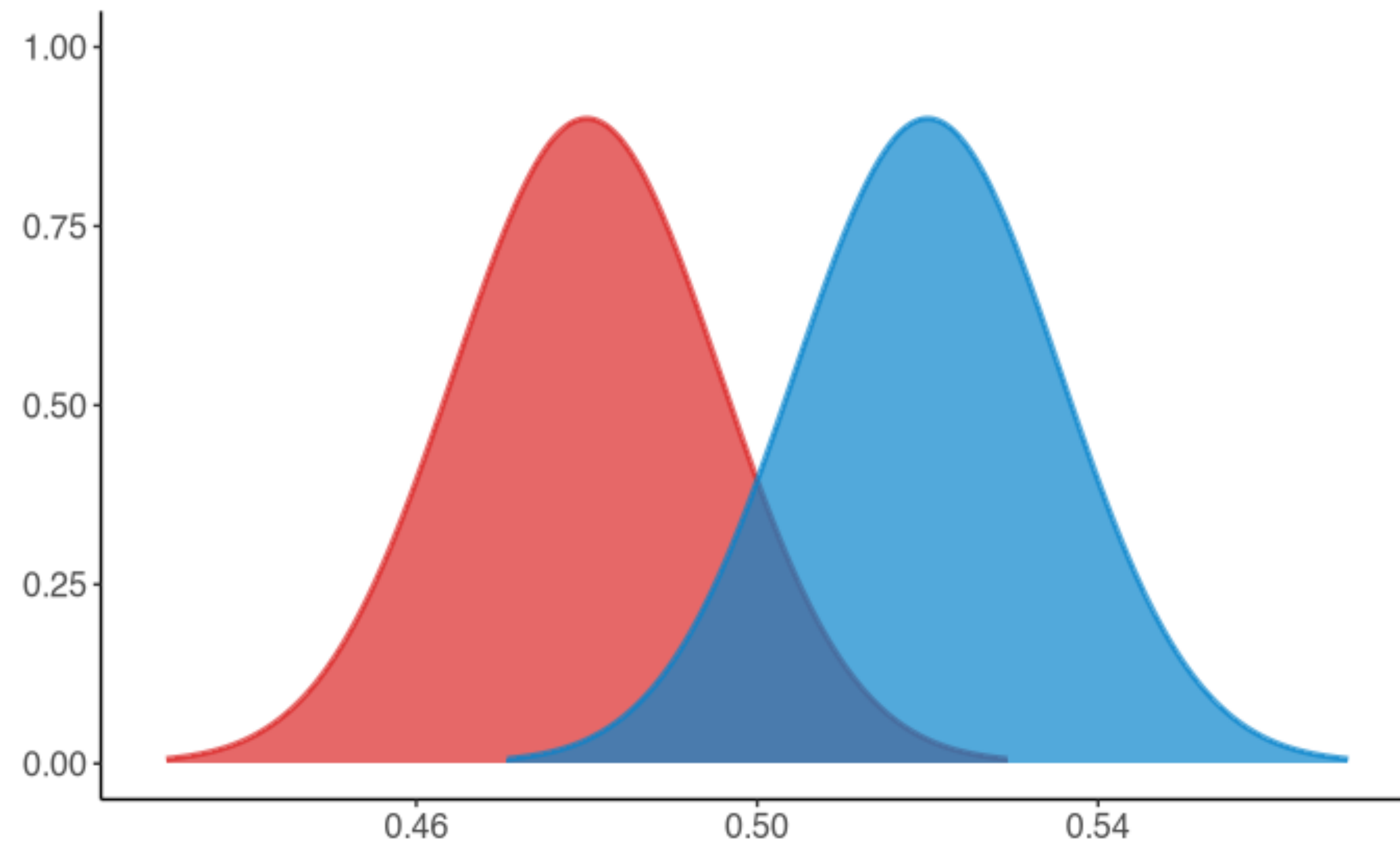
Hypothetical vote share margin in Maryland

Sometimes we just need to show one



Hypothetical vote share in Maryland

Most of the time you see two distributions



the number of D win

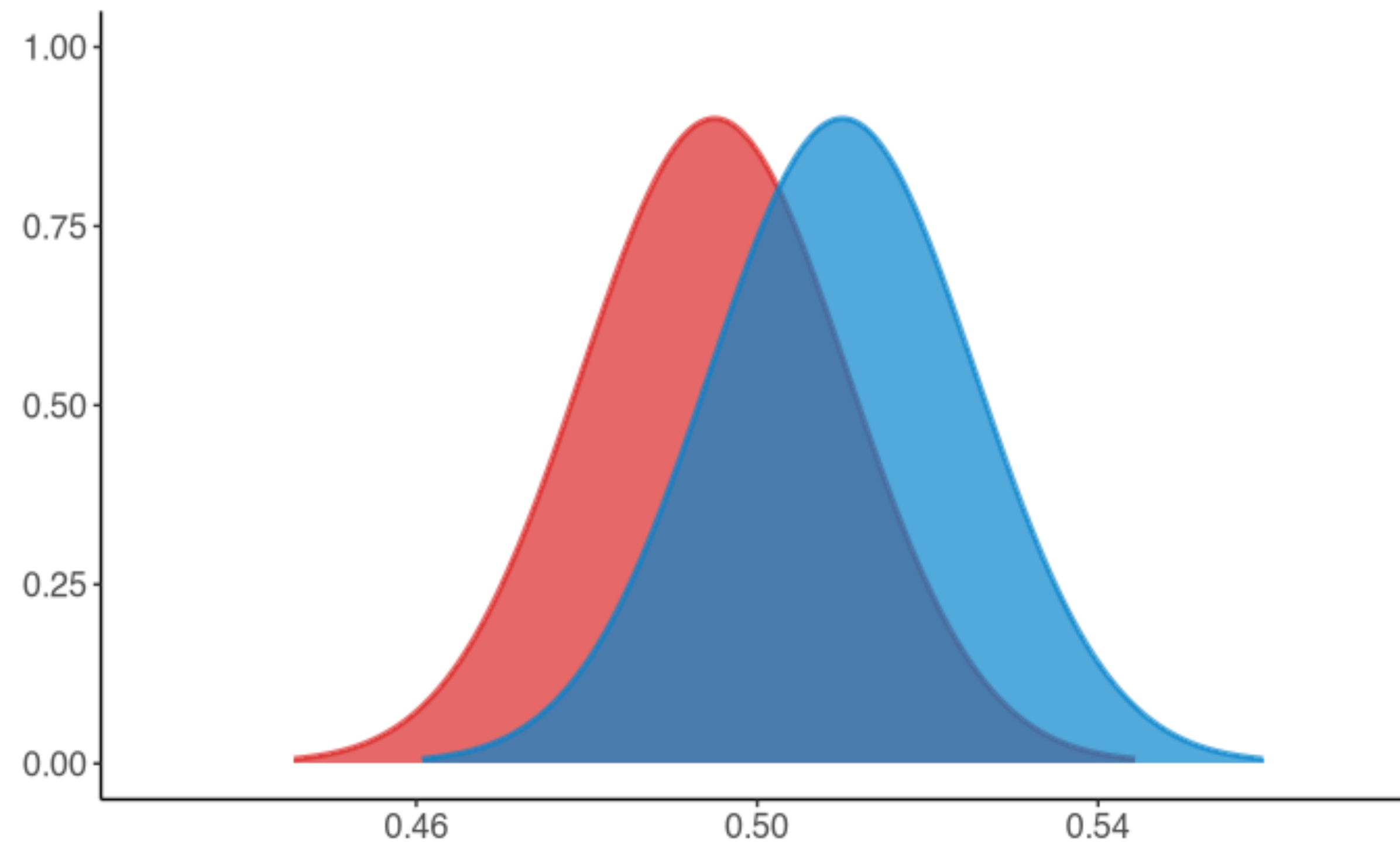
$$P(x_d > 50\%) = 90.9\%$$

the number of R win

$$P(x_r > 50\%) = 9.1\%$$

Hypothetical vote share in Maryland

51% vs. 49%



although the difference in
win probabilities are larger!

the number of D win

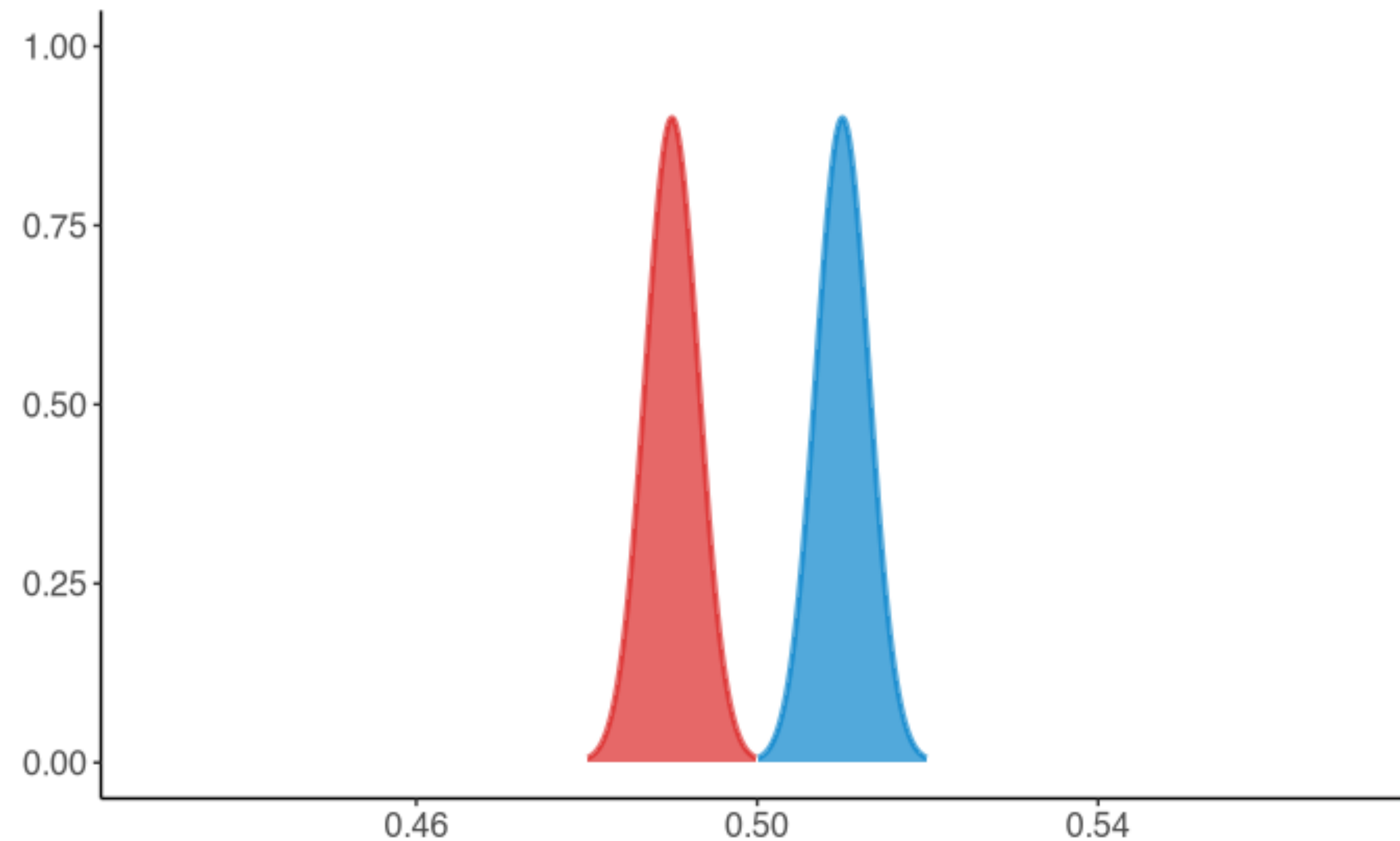
$$P(x_d > 50\%) = 74.8\%$$

the number of R win

$$P(x_r > 50\%) = 25.2\%$$

Hypothetical vote share in Maryland

51% vs. 49%



the number of D win

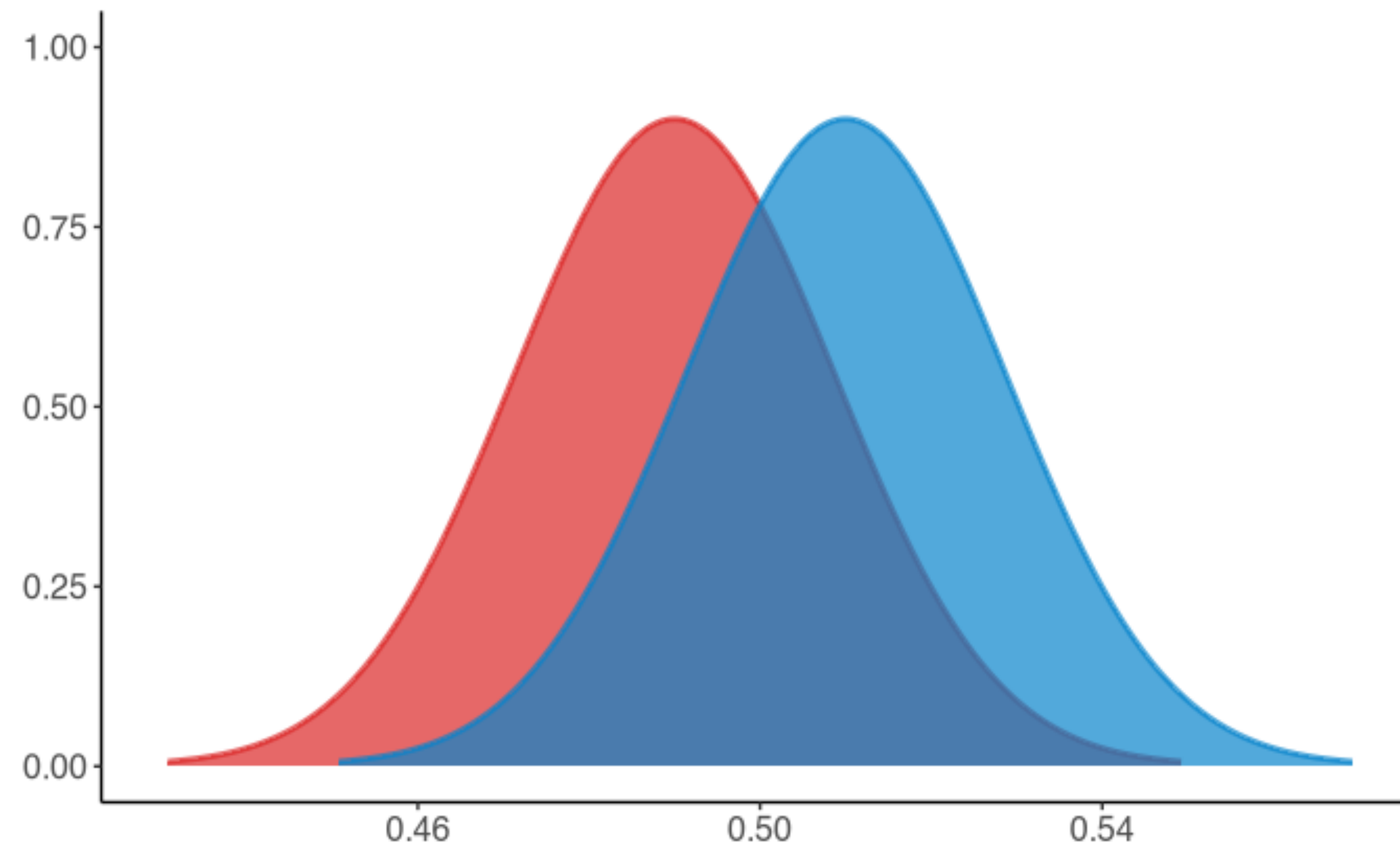
$$P(x_d > 50\%) = 100\%$$

the number of R win

$$P(x_r > 50\%) = 0$$

Hypothetical vote share in Maryland

51% vs. 49%



the number of D win

$$P(x_d > 50\%) = 71.1\%$$

the number of R win

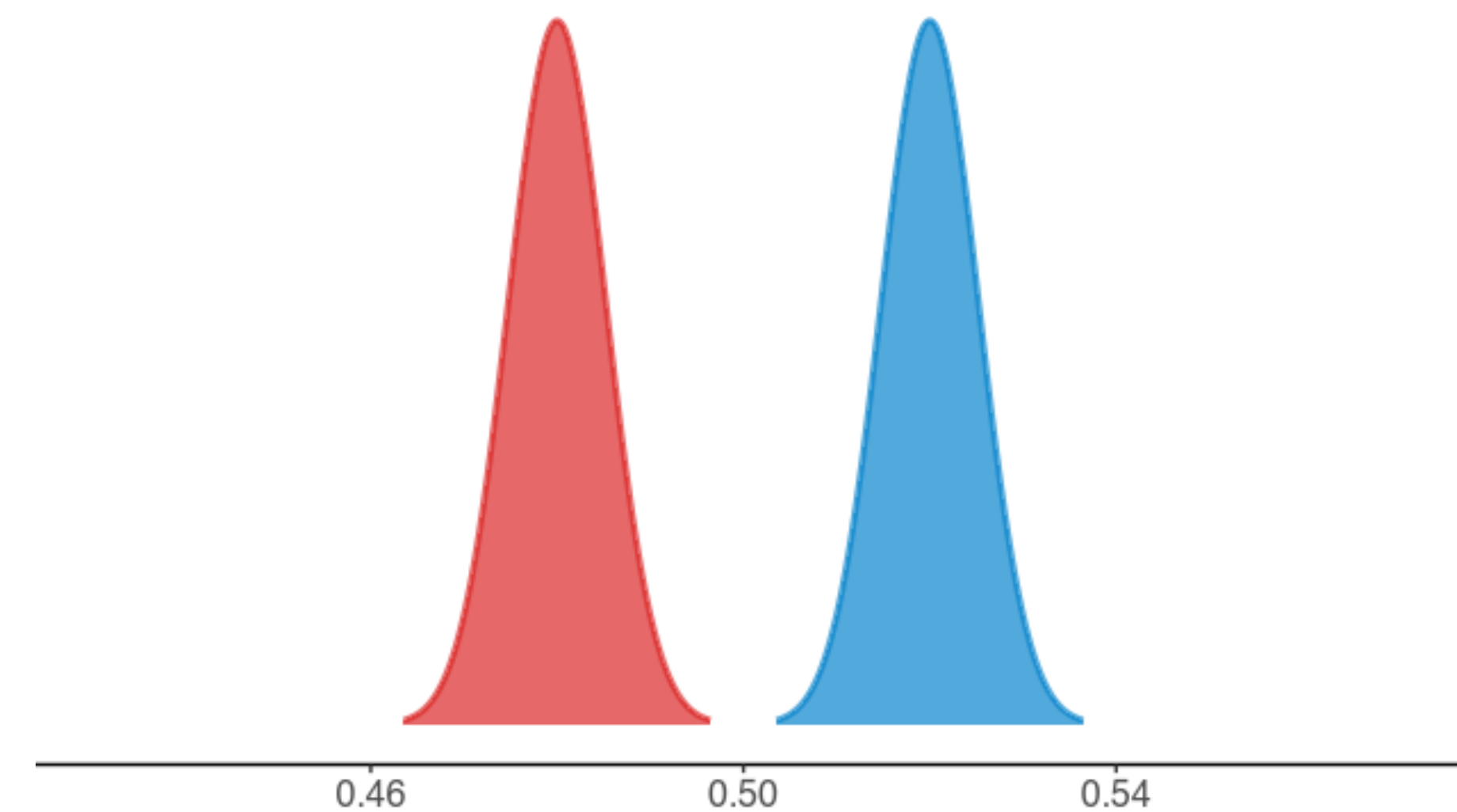
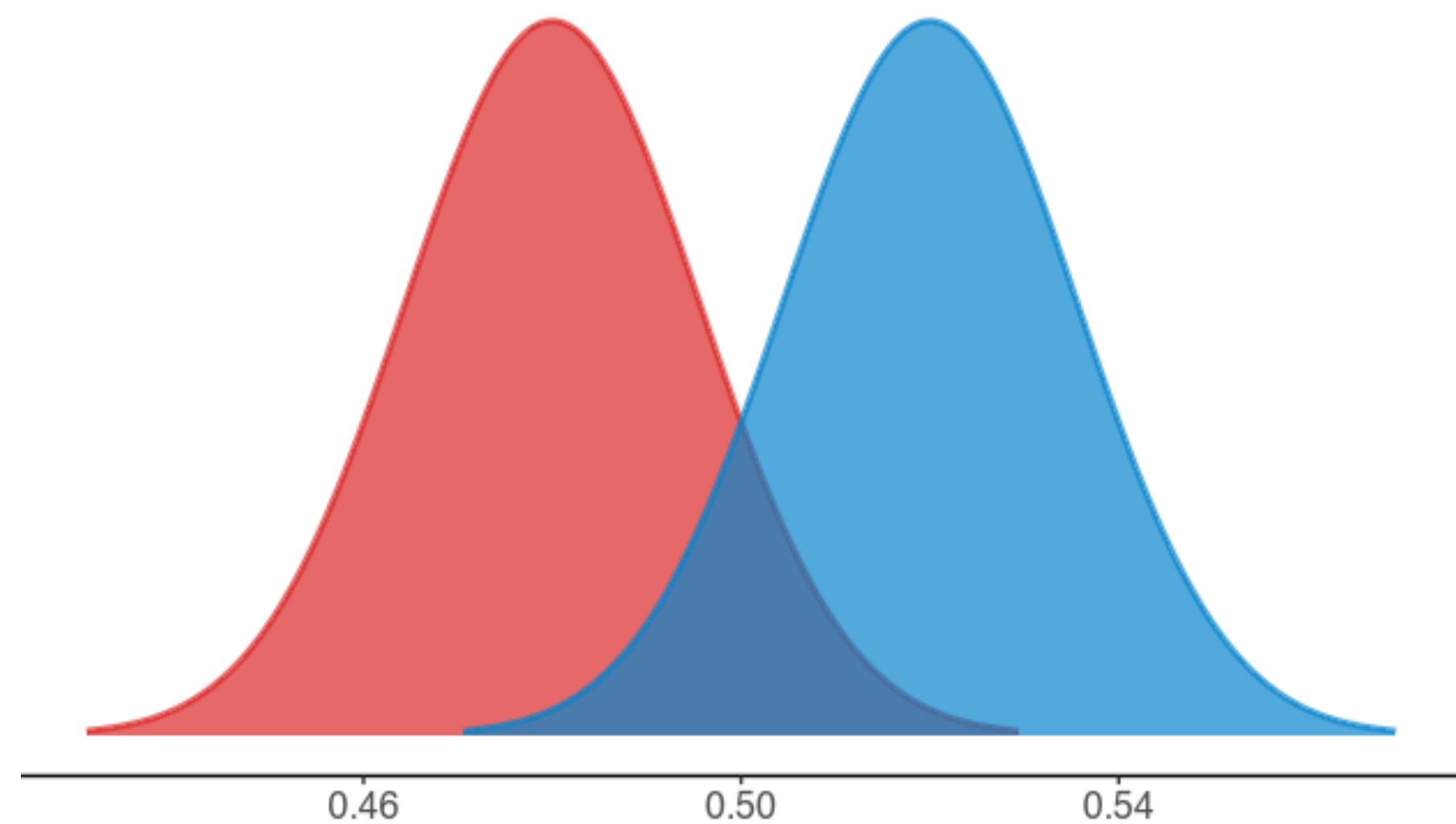
$$P(x_r > 50\%) = 28.9\%$$

**Mean and variance together
determine probabilities**

We roughly know US vote share

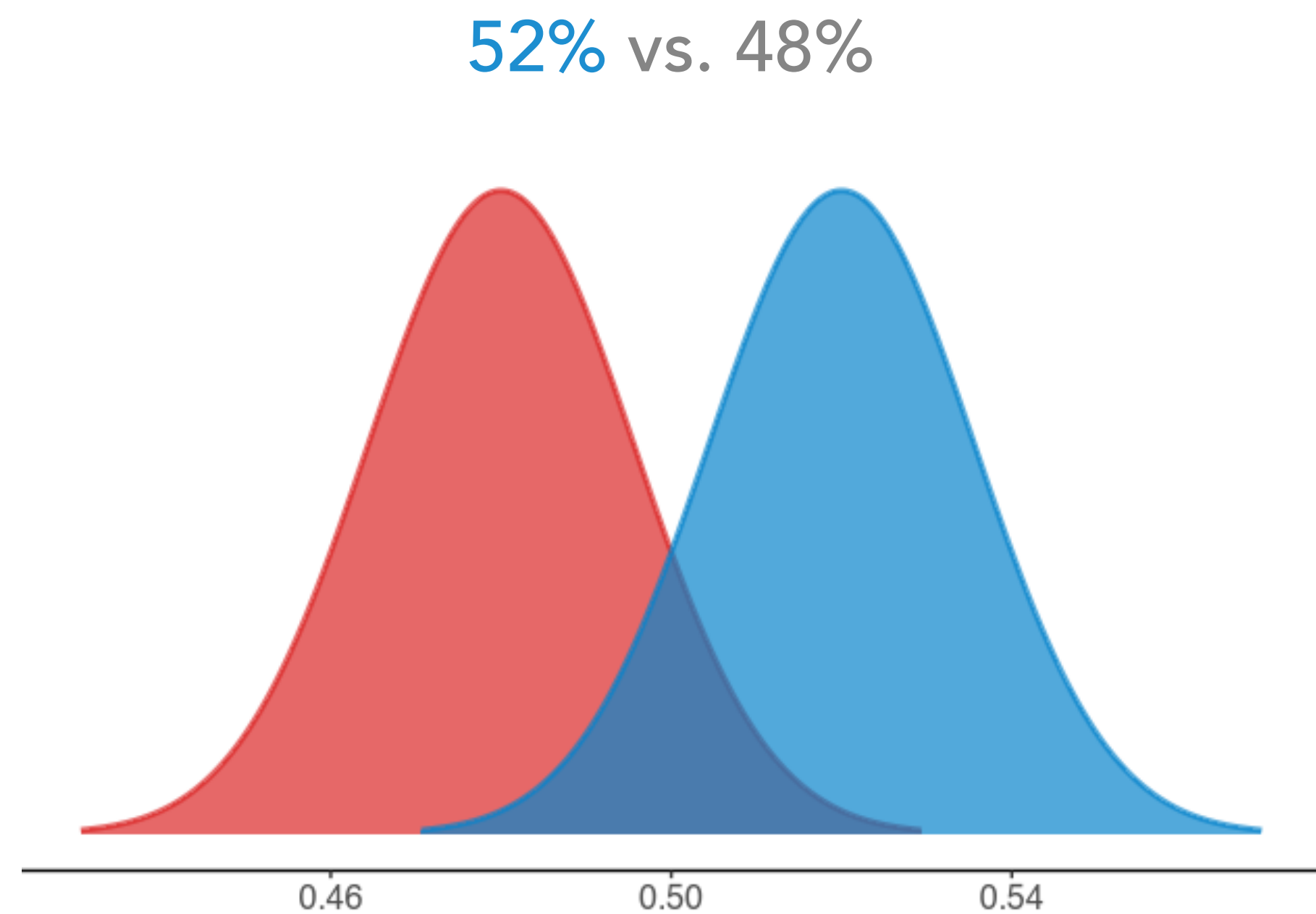
But how about **variance** in real-world election forecasts?

Variance caused by imperfect knowledge, e.g., polling errors, imperfect models



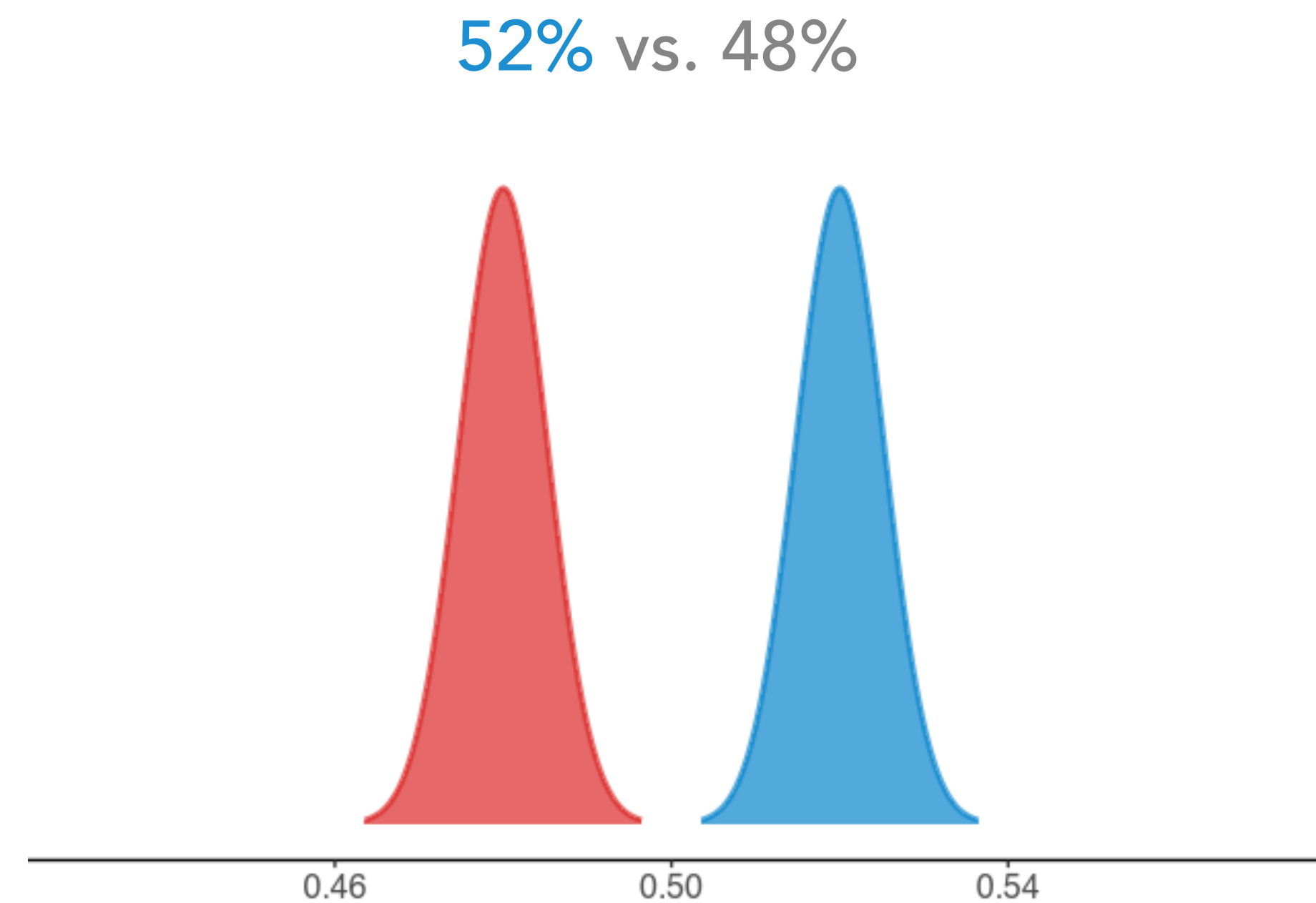
Variance in these forecasts

Typically, $\pm 12\%$ or more...



$\pm 8\%$

$$P(x_d > 50\%) = 90.9\%$$



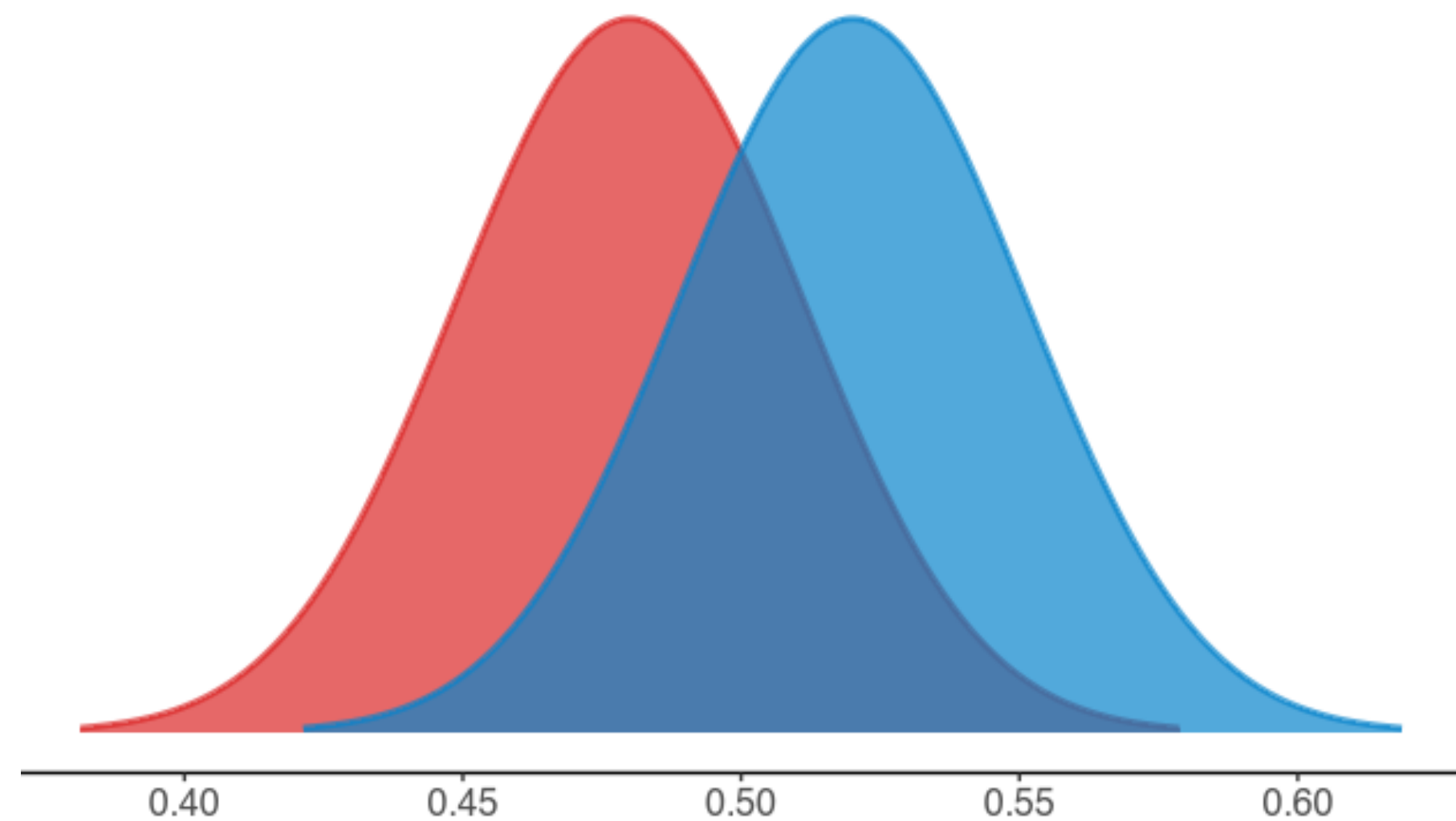
$\pm 2\%$

$$P(x_d > 50\%) = 100\%$$

Variance in these forecasts

Typically, $\pm 12\%$ or more...

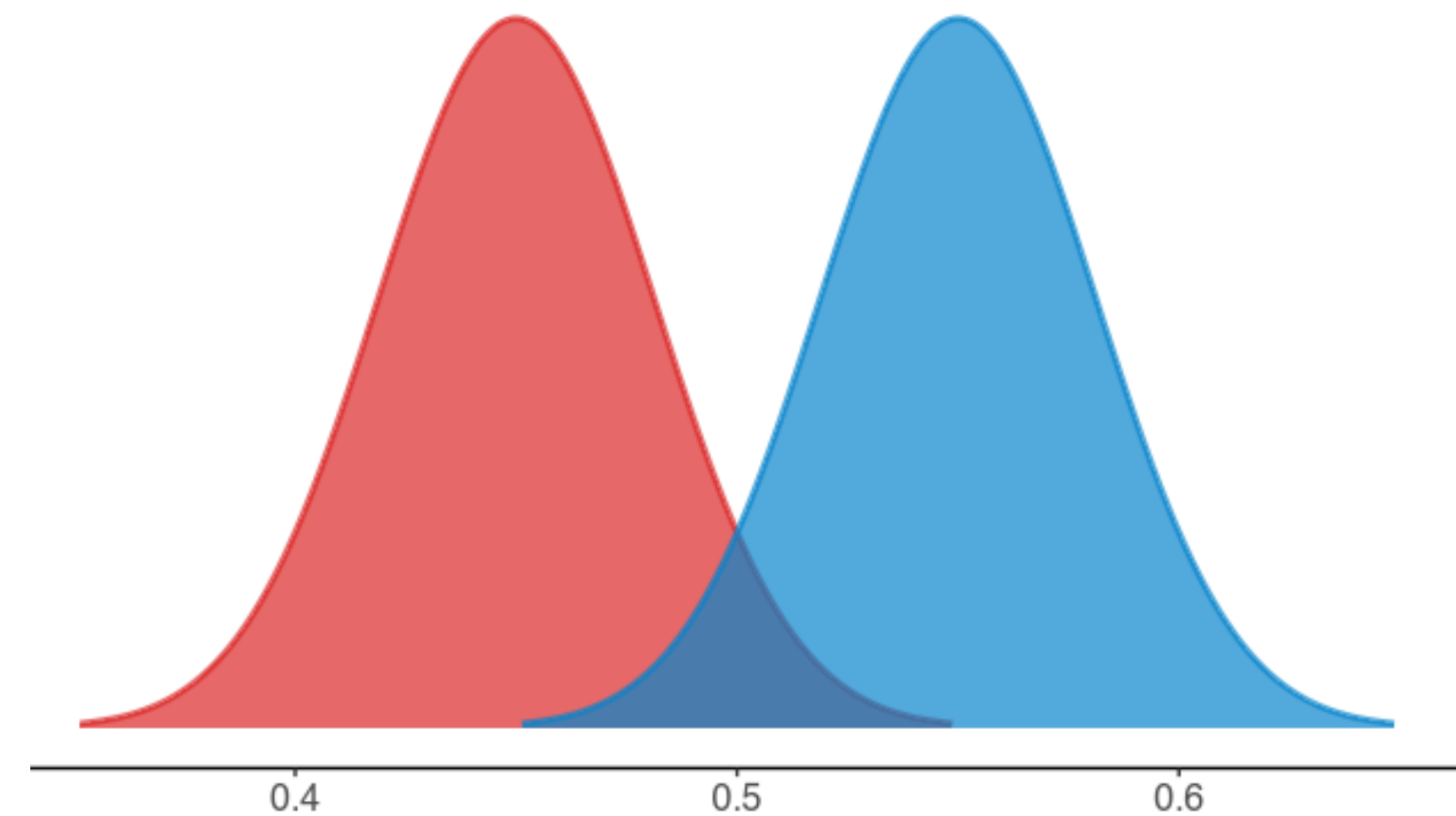
52% vs. 48%



$\pm 11\%$

$$P(x_d > 50\%) = 74.8\%$$

55% vs. 45%



$\pm 11\%$

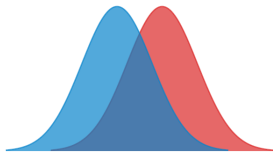
$$P(x_d > 50\%) = 95.2\%$$

Variance and mean in these forecasts

Typically, $\pm 12\%$ or more but deep blue/red states are more certain

52% vs. 48%

New Mexico
Virginia

52% vs. 48% 

Florida (2020)
Texas (2020)

$P(x > 50\%) = \sim 75\%$

55% vs. 45%

Rhode Island
Illinois

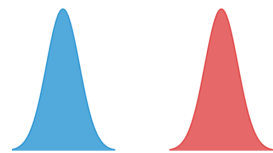
55% vs. 45% 

Florida (2024)
Texas (2024)

$P(x > 50\%) = \sim 90\%$

60+% vs. 40-%

Massachusetts
Maryland

60+% vs. 40-% 

Utah (2024)
Oklahoma (2024)

$P(x > 50\%) = 100\%$

Electoral college votes

You add all states together.

Take a sample from your prediction,

e.g., if the D candidates wins Maryland, assign 10 ECVs to D.

A similar **uncertain** situation

How can I communicate this to my friends who aren't in this class...

Predictions from 2016 US presidential election

FiveThirtyEight

28%

NYT Upshot

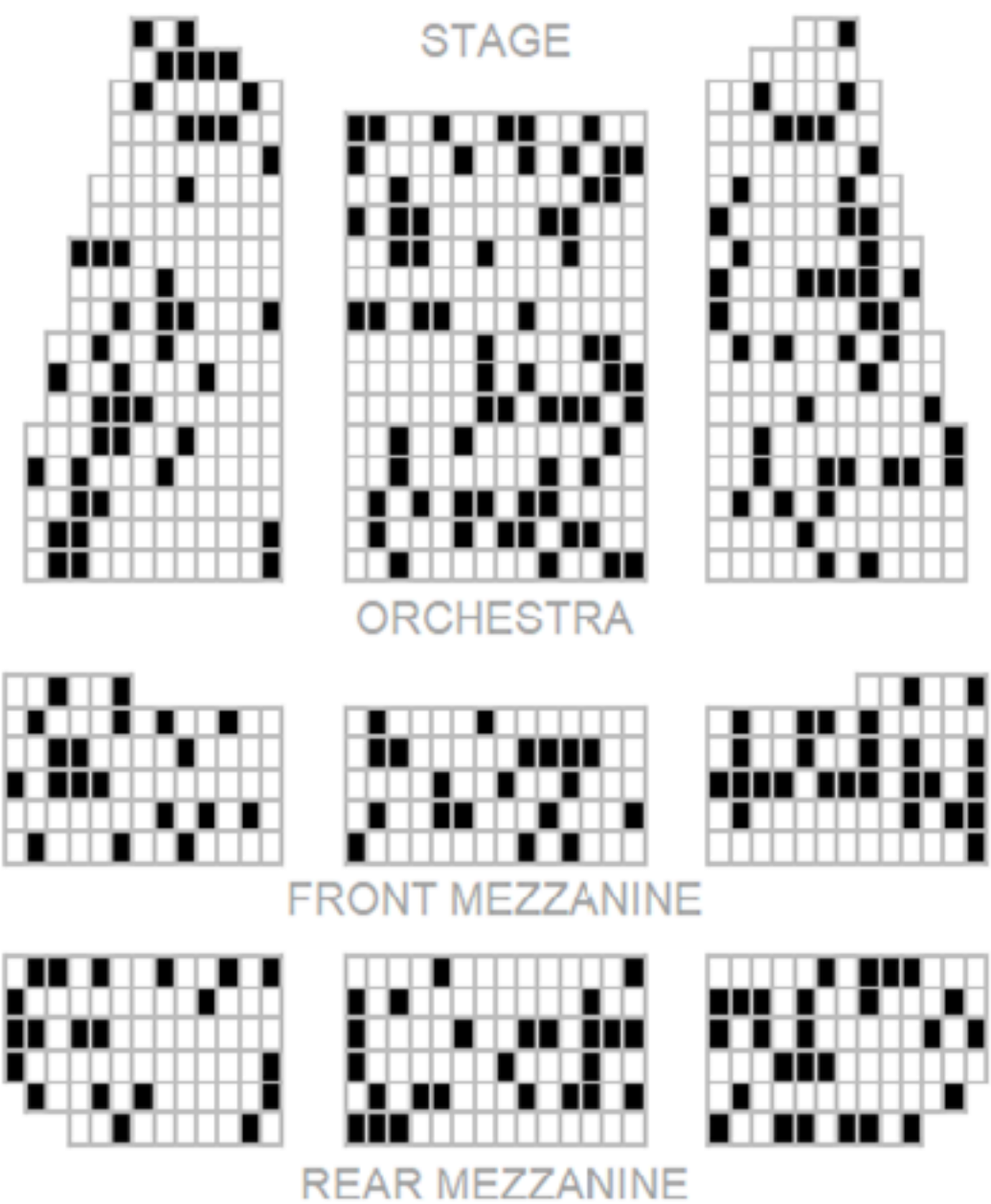
15%

HuffPo Pollster

2%

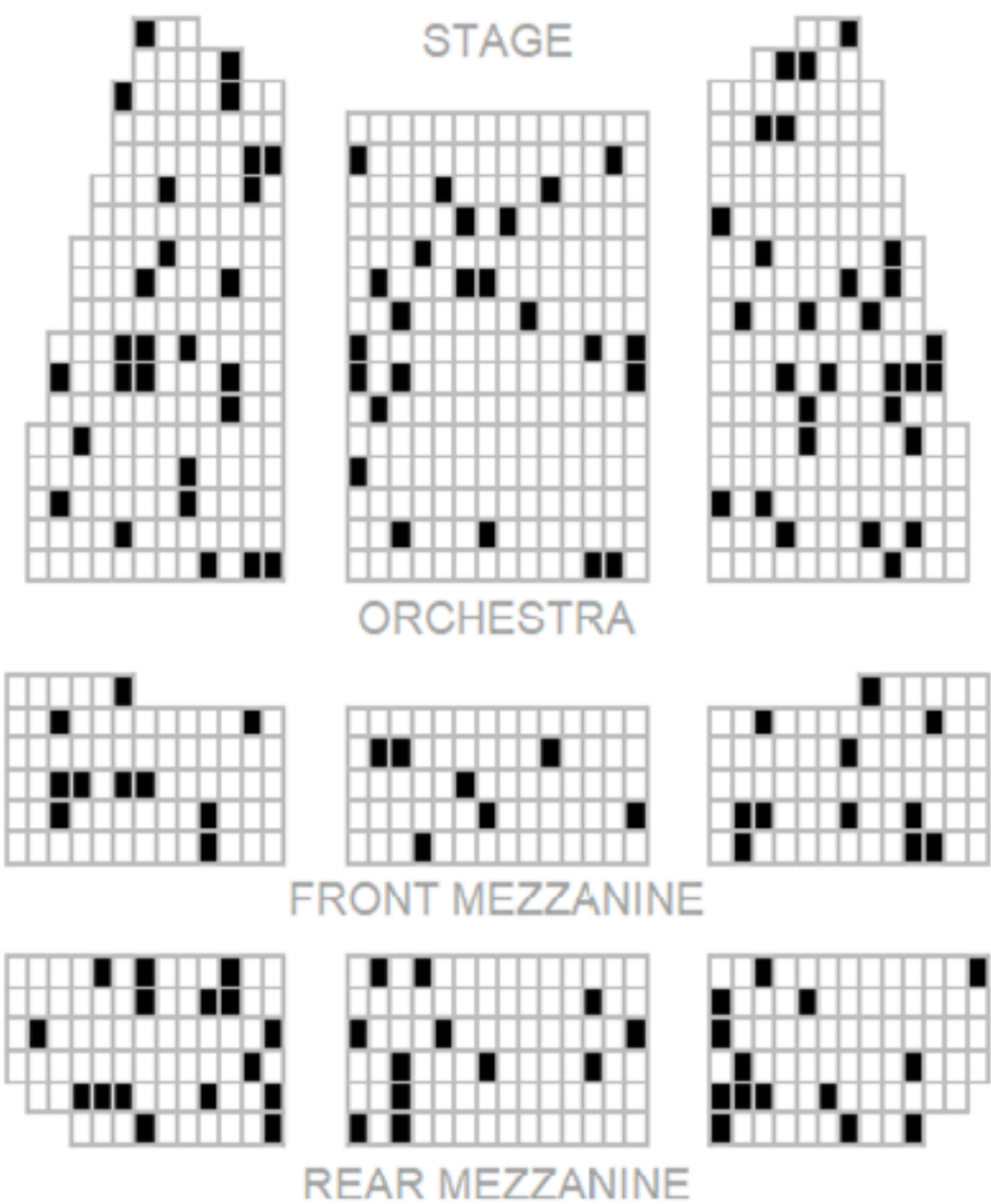
Predictions from 2016 US presidential election

FiveThirtyEight



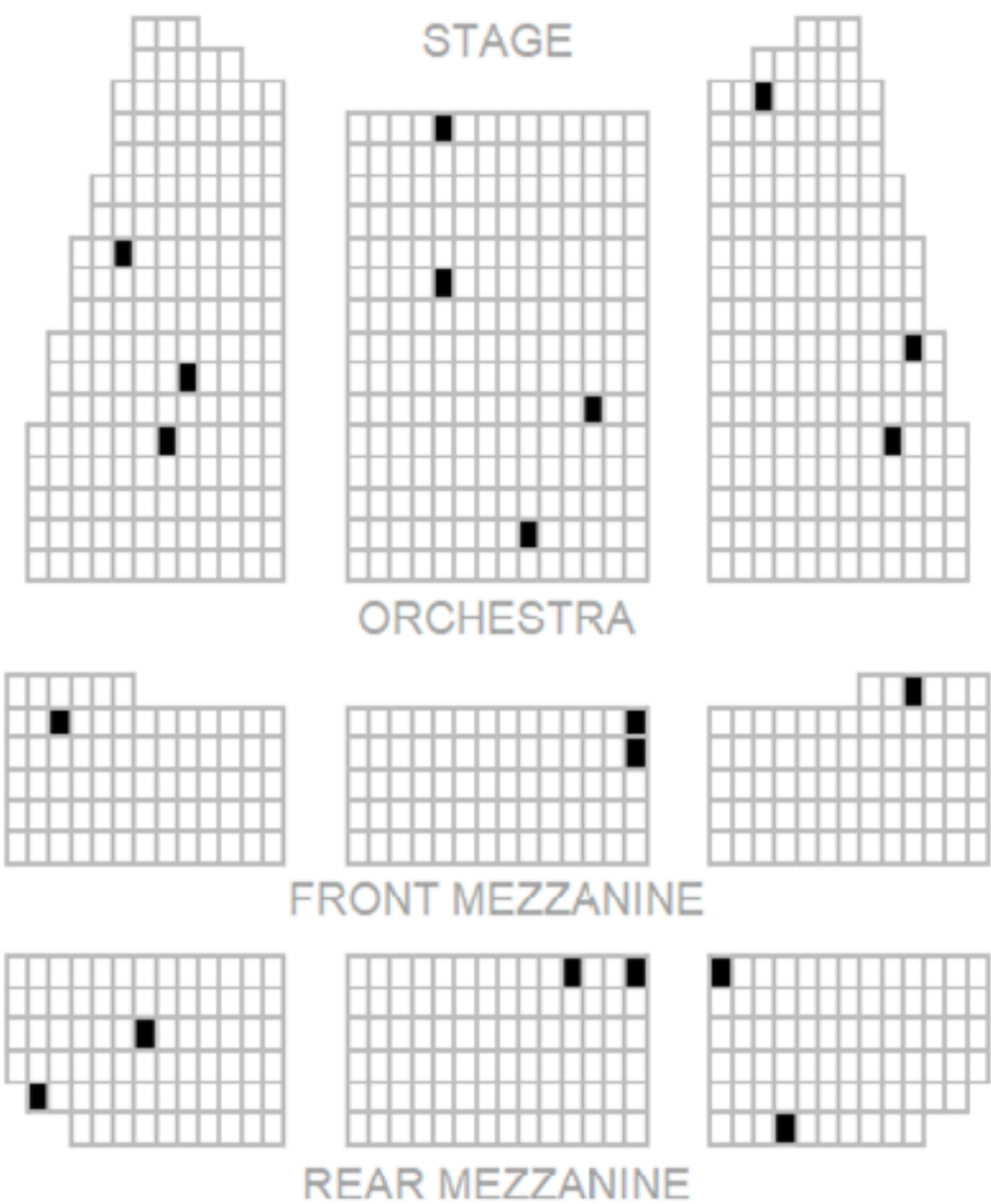
286 in 1,000

NYT Upshot



150 in 1,000

HuffPo Pollster



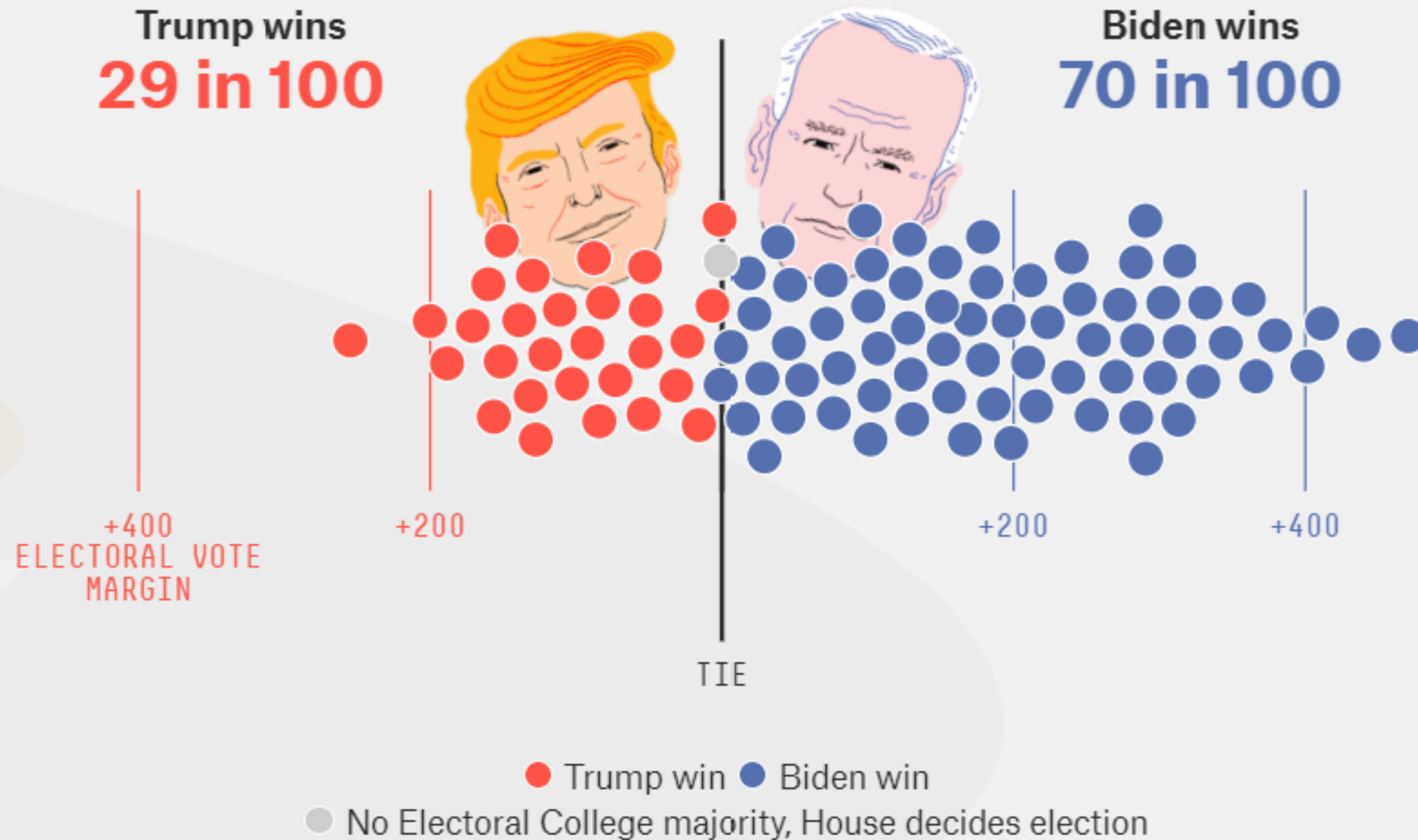
20 in 1,000

Biden is *favored* to win the election

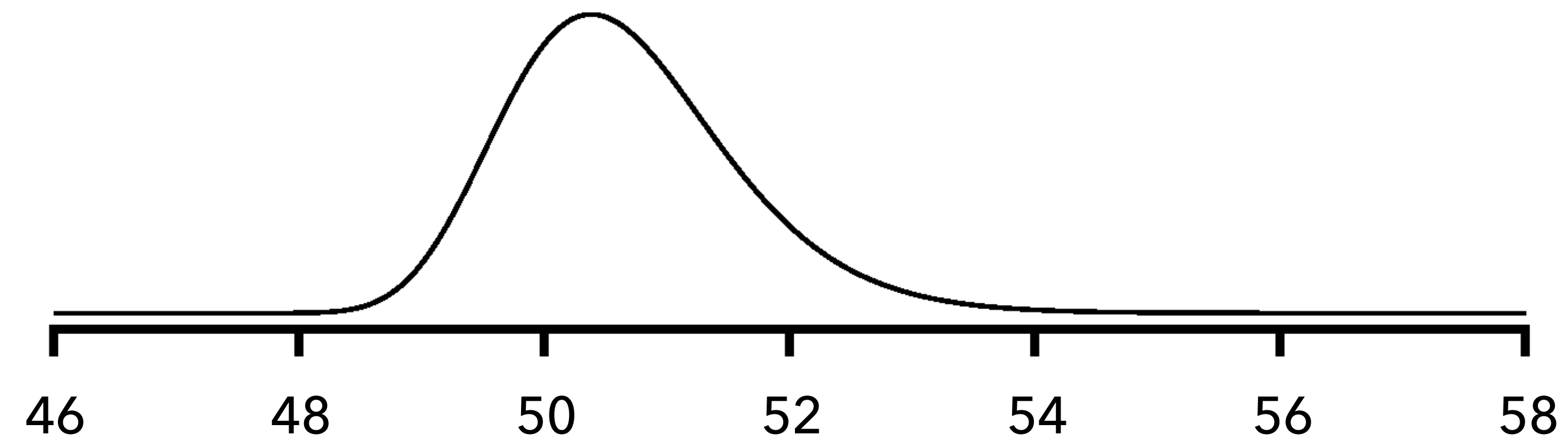
We simulate the election 40,000 times to see who wins most often. The sample of 100 outcomes below gives you a good idea of the range of scenarios our model thinks is possible.

Trump wins
29 in 100

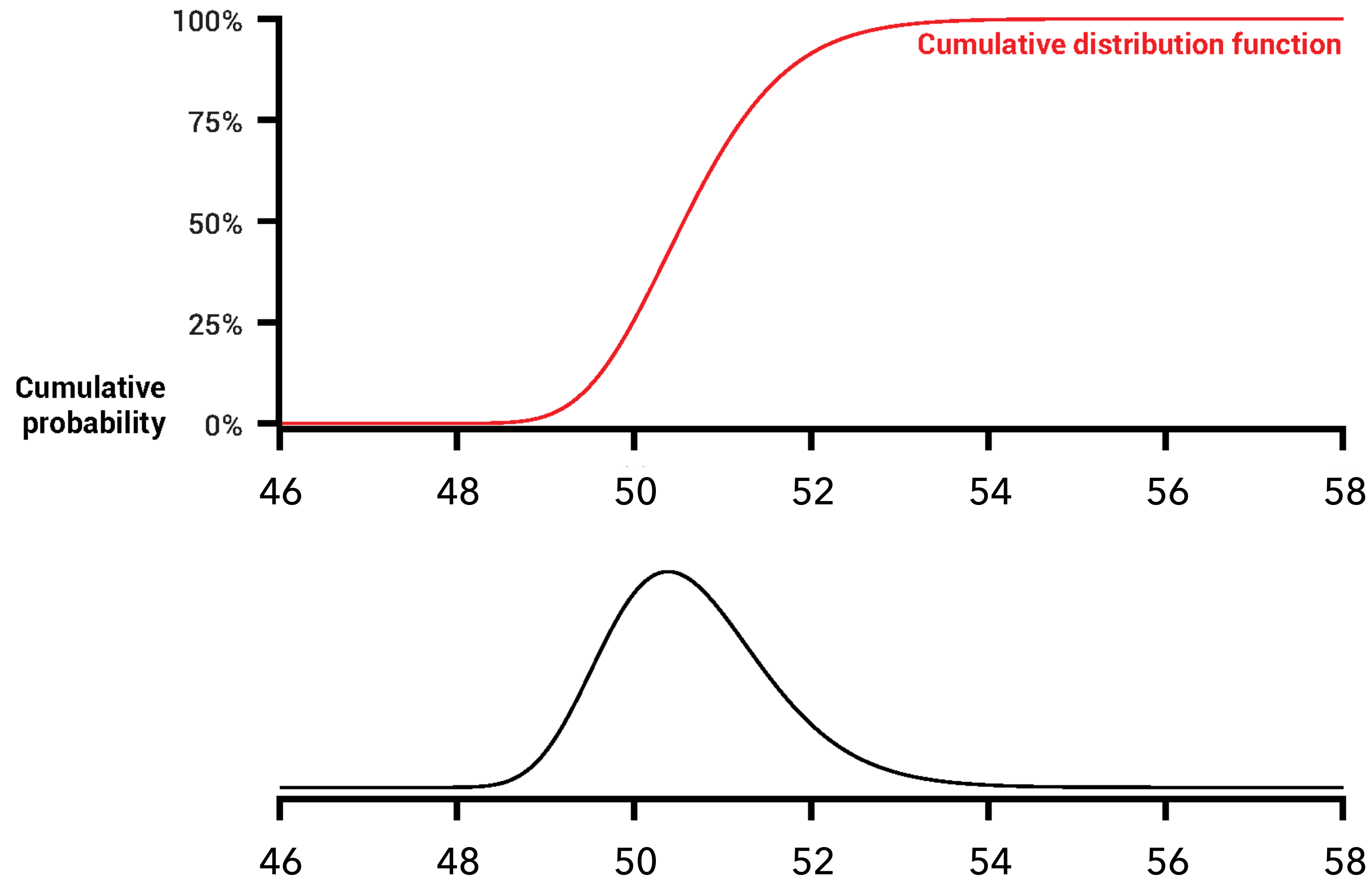
Biden wins
70 in 100



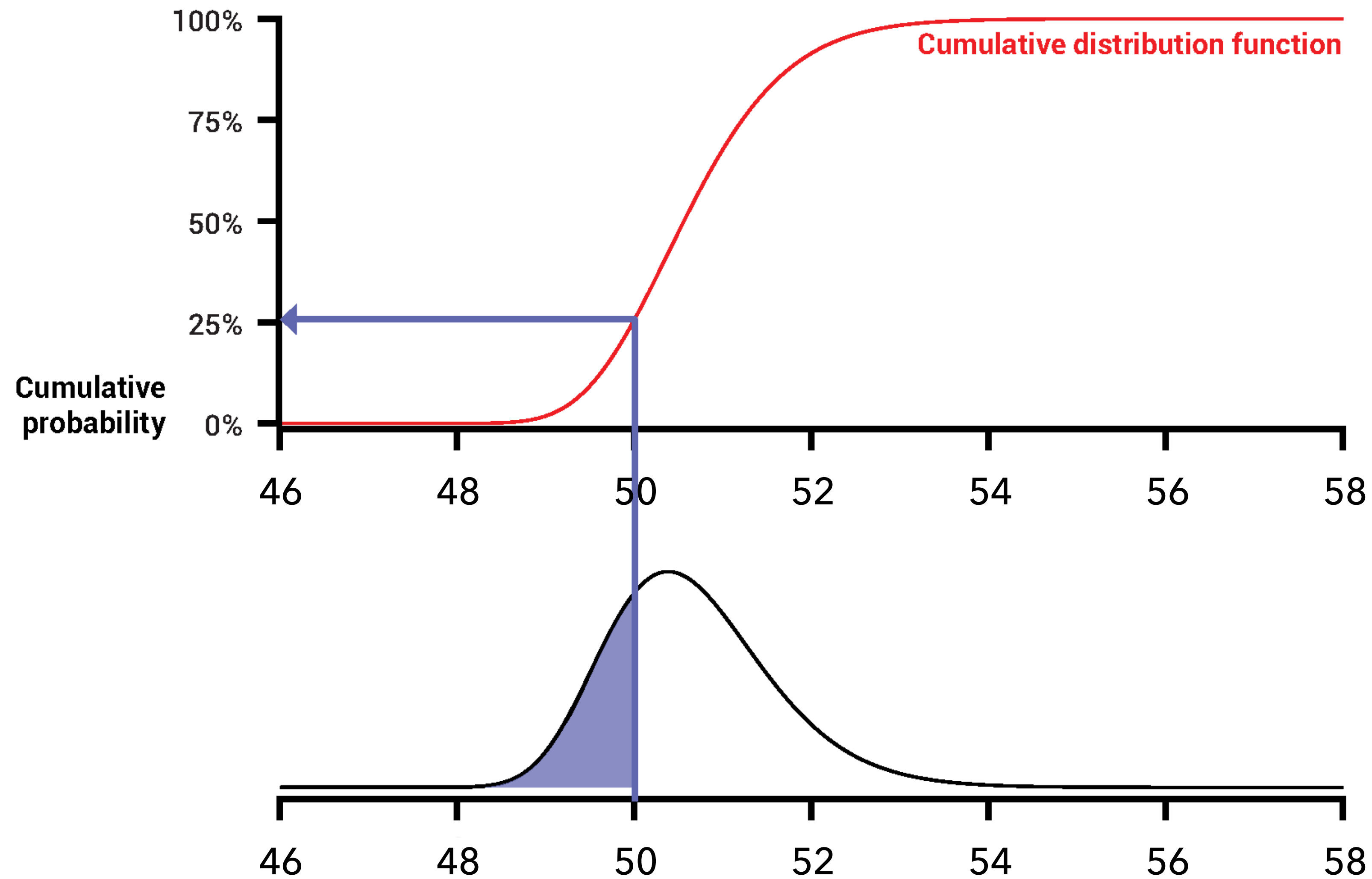
Don't count the underdog out! Upset wins are surprising but not impossible.



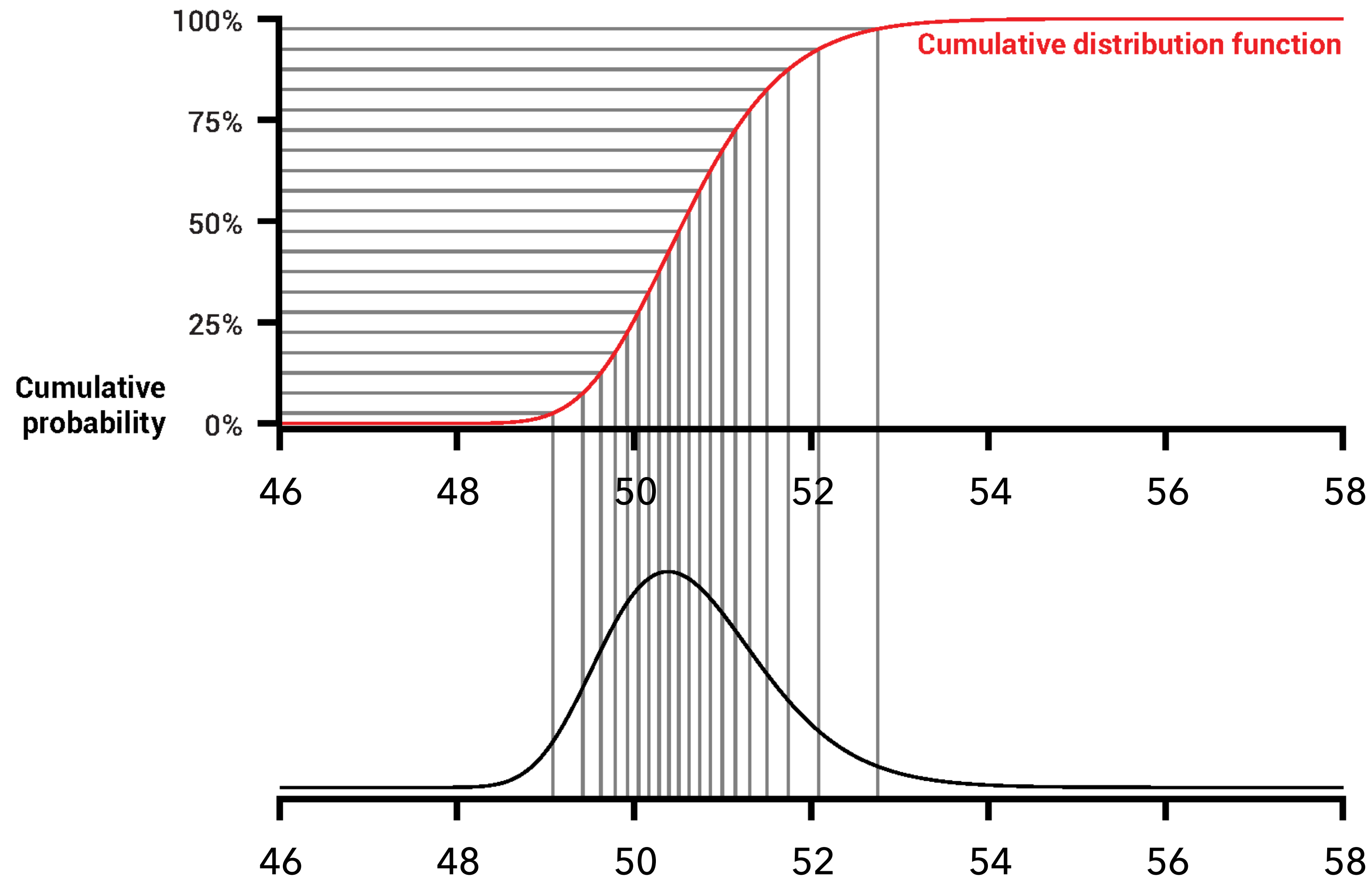
Vote share D can get



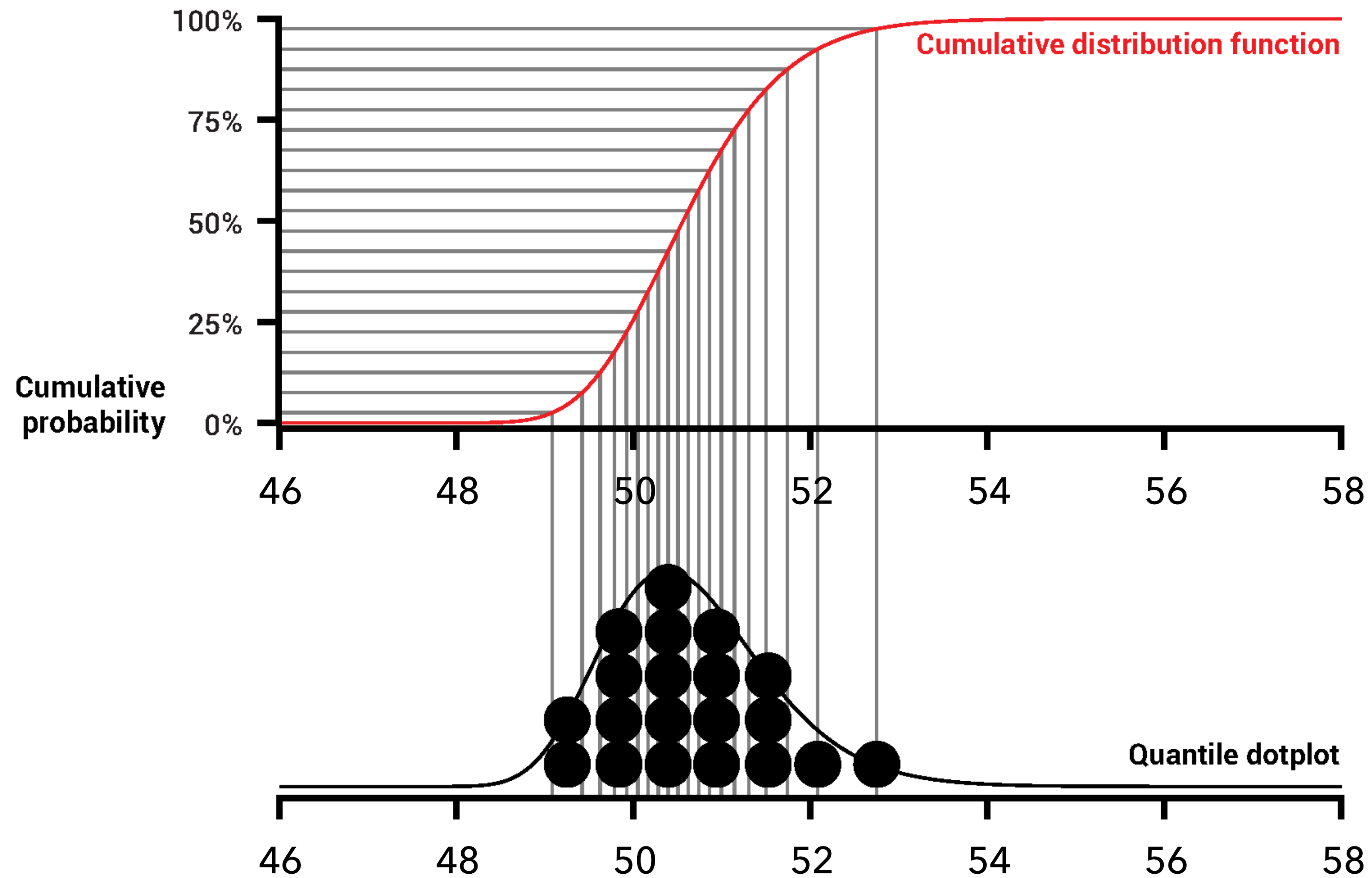
Vote share D can get



Vote share D can get

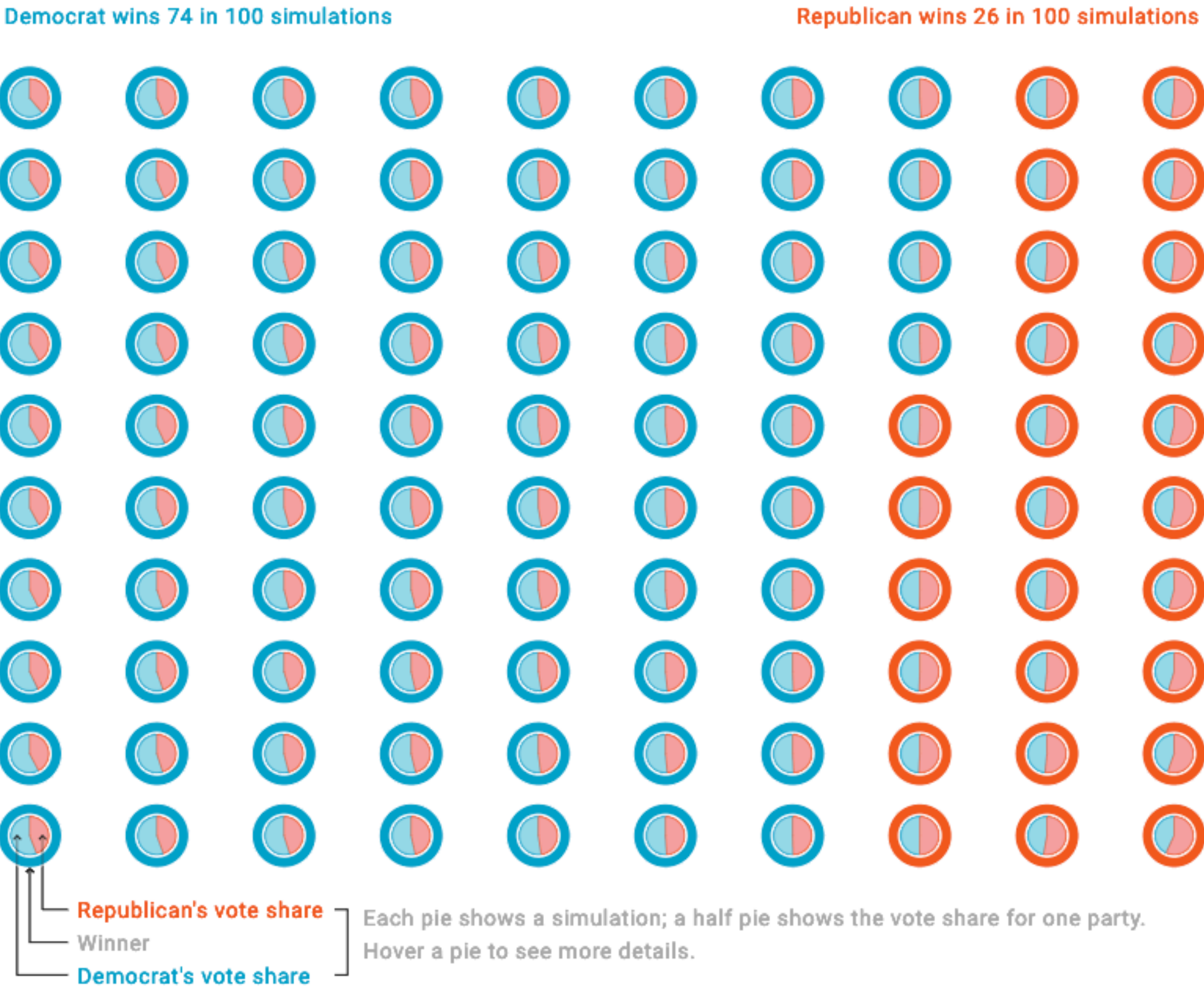
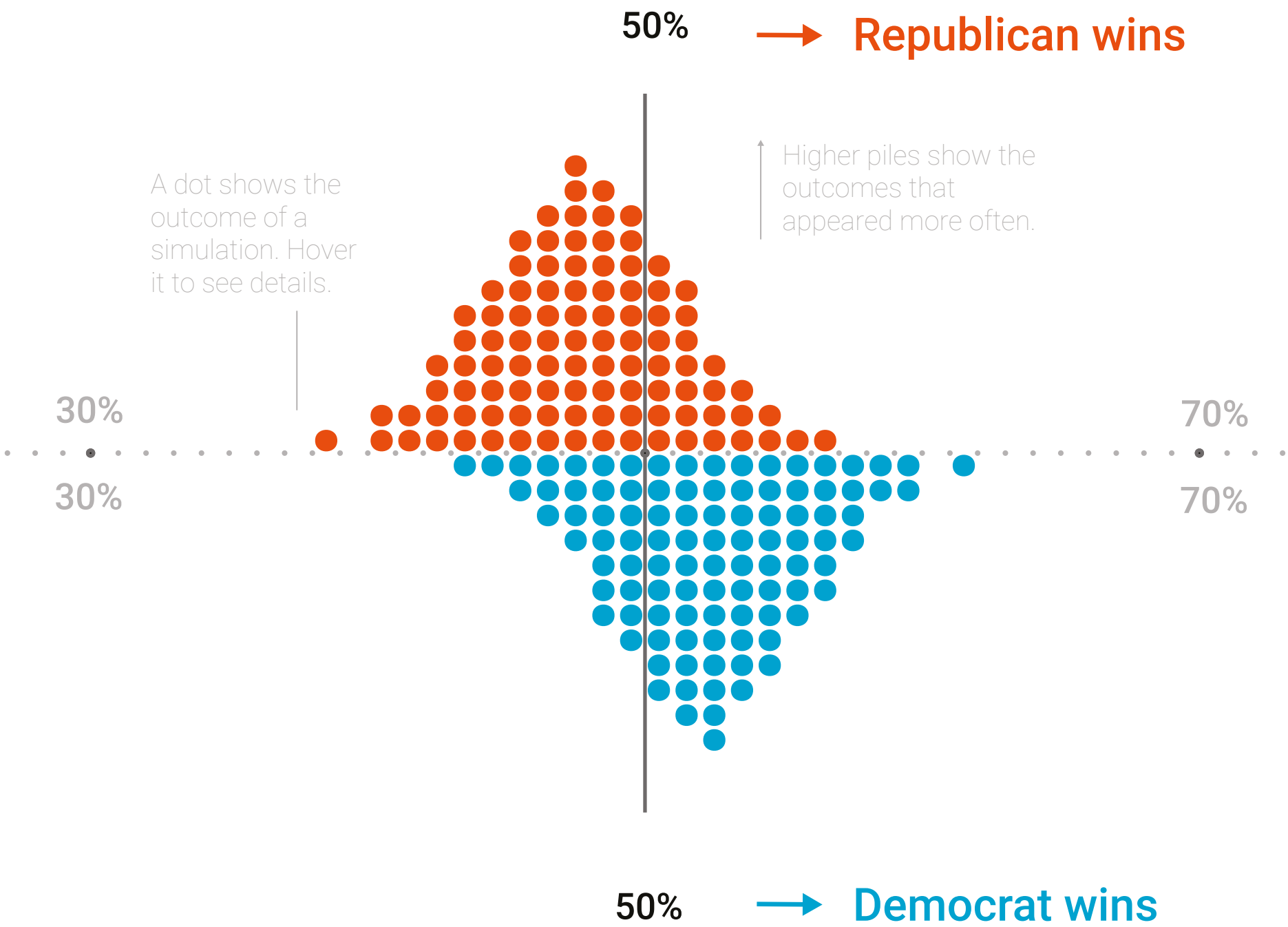


Vote share D can get



Vote share D can get

More ideas (that didn't work)

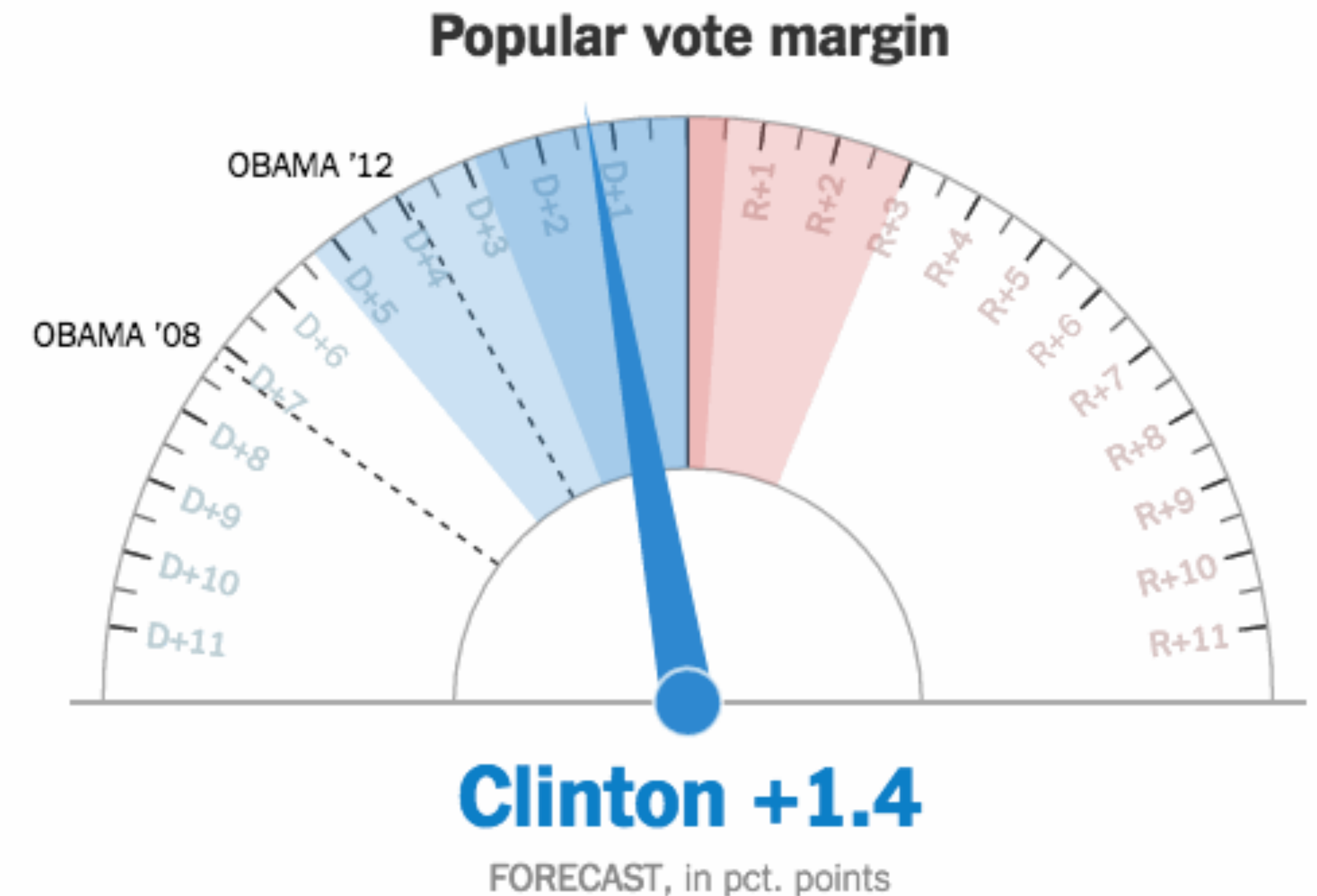


Animation - Needle

Convey that the forecast was “live” (?)

Show uncertainty around the forecast (?)

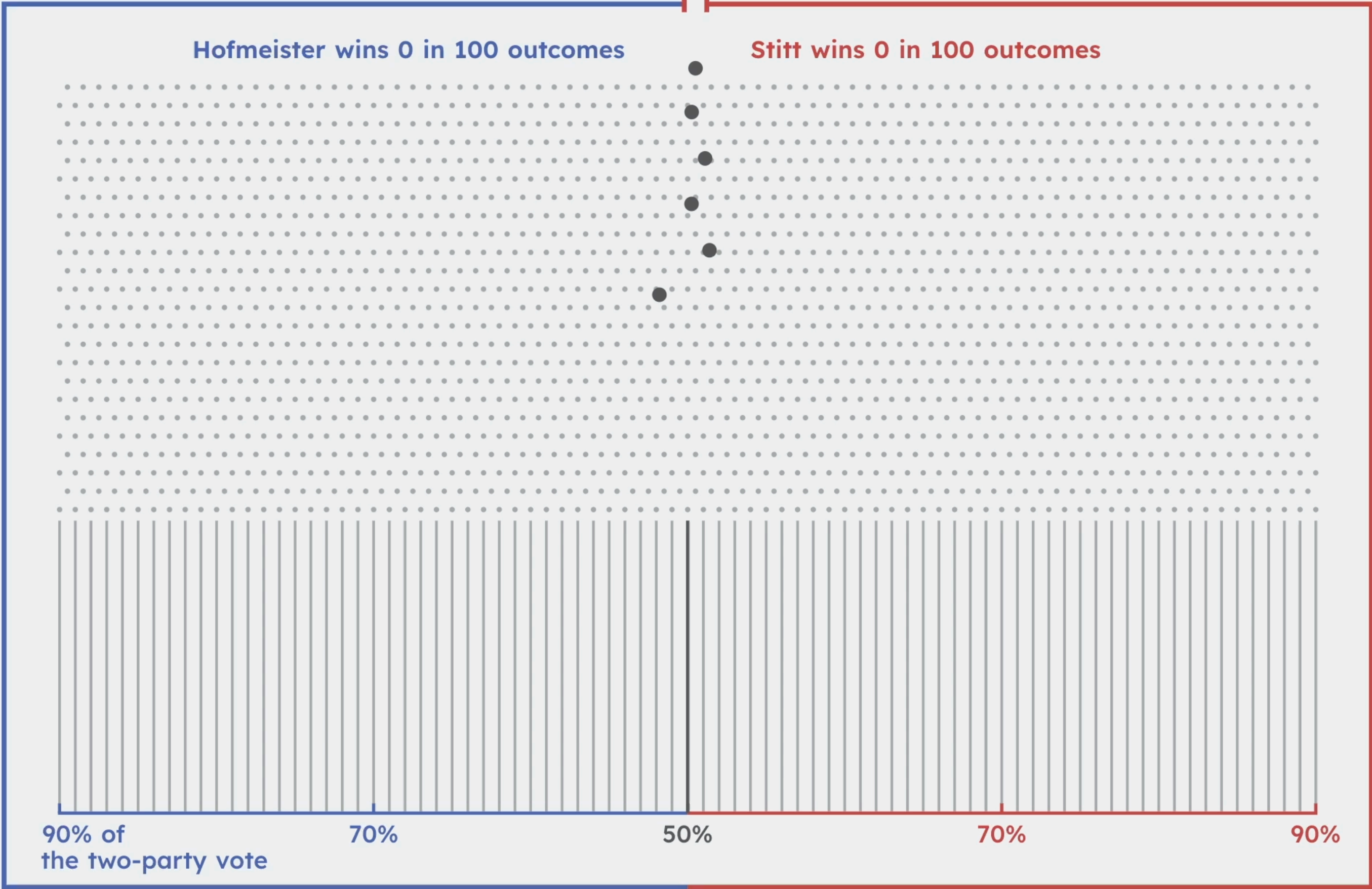
But add changes that don't exist → nervous



Animation - Plinko

This **1 minute** animation shows **100** balls, where
1 ● = 1 election outcome
○ = the actual result (added after election day)

Animating

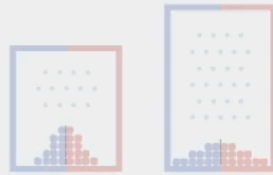


Balls are dropped at the location that represents the most likely election outcome.



Each ball has many possible paths. Some are more or less likely than others.

These paths roughly match the uncertainty in our forecast.



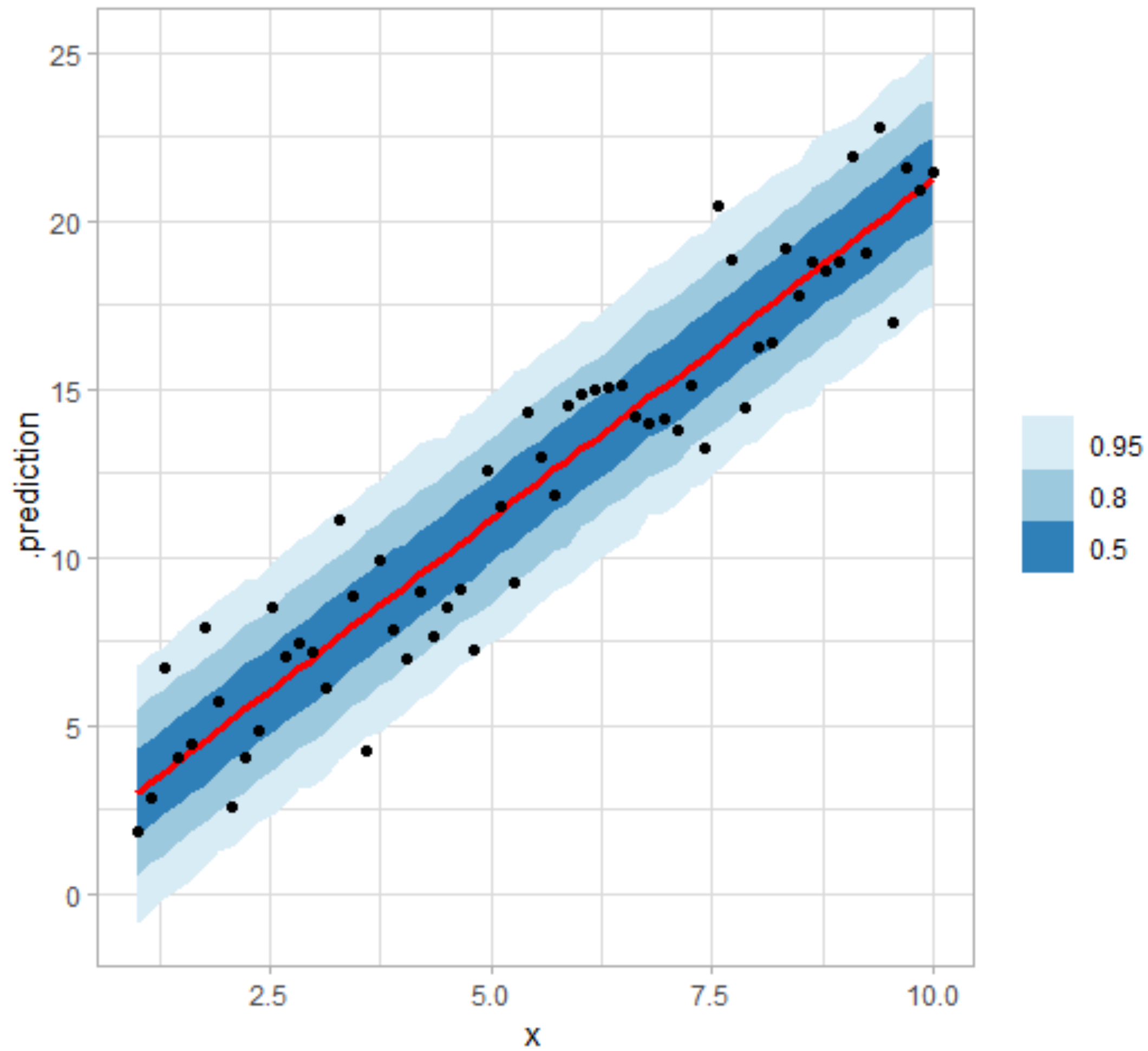
The taller the Plinko board is, the more uncertain the election outcome is.

Communicating uncertainty

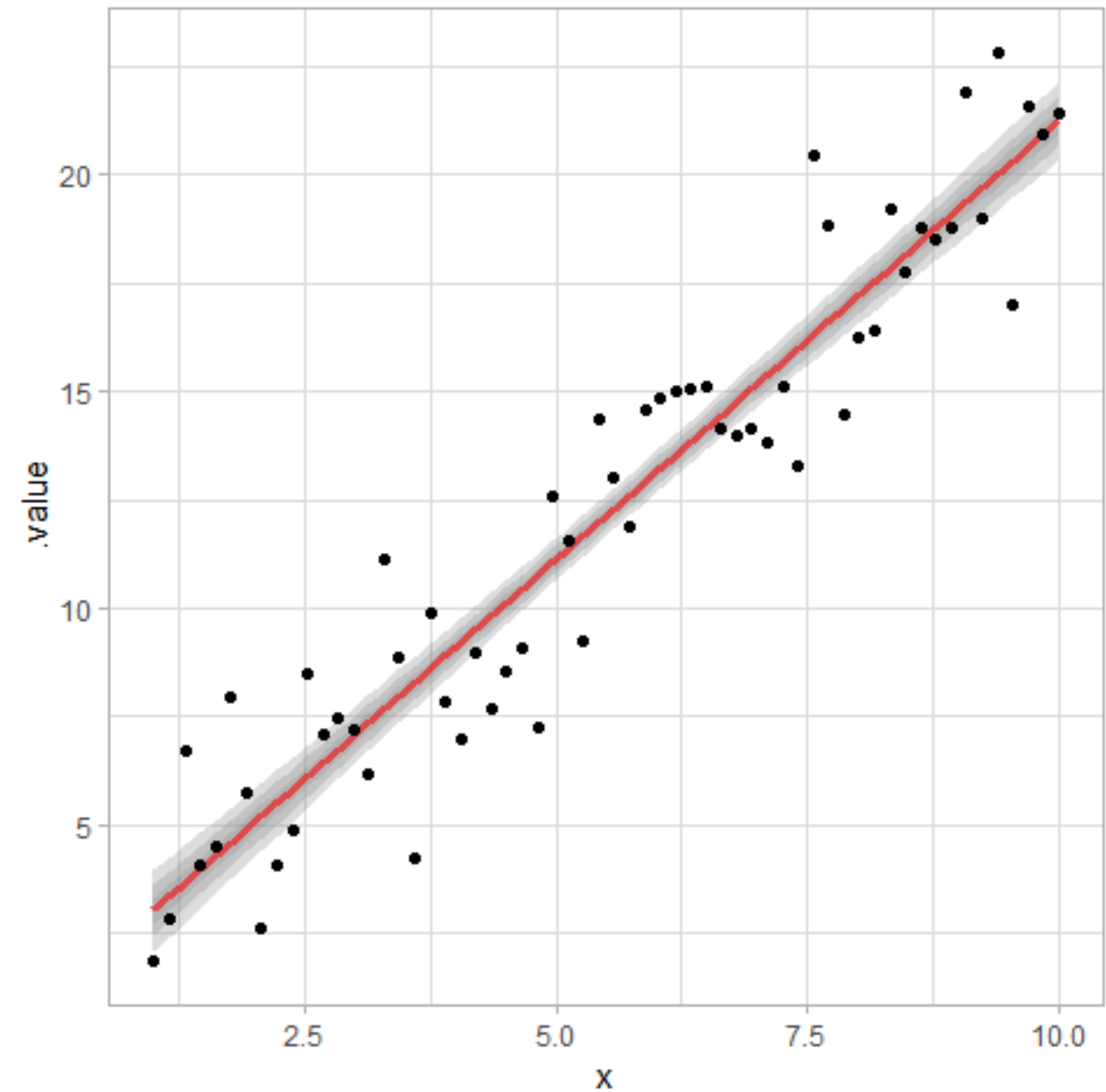
Uncertainty visualization

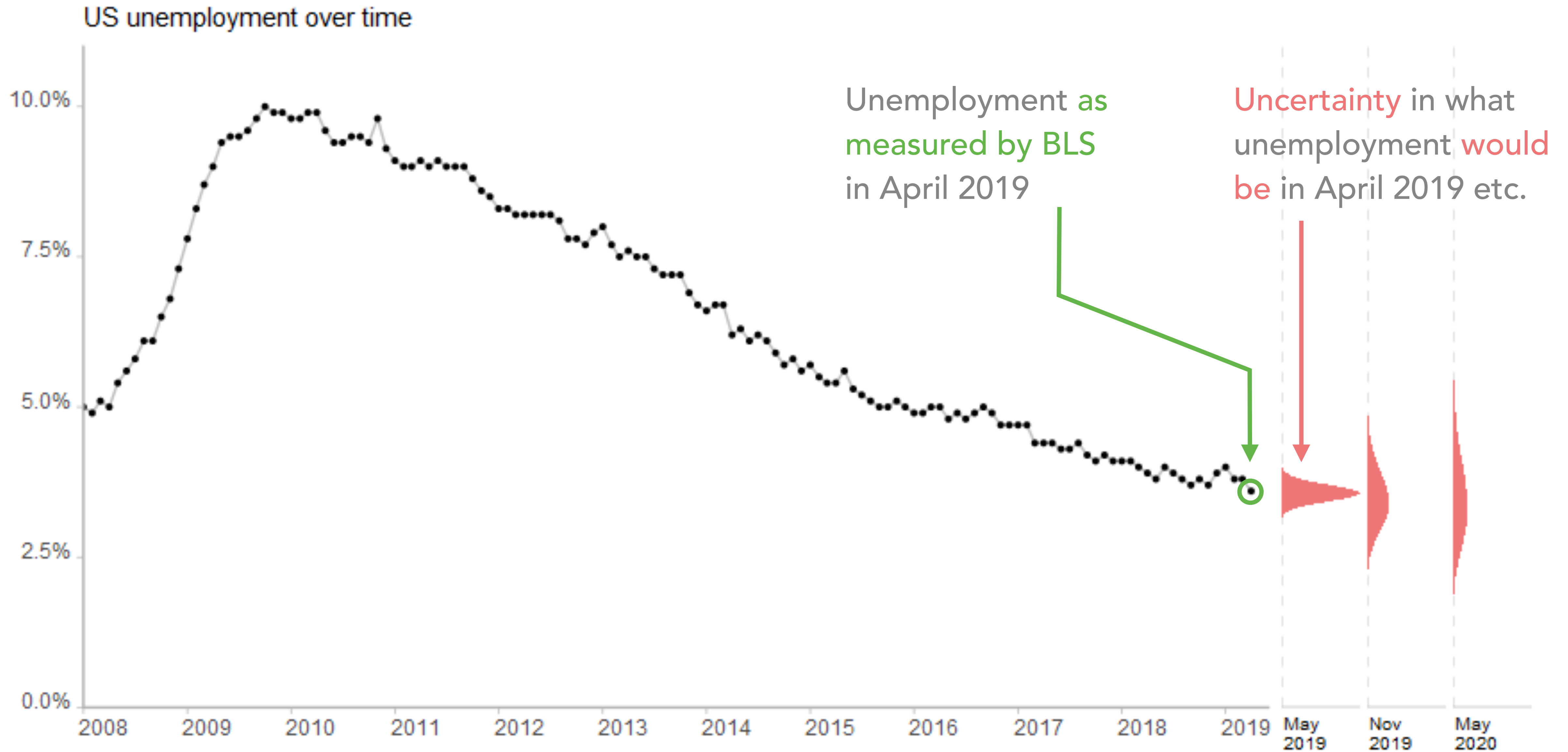
1. Derive a **distribution** describing your uncertainty:
a **confidence distribution**, etc.
2. Map properties of the distribution (**location**, **scale**,
quantiles, **density**) onto visual channels

Predictive uncertainty

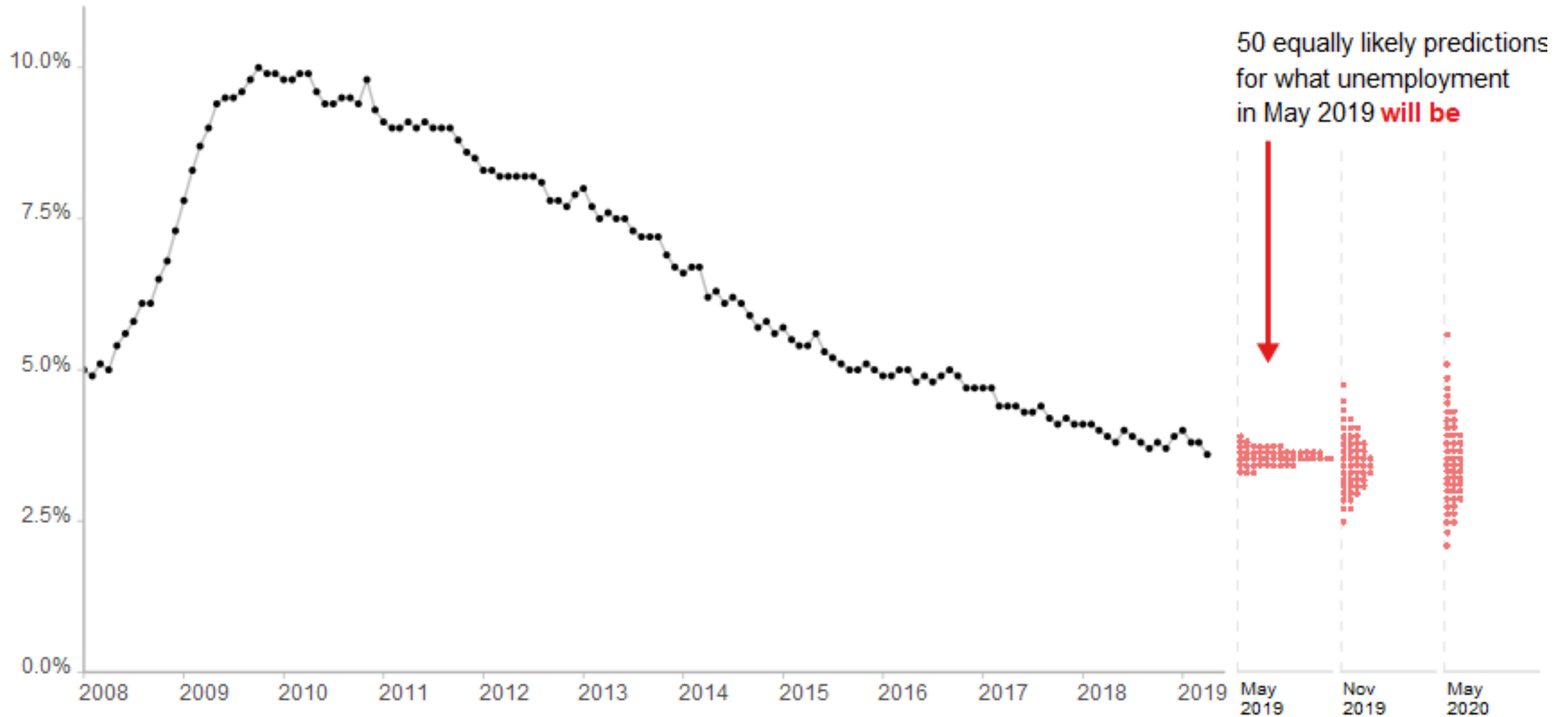


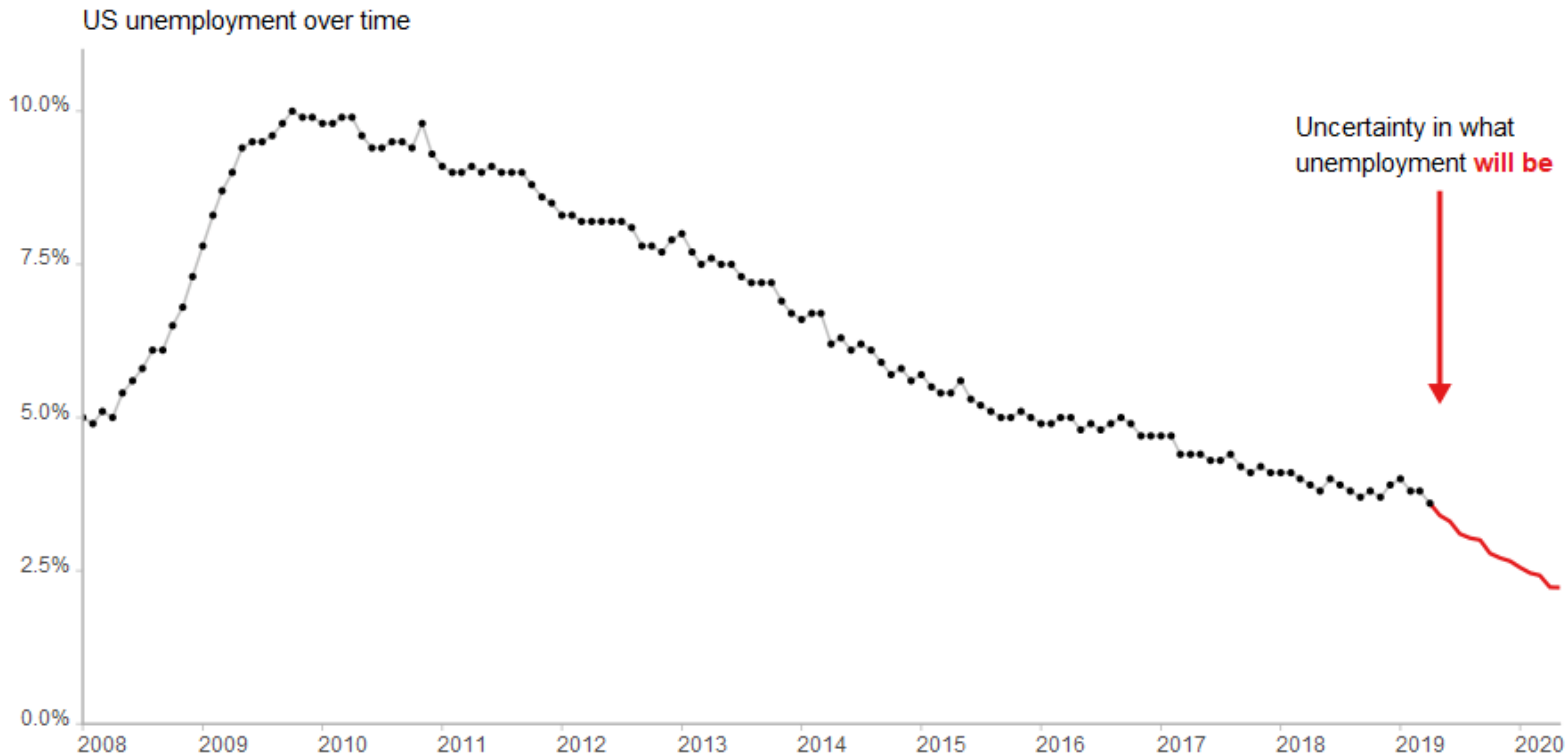
Parameter uncertainty

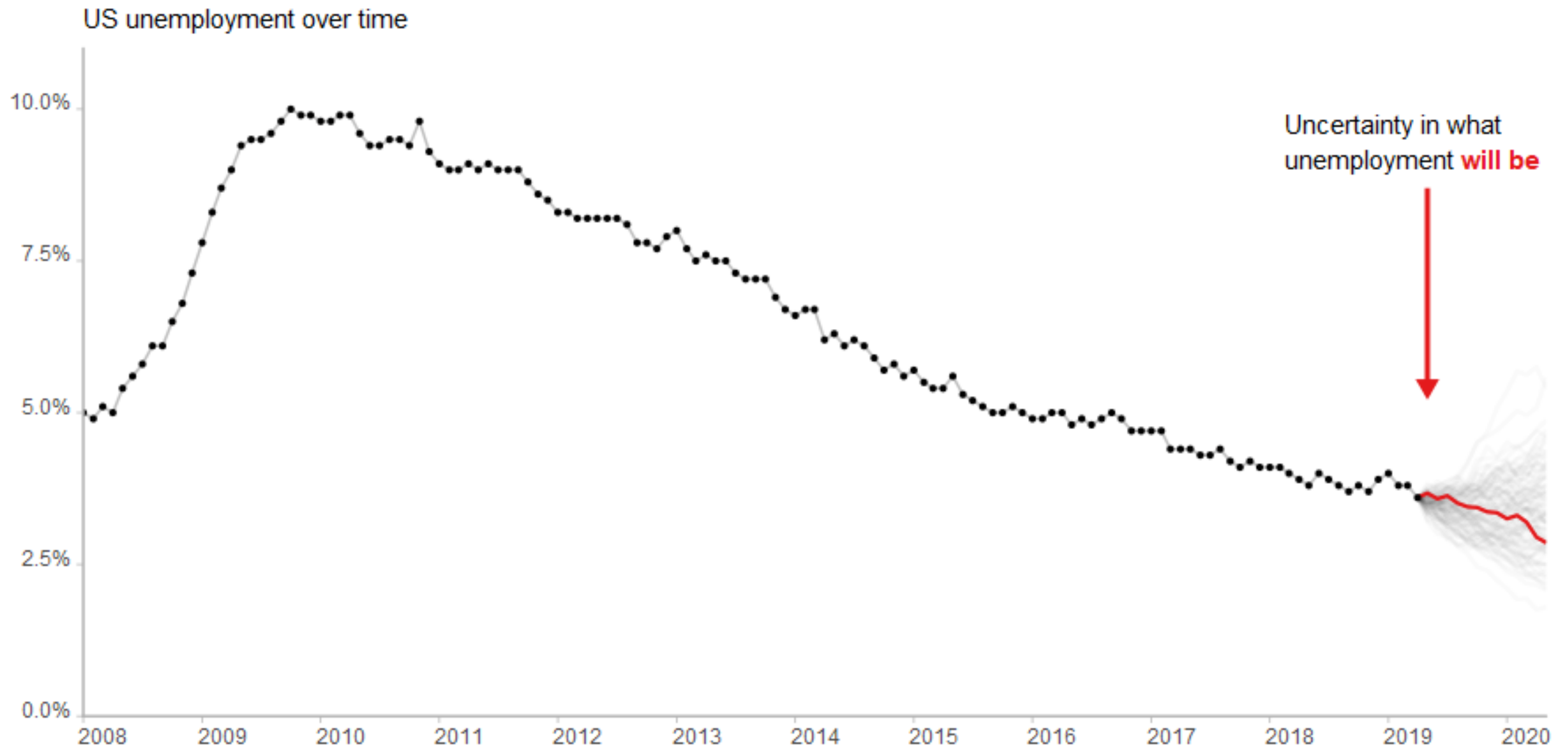




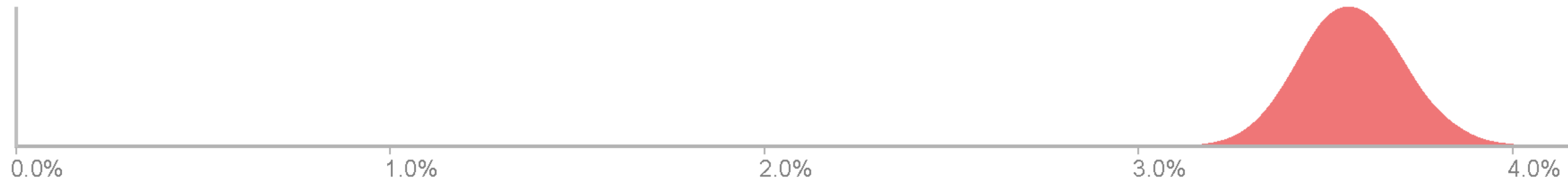
US unemployment over time



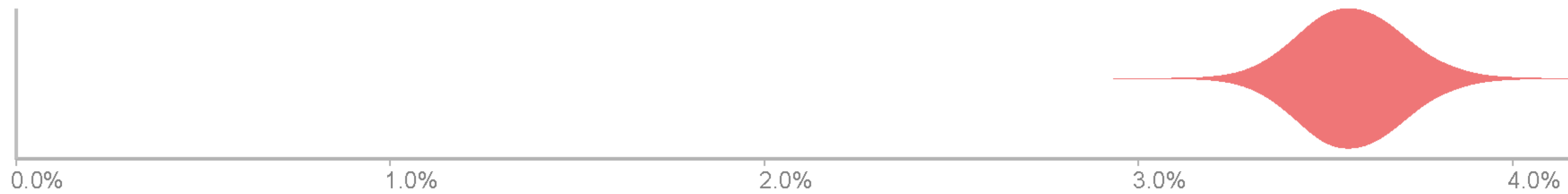




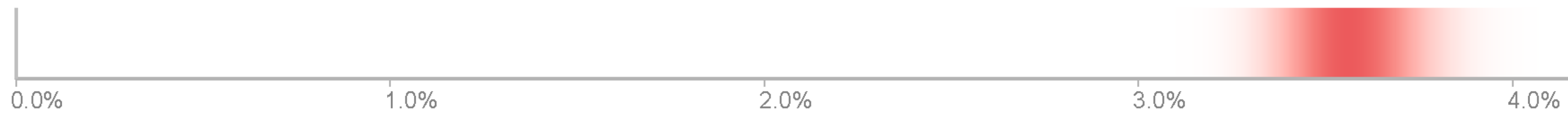
Uncertainty in what US unemployment will be in May 2019: Continuous encodings



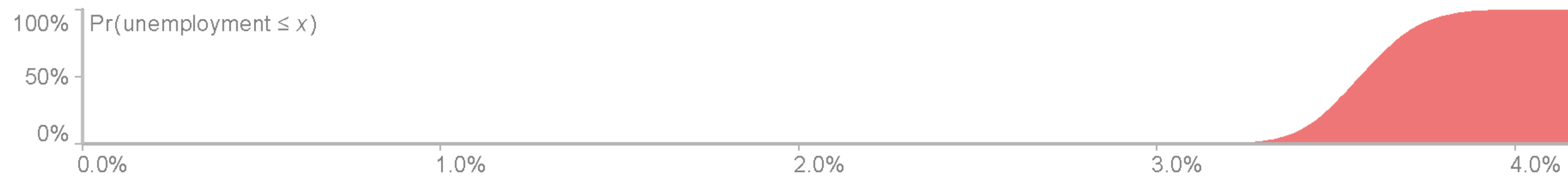
Density plot



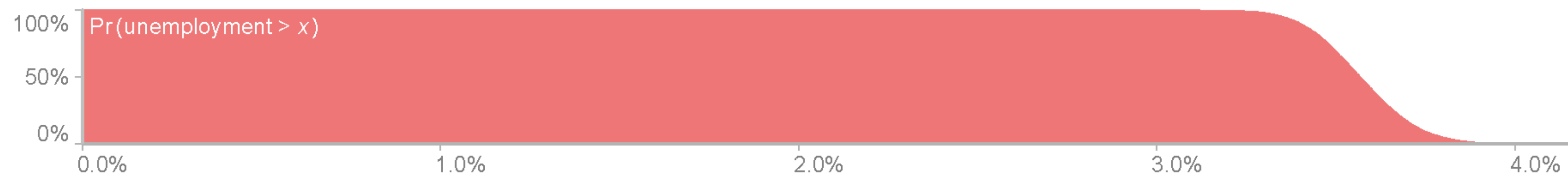
Violin plot



Gradient plot



Cumulative Density F



Complementary CDF

**Important to consider tasks, audience,
tradeoffs in resolution / accuracy**

More examples (and pitfalls)

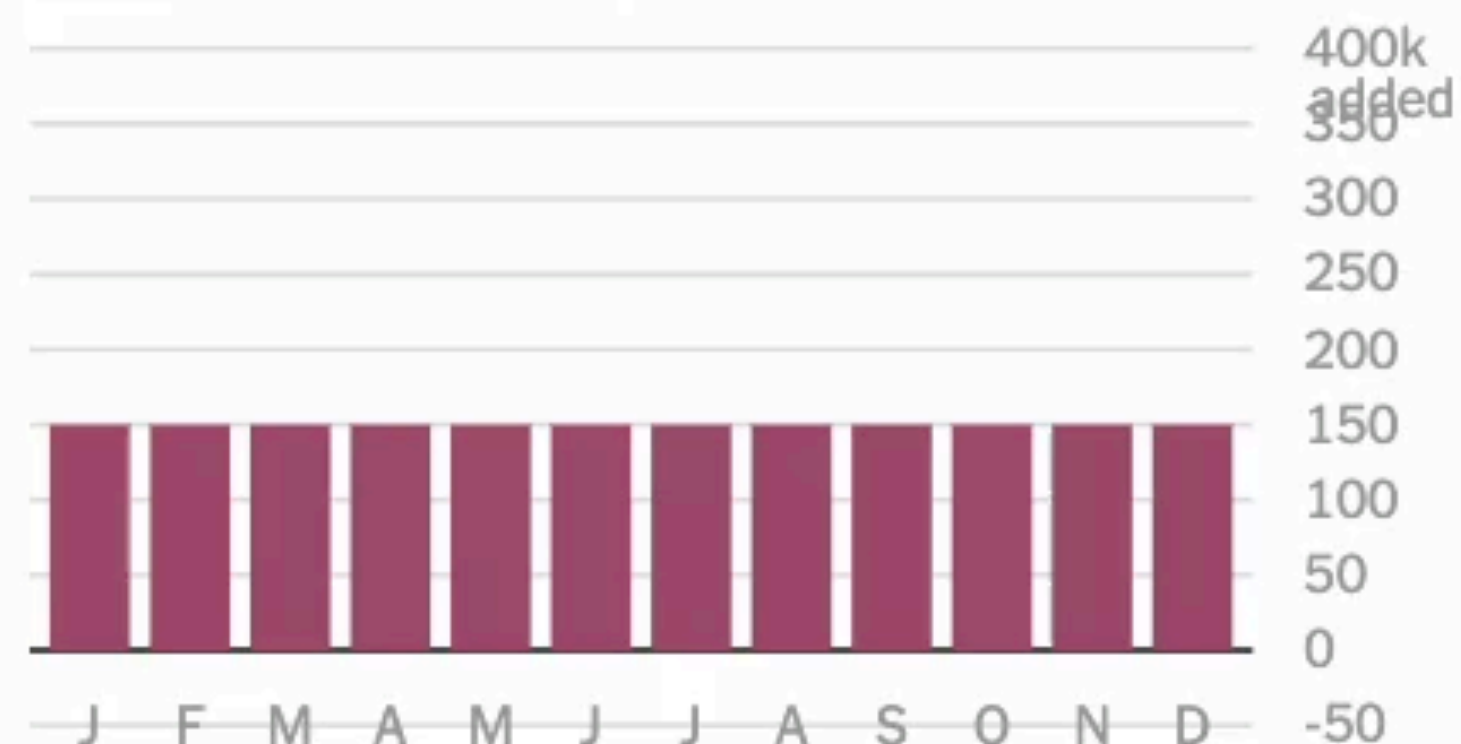
Have seen this

Show uncertainty by
looping over likely
outcomes

How Not to Be Misled by the Jobs Report

Let's consider how statistical noise might affect our perception of the jobs report in two scenarios: one with steady job growth and one with accelerating job growth.

If job growth **were actually steady**
over the last 12 months...

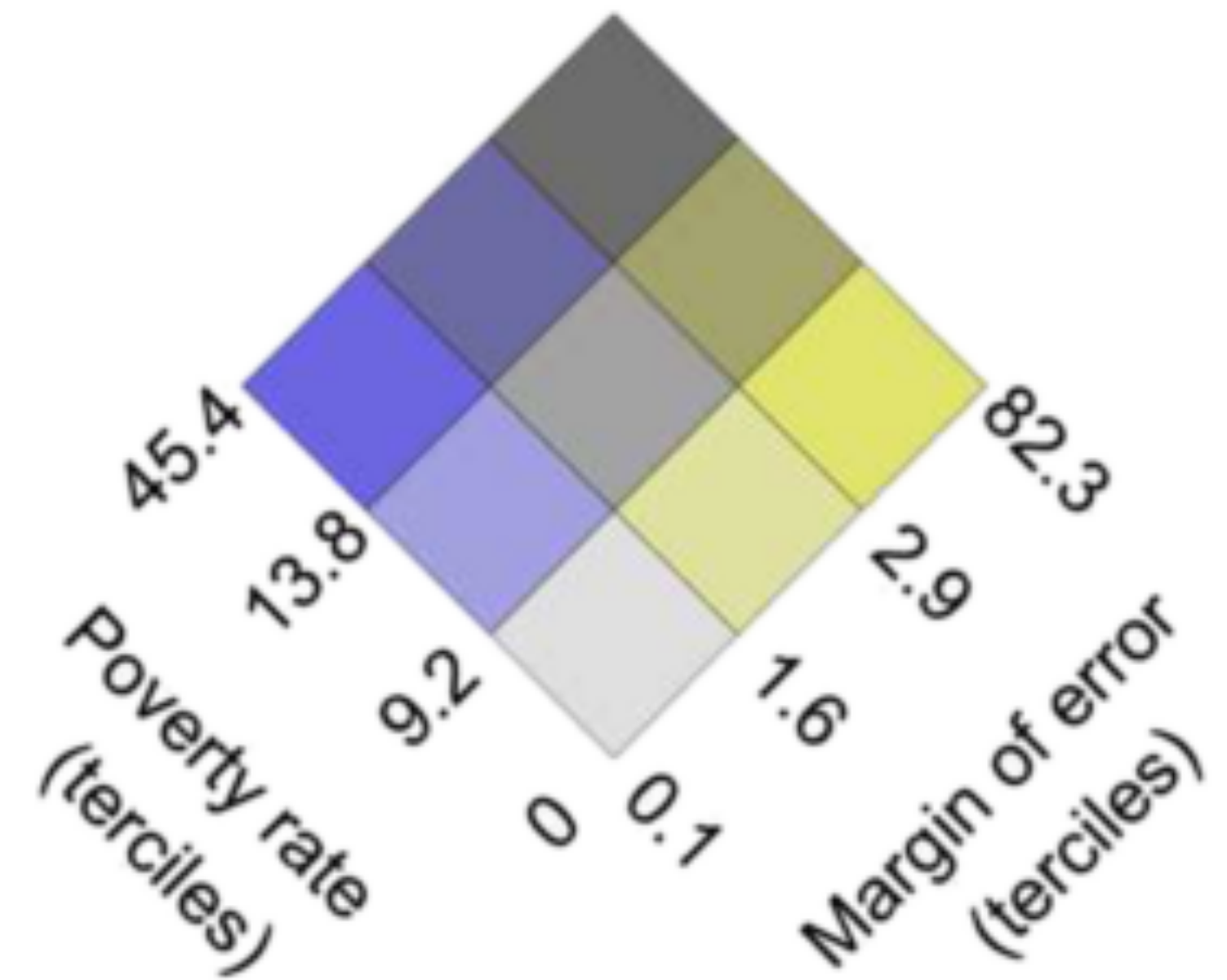
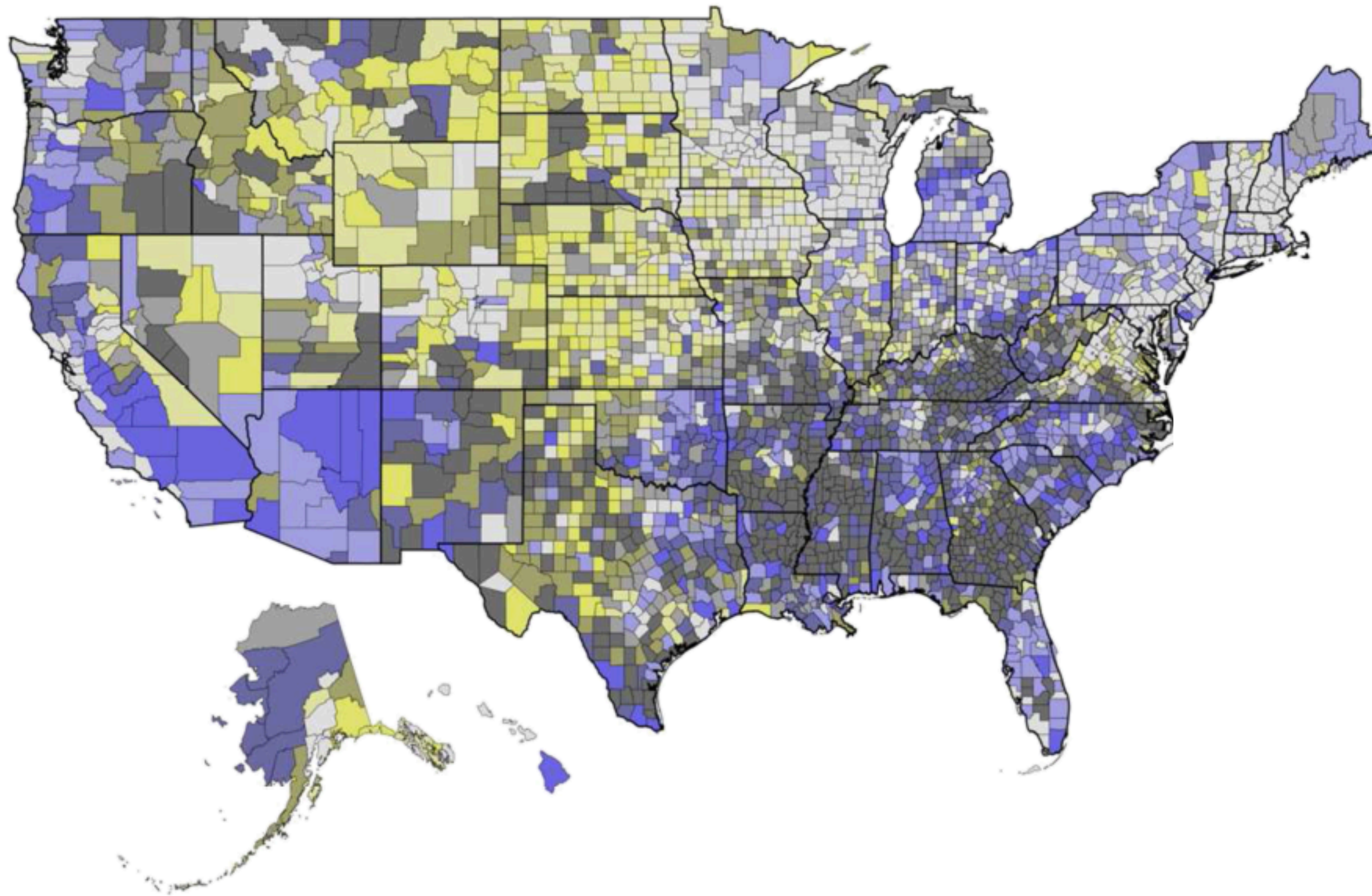


...the jobs report
could look like this:

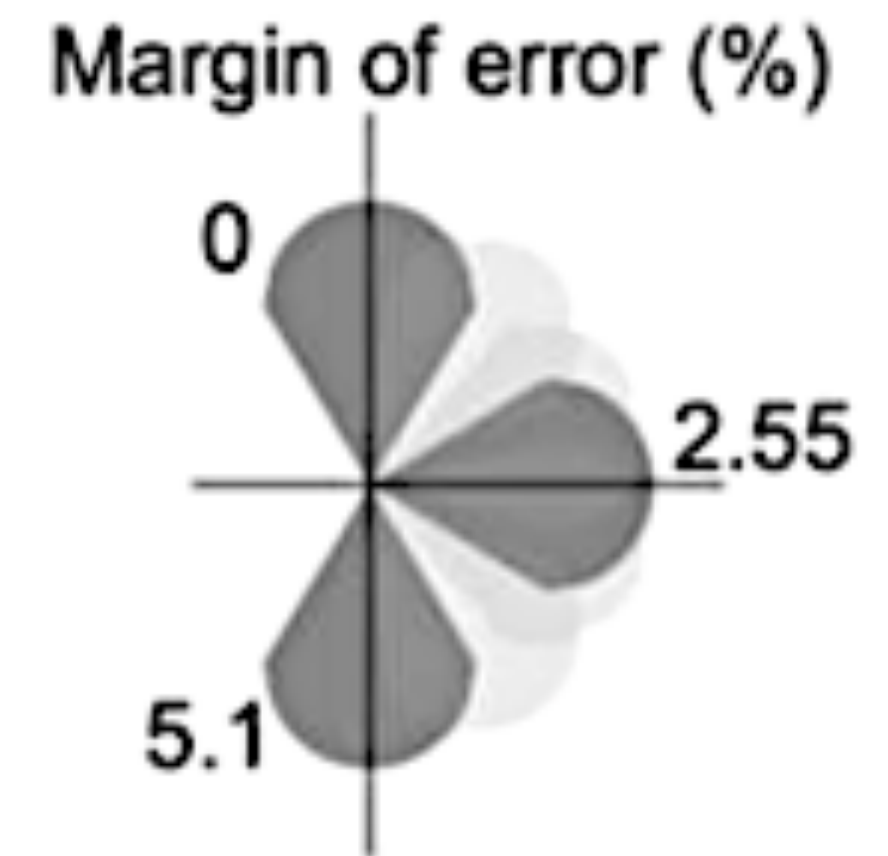
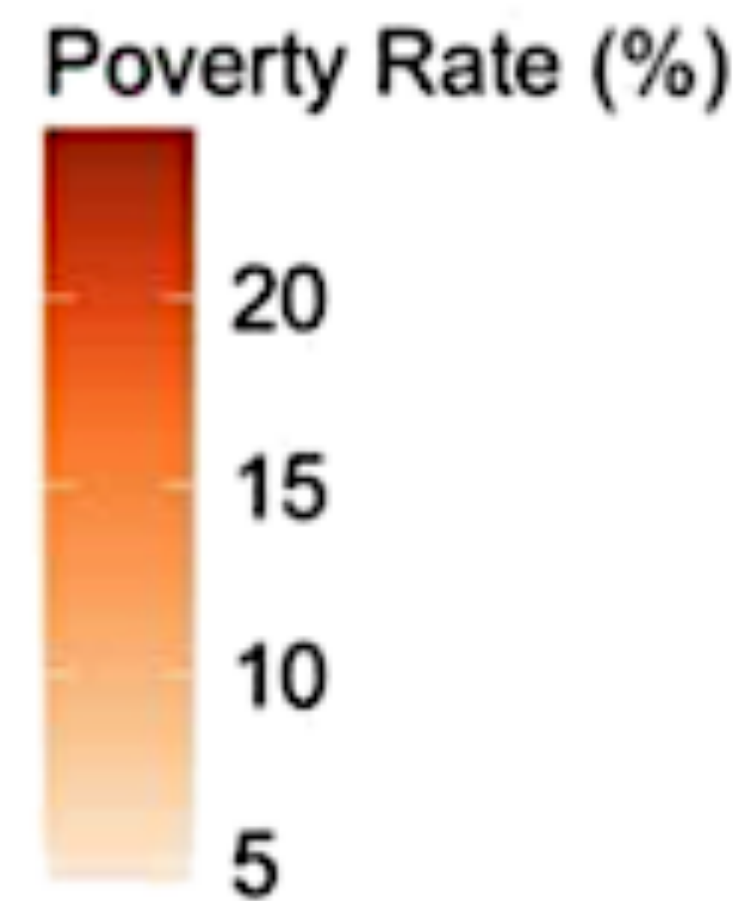
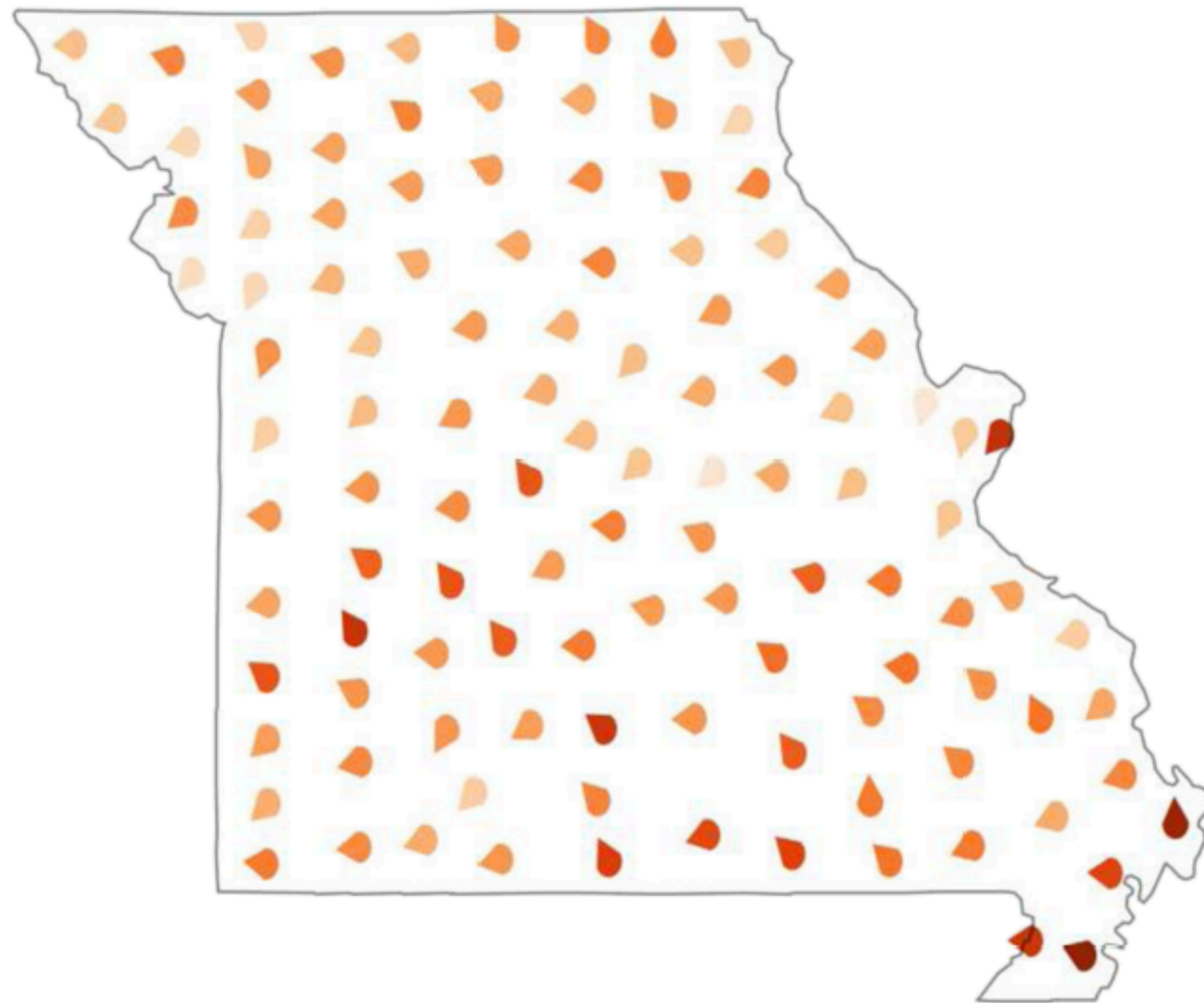


Did you infer a pattern from the chart on the right? If you did, you were reading too much into the jobs numbers. That chart should be a straight line, but we've added the same amount of sampling error that the jobs report has.

Map to another visual channel



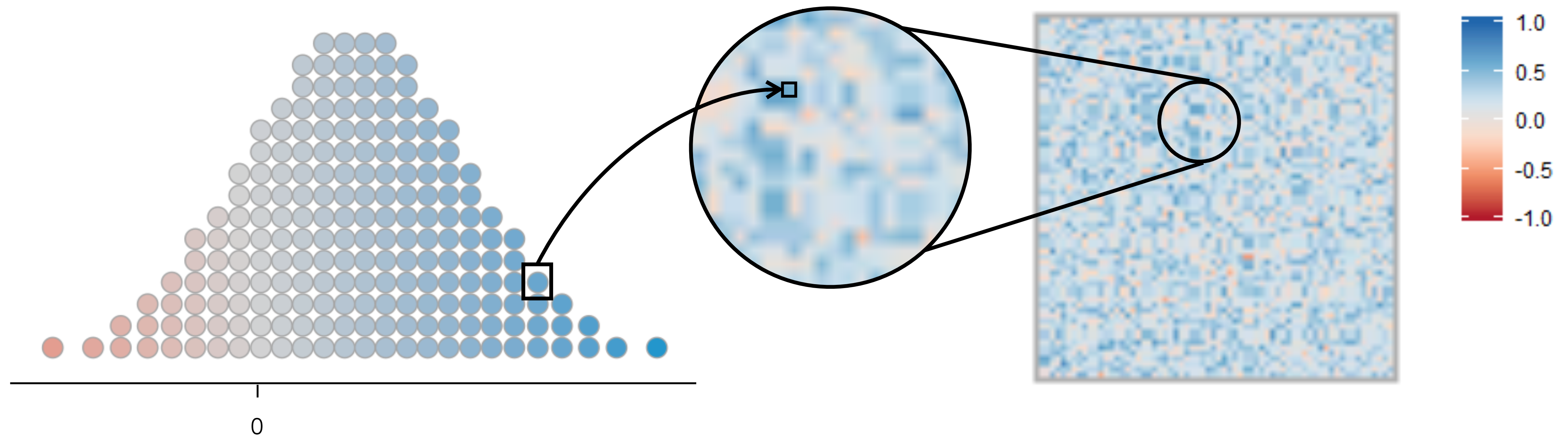
Map to another visual channel



Creative but maybe imaginary....

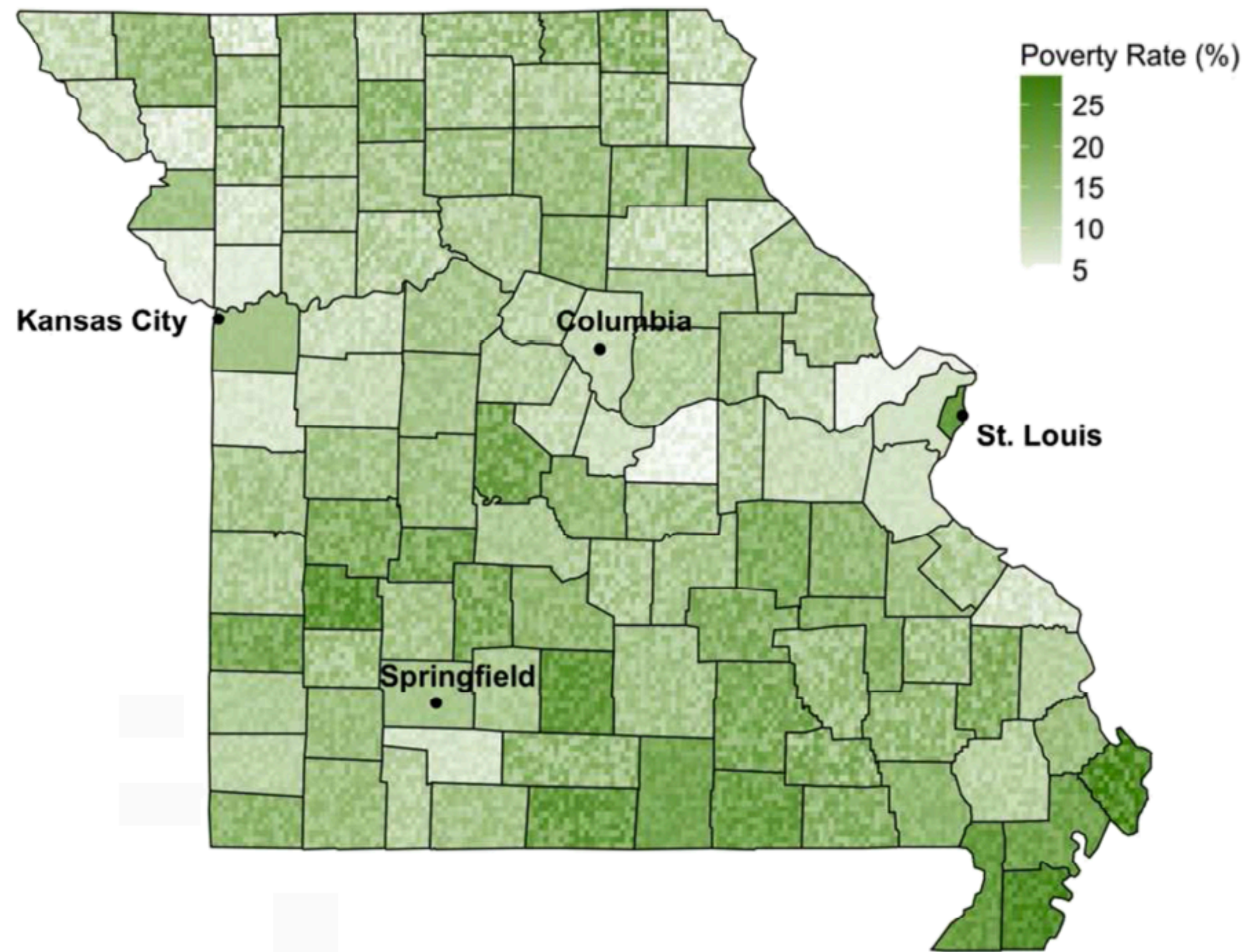
Sample, and map to another visual channel

Illustration purposes. Values might be inaccurate.



This doesn't feel like a bad idea?

Uncertainty → **~dither** (samples from dist)



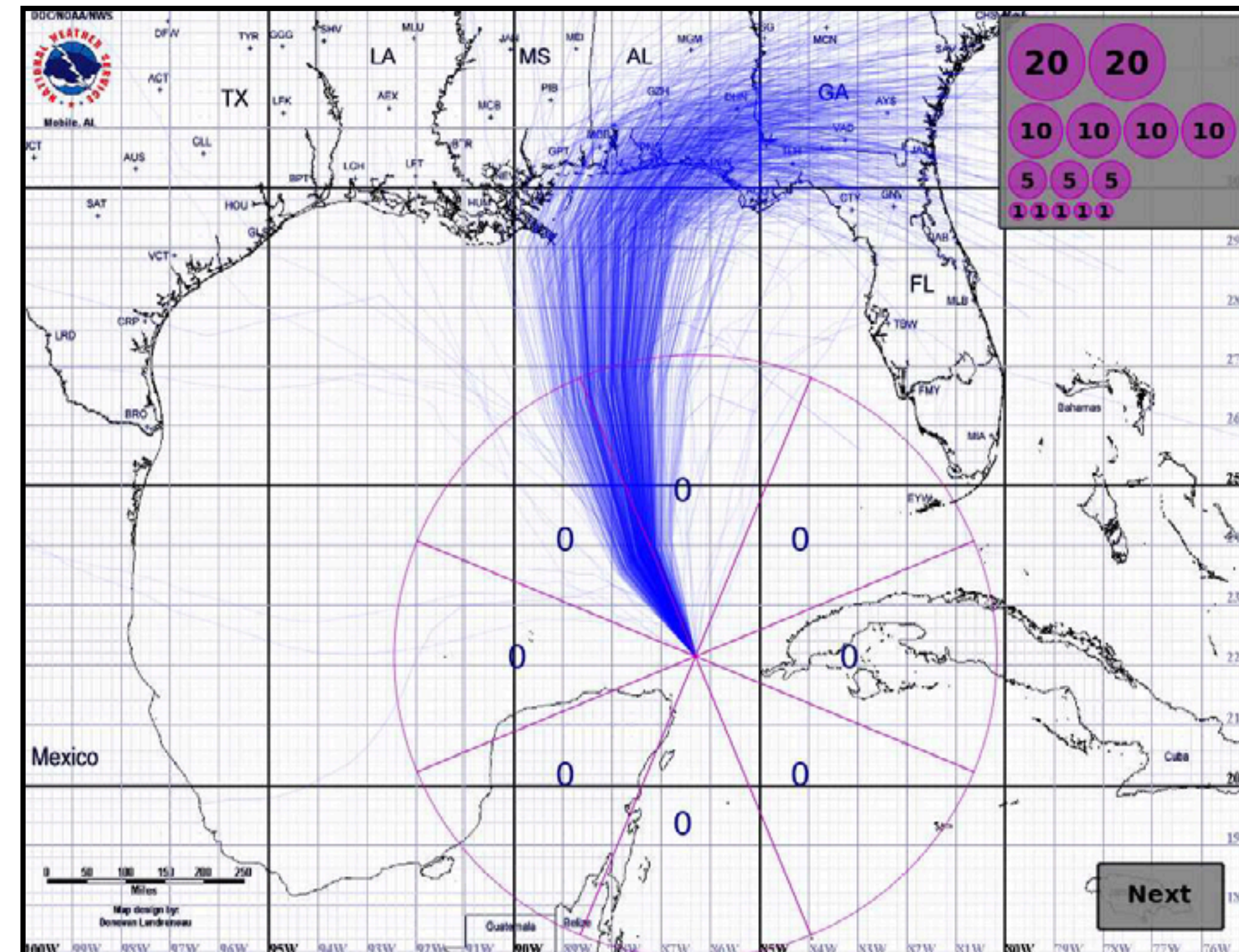
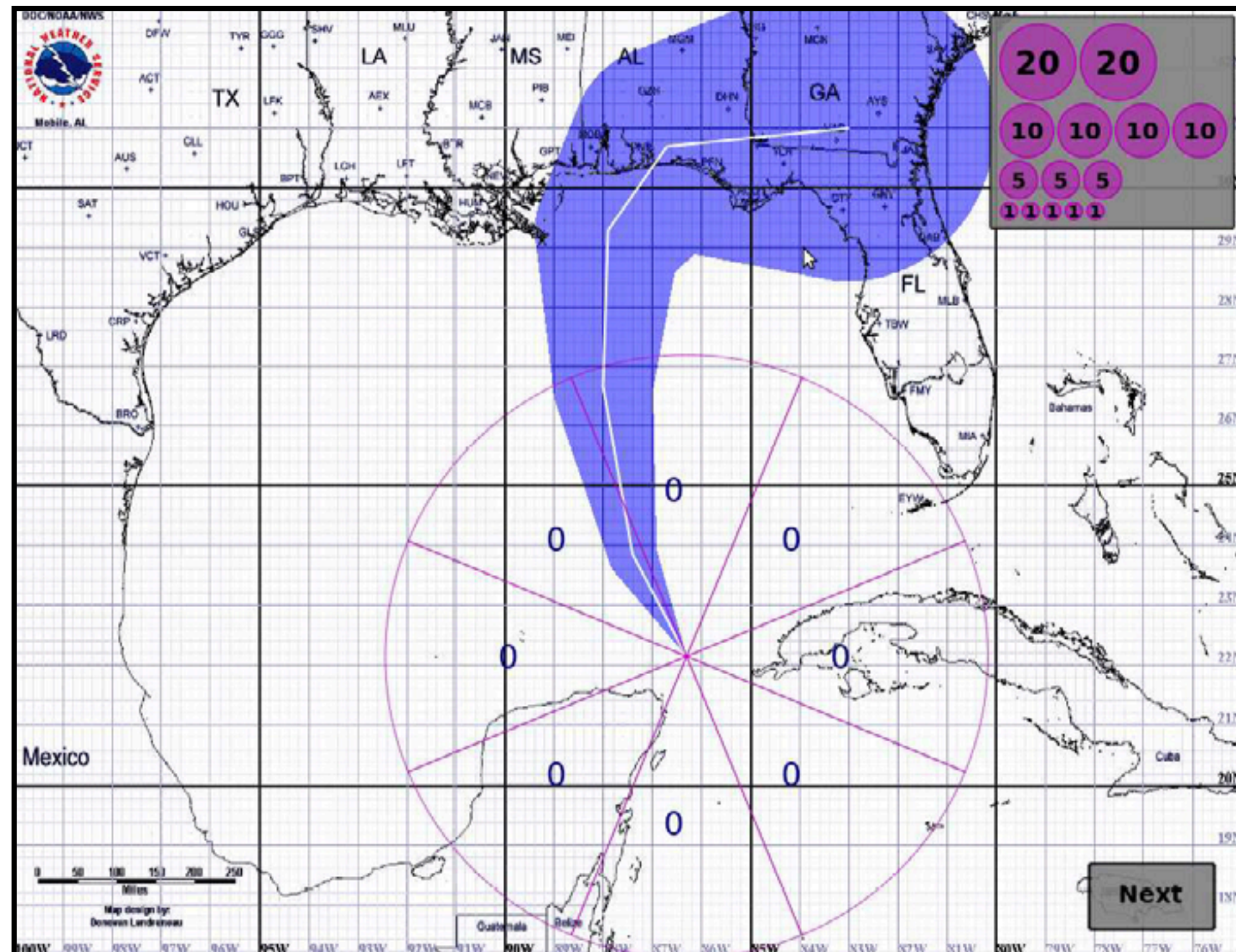
Discrete outcomes

Maybe more intuitive,
maybe less?

Possible **deterministic**
construal errors

Deterministic construal errors

Interpret information about uncertainty as if it were deterministic rather than uncertain

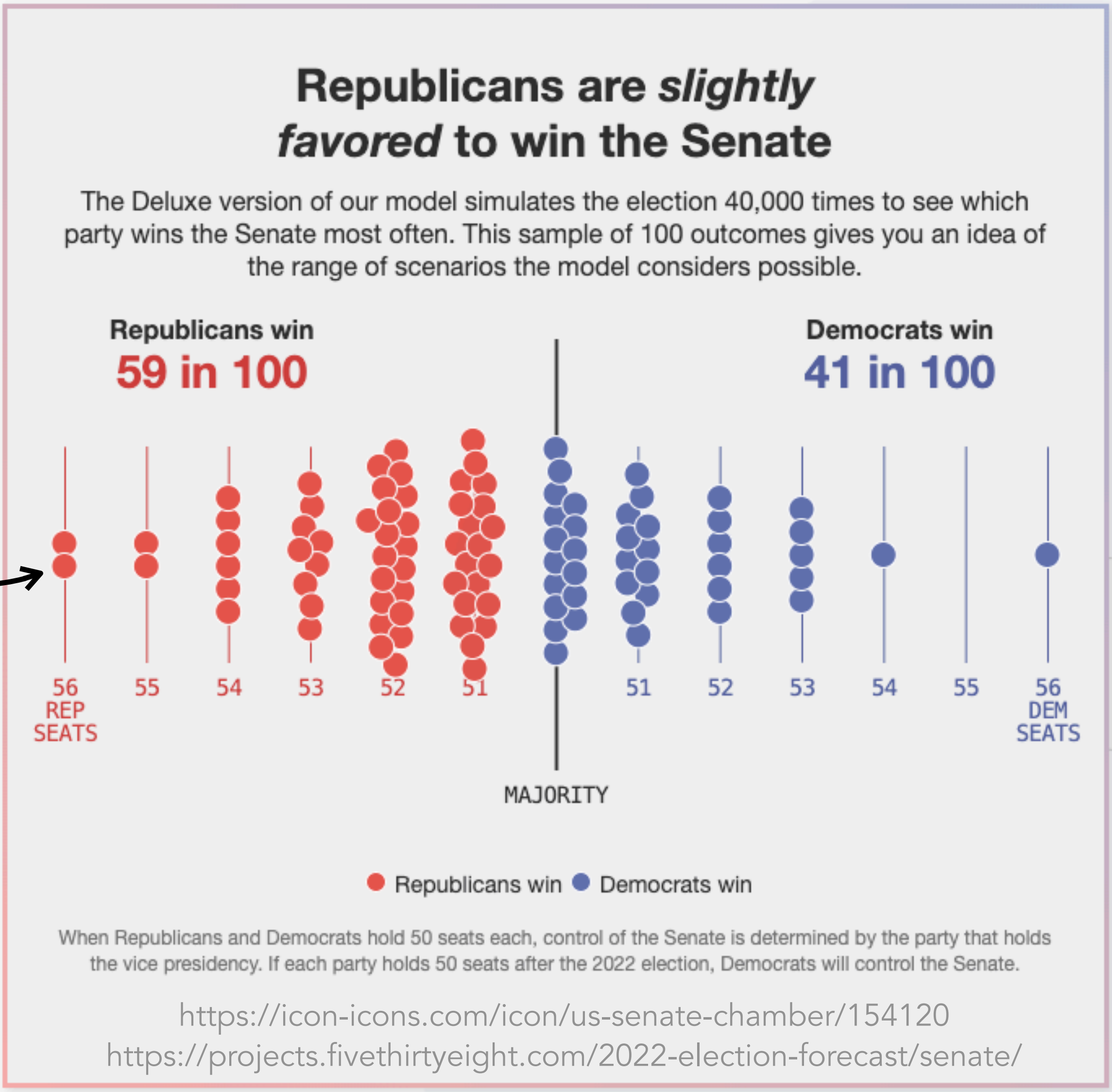


Hurricane error cones: Path? Area?

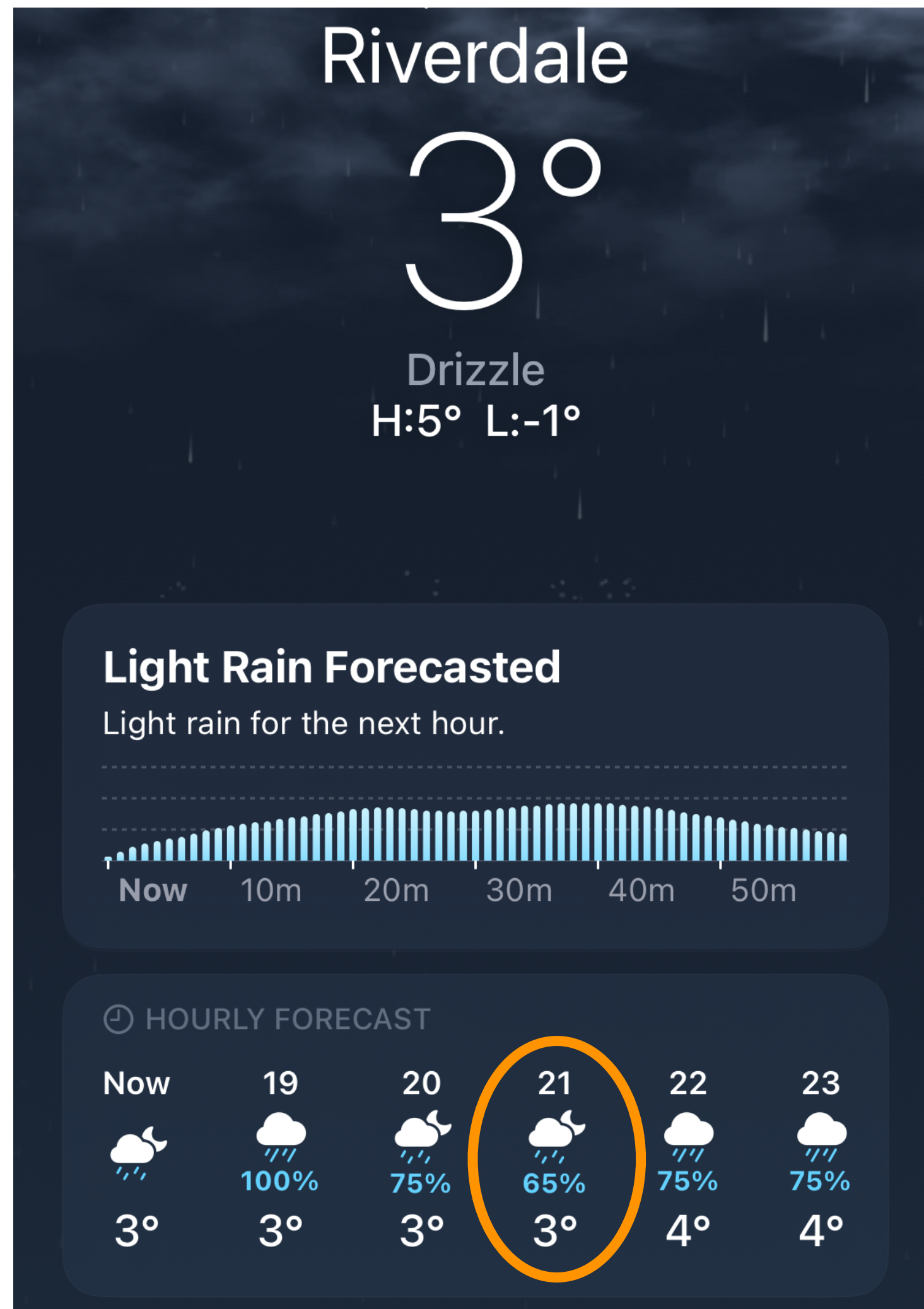
[Joslyn & LeClerc. Decisions With Uncertainty: The Glass Half Full. Current Directions in Psych. Science, 22(4), 2013]

Deterministic construal errors

● = a seat in Senate?



The Probability of Precipitation (PoP)

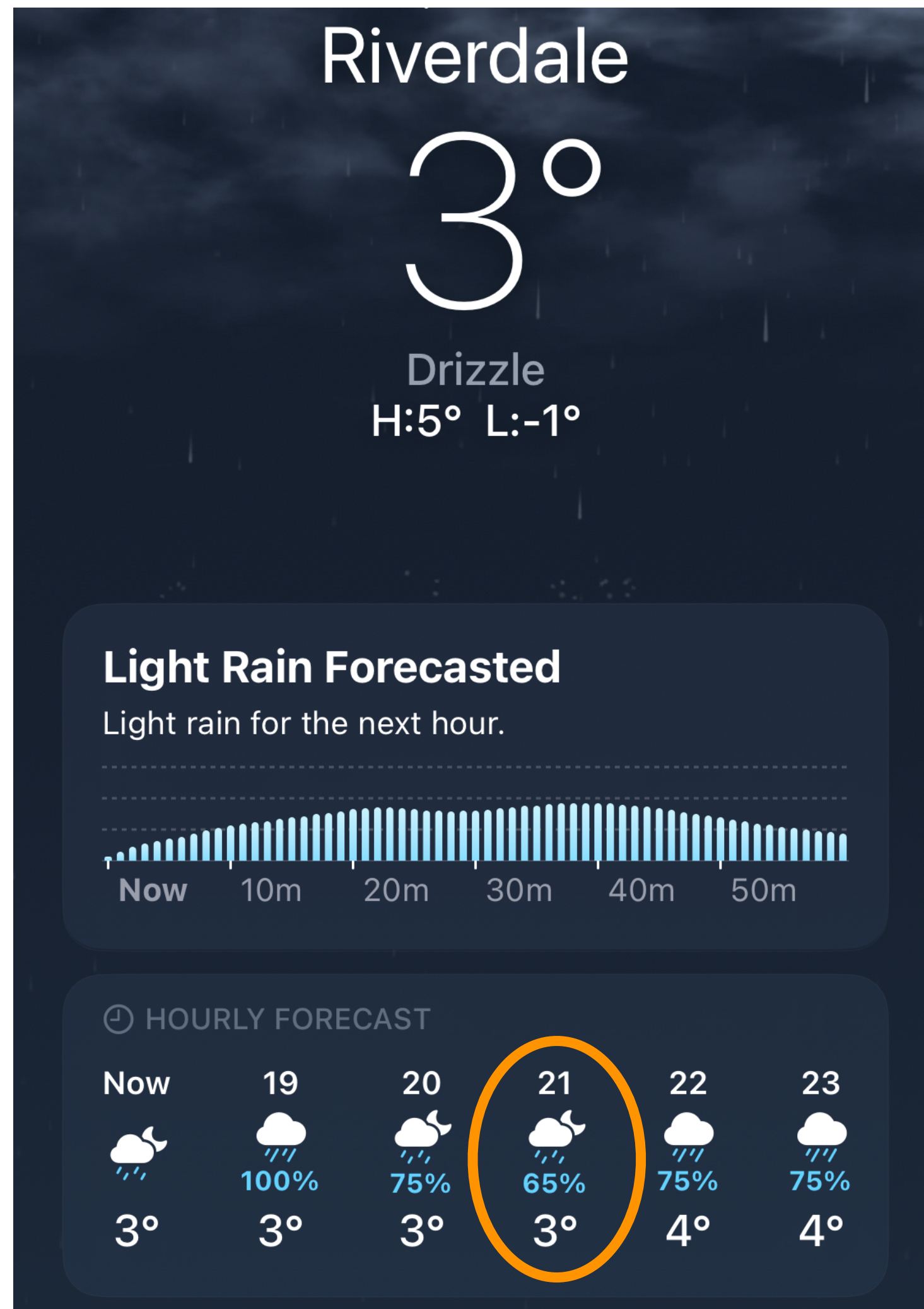


How would you interpret this?

65% of the area will be covered by precipitation at 9pm in Riverdale

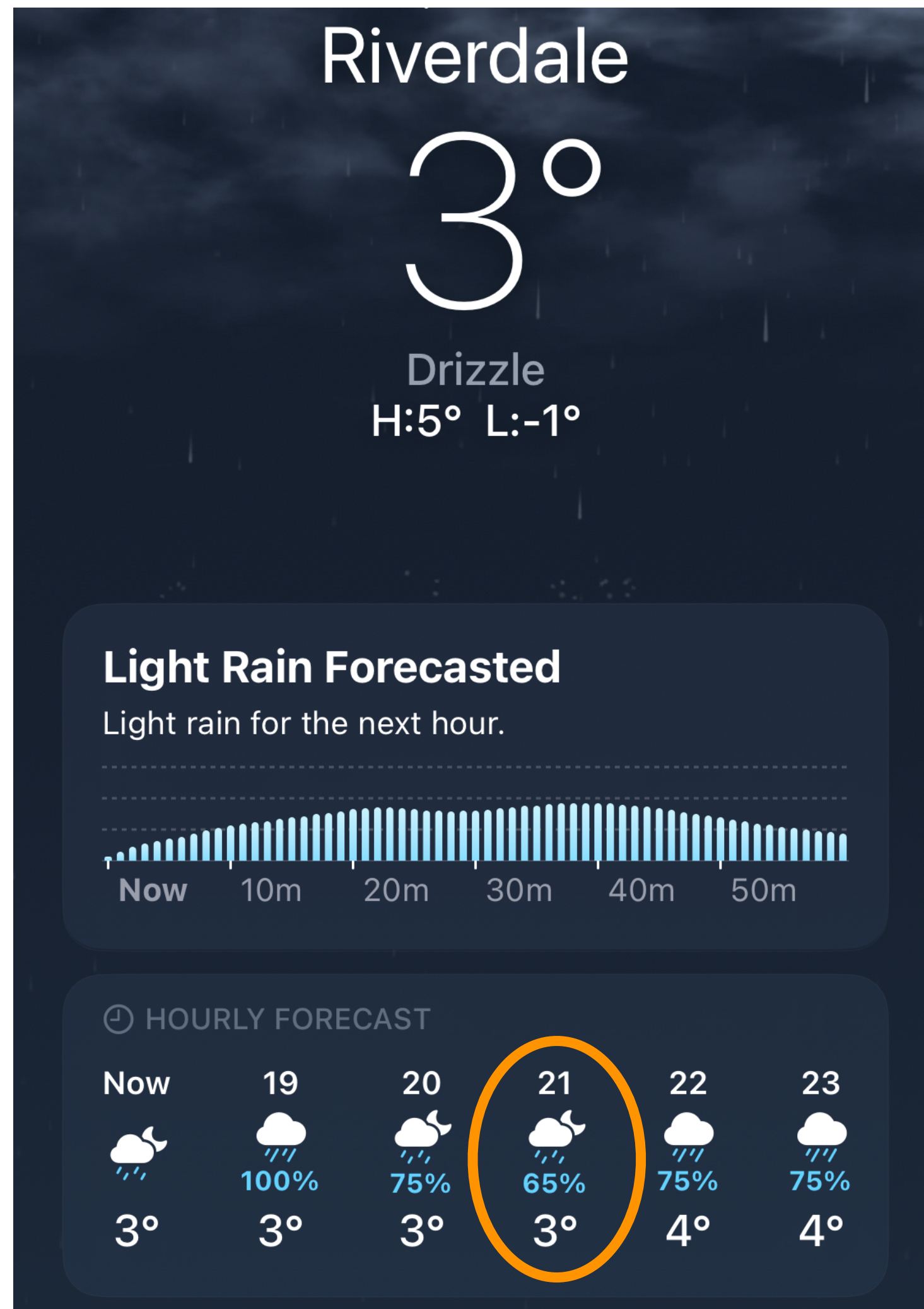
I will be seeing precipitation 65% of the time in Riverdale at 9pm

The Probability of Precipitation (PoP)



The basic definition is the probability of measurable precipitation (0.01 inches or 0.254 mm) occurring at a **location** during a specific **timeframe**.

The Probability of Precipitation (PoP)



PoP = forecaster confidence x area

The forecaster is 100% confident that rain will fall over 40% of the forecast area. ($100\% \times 40\% = 40\%$)

The forecaster is 40% confident that rain will fall over 100% of the forecast area. ($100\% \times 40\% = 40\%$)

The forecaster is 80% confident that rain will fall over 50% of the forecast area. ($80\% \times 50\% = 40\%$)

Well, I keep an umbrella  in my office...

The UncertaintyVis Browser

An Interactive Survey of Uncertainty Visualisation Research Publications.

DOI: 10.1109/PacificVis48177.2020.1014

Styling adapted from Tominski & Aigner's [The TimeViz Browser](#).

of Research Papers: 286

Search:

How to use filters:

- ☒ **Show:** Interested!
- ☐ **Neutral:** I don't care.
- ☐ **Hide:** I'm not interested!

Publication Details

Publication Type

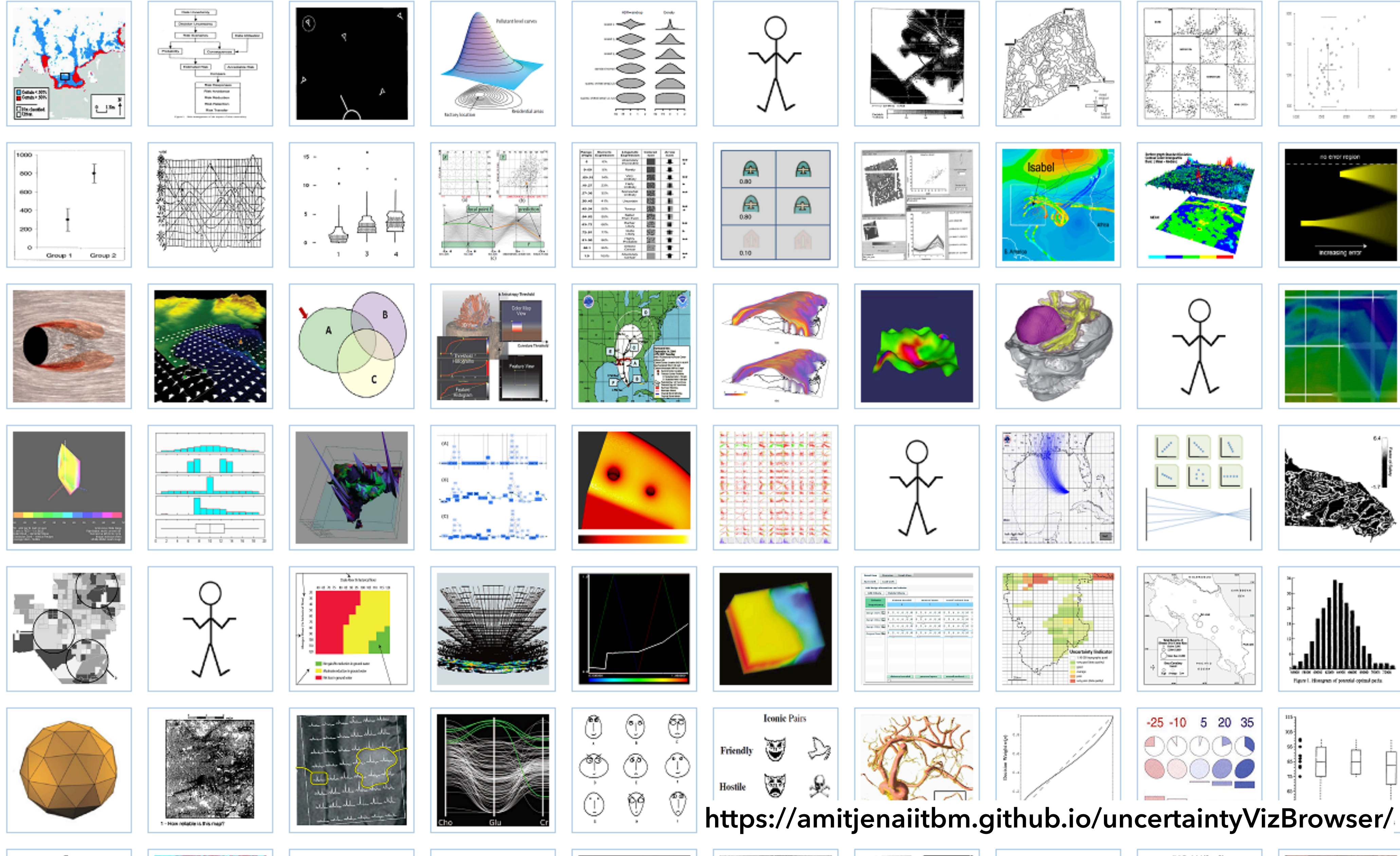
☒ Journal	☒	☒	☒
☒ Conference	☒	☒	☒
☒ Poster	☒	☒	☒
☒ Thesis	☒	☒	☒
☒ Tech Report	☒	☒	☒
☒ Book	☒	☒	☒

Publication Venue

☒ Cartography	☒	☒	☒
☒ Psychology	☒	☒	☒
☒ Health	☒	☒	☒
☒ Human Factors	☒	☒	☒
☒ Decision Scie.	☒	☒	☒
☒ Info Vis	☒	☒	☒
☒ Sci Vis	☒	☒	☒
☒ Cyber Security	☒	☒	☒
☒ Ubi Comp	☒	☒	☒
☒ Statistics	☒	☒	☒
☒ Data Science	☒	☒	☒

Application Domain

☒ Aviation	☒	☒	☒
☒ GIS	☒	☒	☒



<https://amitjenaitbm.github.io/uncertaintyVizBrowser/>

Recap

The difference of win probabilities are large but the race is close

The probability is calculated based on repeated elections/experiments

Quantile dotplot, gradient intervals, spaghetti charts, ...

Lab 12: A different U.S. Election Map

Each cell represents a U.S. state, arranged in a grid layout that approximates their geographic positions. Within each cell, the red line shows the Republican candidate's vote share and the blue line shows the Democrat's vote share across presidential elections from 1976 to 2020.

